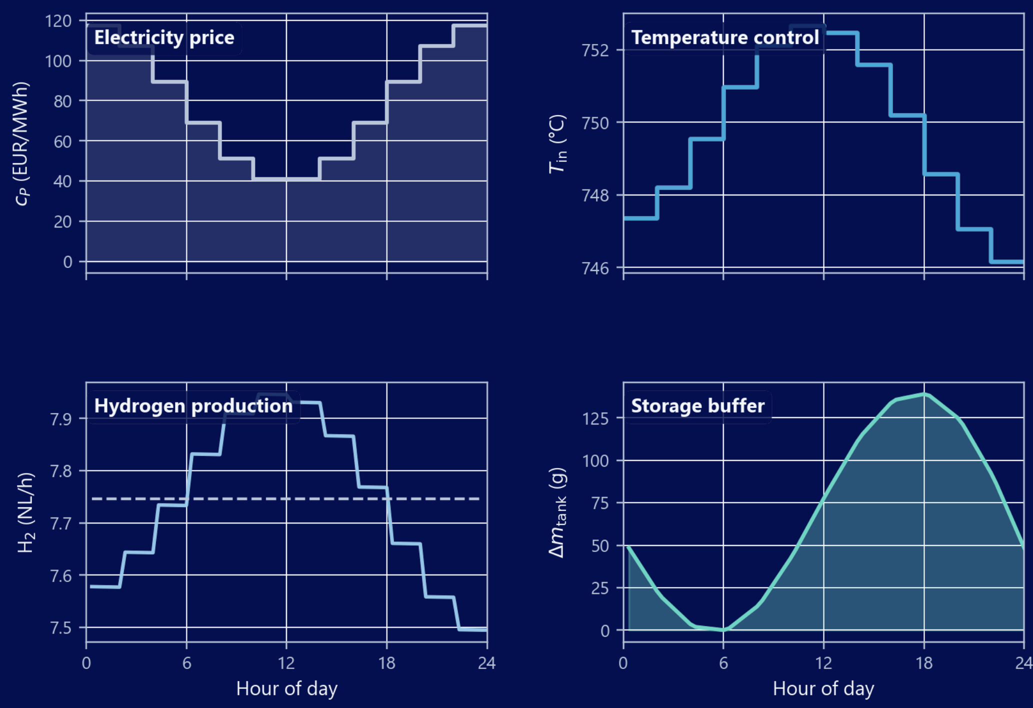
Optimal Control of SOEC's and sensitivity towards degradation

Analysis of the optimal control of SOEC's with physically modelled degradation in a one-dimensional flow model.



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Preface

This thesis was written as part of the MSc programme in Mathematical Modelling and Computation at DTU Compute, Technical University of Denmark (DTU), in collaboration with Dynelectro and DTU Energy. The work combines physical modelling of solid oxide electrolysis cells with optimal control, with the aim of understanding how degradation and market conditions shape economically sensible operating strategies.

I would like to thank my supervisors for their guidance, feedback, and support throughout the project. Henrik Lund Frandsen (DTU Energy) provided overall direction, kept the work grounded in realistic engineering practice, and gave substantial support in the final stretch before submission. Rafael Nogueira Nakashima (DTU Energy) helped me model and understand the physical aspects of the project, in particular the SRU model. Tobias Kasper Skovborg Ritschel (DTU Compute) guided me through the numerical methods, including the finite volume discretization and the adjoint gradients used in the optimization. Finally, Thomas Erik Lyck Smitshuysen (Dynelectro) helped me relate the results to new laboratory data and was a valuable sparring partner in interpreting the optimizer's choices. Their expertise in electrochemical modelling, optimal control, and industrial context has been invaluable.

The source code, simulation scripts, and report build files for this project are available on GitHub at:

https://github.com/LePies/OM_Master2026

The repository will be made public upon submission of this thesis.

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List of Symbols

Symbol	Unit	Description
Geometry, discretization, and time		
A_{cell}	m^2	Active electrode area
A_{int}	m^2	Flow-channel cross-sectional area
d	m	Cell thickness (channel height)
L	m	Channel length
w	m	Channel width
z	m	Axial coordinate along the channel
N	—	Number of axial channel segments
N_{cell}	—	Number of FVM control volumes along z
N_t	—	Number of control / time segments in the NLP
Δz	m	FVM mesh spacing
z_k	m	Centre of FVM cell k
Ω_k, V_k	m^3	FVM cell k and its volume
t	s	Time
t_{horizon}	s or days	Optimization horizon
Δt_k	s	Duration of control interval k
Species, flow, and thermodynamics		
$\mathcal{S}$	—	Gas species $\{H_2O, H_2, O_2\}$
$\mathcal{S}_\zeta$	—	Species on side $\zeta \in \{a, f\}$ (anode / fuel)
$\mathcal{M}$	—	Species index set for mass fractions y_k
c_i, μ_i	mol/m^3	Molar concentration; μ_i along channel (state)
x_i, y_k	—	Molar / mass fraction of species i or k
N_i, $N_{c,i}$	$mol/(m^2{\cdot}s)$	Molar flux; electrochemical source flux
$\dot{n}_i$, $\dot{V}_\zeta$	mol/s, m^3/s	Molar flow rate; volumetric flow on side ζ
v_{fuel}, v_{air}	m/s	Gas velocity on fuel / air side
p_i, P	Pa	Partial / total pressure
p_{ref}, p_{air}	Pa	Reference and air-side pressures
ρ	kg/m^3	Gas density
$c_{p,i}$, c_p, C_p	J/(mol·K), $J/(K{\cdot}m^3)$	Species, mixture, and volumetric heat capacity
$h_i(T)$, Δh	J/mol	Species enthalpy; reaction enthalpy change
Q_T	W/m^3	Volumetric heat source
Electrochemistry		
F	C/mol	Faraday constant
n_e	—	Electrons per reaction ($n_e = 2$)
ν_i	—	Stoichiometric coefficient of species i
i	A/m^2	Current density
I, I_{cell}	A	Total cell current
U	V	Applied cell voltage (control)
U_{th}	V	Thermal (reversible) cell voltage

Symbol	Unit	Description
η_{cell}	V	Cell overpotential ($\eta_{\text{cell}} = U$)
$\eta_{\text{nernst}}, \eta_{\text{act}}, \eta_{\text{ohm}}$	V	Nernst, activation, and ohmic overpotentials
$\eta_{\text{act}}^{\zeta}$	V	Activation overpotential on side ζ
i_0^{ζ}	A/m^2	Exchange current density on side ζ
$R_{\text{deg}}, R_{\text{ohm}}$	$\Omega\cdot\text{m}^2$	Degradation and total ohmic resistance
R_{contact}	$\Omega\cdot\text{m}^2$	Contact resistance
B_{ohm}	$\Omega\cdot\text{m}^2/\text{K}$	Base factor in intrinsic ohmic term
$E_{\text{act}}^{\text{ohm}}$	J/mol	Activation energy in intrinsic ohmic term
ΔG	J/mol	Gibbs free energy change
Degradation		
$\mathcal{D}$	—	Set of slow degradation mechanisms
$\mathcal{D}_{\text{ohm}}$	—	Subset in R_{ohm} (Ni-mi, IC-ox)
$\mathbf{d}$, d_i	—	Degradation state vector and component i
$h_{\text{Ni-mi}}, h_{\text{IC-ox}}$	m	Ni-migration gap; interconnect oxide thickness
$s_{\text{IC-ox}}$	m^2	Squared IC-oxide thickness (ODE state)
$d_{i,\text{critical}}$	m	Critical degradation limit for mechanism i
$R_{\text{Ni-mi}}, R_{\text{IC-ox}}$	$\Omega\cdot\text{m}^2$	Ohmic contributions from Ni-mi and IC-ox
$\sigma_{\text{elec}}, \sigma_{\text{elec},0}$	S/m	Electrode conductivity and pre-factor
$\sigma_{\text{IC-ox}}$	S/m	Oxide-layer conductivity
$E_{a,\text{elec}}, E_{a,\text{ox}}, E_{\text{el}}$	J/mol	Activation energies (electrode, IC ox., IC cond.)
$c_1, \ldots, c_4$	various	Fitted Ni-mi growth-rate coefficients
$k_{\text{mg}}, k_{\text{ox}}$	various	IC-oxidation rate and conductivity prefactors
ε, τ	—	Electrode porosity and tortuosity
Optimal control and economics		
$\mathbf{u}(t), \mathcal{U}_{\text{ad}}$	—	Control vector (U, T_{in}); admissible (constrained) control space
$T_{\text{in}}, T_{\text{out}}, T_{\text{loss}}$	K	Inlet (control), outlet, and thermal recovery loss
$R(\mathbf{u}, \mathbf{d}), R_{\text{tank}}$	EUR/s	Revenue rate; contract rate with tank
$P_{\text{cell}}, P_{\text{heat}}$	W	Stack power; auxiliary sensible heating
$\varphi, \Psi(\mathbf{d}), \Psi_{\text{tank}}$	EUR	Integrated revenue; salvage; tank liquidation
$c_{\text{H}_2}, c_P, c_{P,k}$	EUR/kg, EUR/kWh	Hydrogen and power prices
$\bar{c}_P, a$	—	Mean daily power price and relative swing
c_{cell}	EUR/m^2	Cell capital cost in Ψ
$\alpha(t), \alpha_{\Psi}$	—	Schematic weight; salvage exponent
$\kappa_{\text{buy}}, \kappa_{\text{sell}}$	—	Spot-purchase penalty; liquidation multiplier
$m_{\text{tank}}, n_{\text{purchase},k}$	mol, kg	Tank inventory; shortfall purchase
$\dot{n}_{\text{H}_2,\text{agreement}}, \dot{n}_{\text{prod}}, \dot{n}_{\text{purchase}}$	mol/s	Delivery, production, and purchase rates
$\Delta\dot{n}_{\text{H}_2}$	mol/s	Net H_2 production in R
PDE / FVM notation		
u, Q_u, F_u	various	Generic FVM field; source and flux terms
m_k	kg	Gas mass in FVM volume k
Greek symbols		
R	J/(mol·K)	Universal gas constant
α_{ζ}	—	Butler–Volmer symmetry factor; $\alpha_a = 0.65$, $\alpha_f = 0.59$

Symbol	**Unit**	**Description**
γ_ζ	A/m^2	Exchange-current pre-factor; $\gamma_a = 1.515{\times}10^8$, $\gamma_f = 7.666{\times}10^5$
E^ζ_{act}	J/mol	Electrode activation energy; $E^a_{\text{act}} = 1.40{\times}10^5$, $E^f_{\text{act}} = 1.05{\times}10^5$
m_i	—	Species exponent in i^ζ_0; $m_{H_2O} = 0.33$, $m_{H_2} = -0.1$, $m_{O_2} = 0.22$
λ_a, λ_c	—	Exchange-current multipliers ($\lambda_a = 1$; $\lambda_c \approx 0.55$)
λ_T	W/(m·K)	Thermal conductivity (neglected)
Ω	m	Spatial domain
x_0	—	Reference molar-fraction scale ($p_{\text{ref}}/p_{\text{air}}$)

Greek parameter values follow table 2.1 and table 2.2 unless stated otherwise.

Abstract

This thesis studies the sensitivity of optimal control for a solid oxide electrolysis cell (SOEC) over its operating lifetime, with degradation included as an economic parameter. The plant is represented at system scale by an average stack repeat unit (SRU) with a lumped heat exchanger: Steady-state electrochemical and thermal fields along the gas channels (solved with a finite volume method) are coupled to differential degradation laws for nickel migration and interconnect oxidation. The model uses implicit Euler for forward time integration. In addition to this, an optimal control problem is formulated to maximize integrated revenue (hydrogen sales minus electricity cost) plus a terminal *salvage* term penalizing degradation. Cell voltage U and inlet temperature T_{in} are the primary controls; the resulting nonlinear program is solved by gradient-based optimization (SLSQP/Trust-Constr) using single-shooting forward replay.

Three scenarios examine how much temporal control freedom matters and what drives the optimum. In an unrestricted 200-day baseline, the optimizer selects constant potentiostatic-isothermal (pot-iso) operation at an exothermal set-point ($U^* \approx 1.32\,\text{V}$, $T_{\text{in}}^* \approx 751\,^\circ\text{C}$); granting additional segment-wise freedom in U and T_{in} does not change the optimal trajectory. Nickel migration reaches its critical degradation limit at the horizon end, while interconnect oxidation remains secondary. When a fixed hydrogen delivery contract and buffer tank are introduced, inventory-aware operation becomes relevant: The optimizer alternates gentle pot-iso periods with galvanostatic intervals dictated by tank inventory and penalty prices, at the cost of lower average net economy. A final 24-hour case with a cyclic power price and cyclic tank constraint shifts production toward low-price midday intervals mainly by modulating T_{in} while voltage stays essentially fixed.

Current-voltage validation, comparison of degradation trajectories with published reference simulations, and gradient checks support using the toolchain for qualitative optimal-control studies at engineering fidelity. The main conclusion is that optimal SOEC operation is scenario-dependent: Unrestricted operation favors steady potentiostatic-isothermal controls at exothermal voltage; a fixed delivery contract introduces tank-mediated galvanostatic intervals at thermoneutral voltage; and a cyclic power price is met mainly by modulating inlet temperature while voltage stays near the thermoneutral value.

Resumé

Optimal styring af SOEC'er og følsomhed over for degradering

Denne afhandling undersøger følsomheden af optimal regulering for en fastoxid elektrolysecelle (SOEC) over celles levetid, hvor degradation indgår som en økonomisk parameter. Anlægget modelleres på systemniveau som en gennemsnitlig stack repeat unit (SRU) med lumpet varmeveksler: Stationære elektrokemiske og termiske felter langs gaskanalerne (løst med finit-volumen-metoden) kobles til differentialligninger for nickelmigration og interconnect-oxidation. Modellen anvender implicit Euler til fremadrettet tidsintegration. Derudover formuleres et optimalt reguleringsproblem, der maksimerer integreret indtægt (brintsalg minus elomkostning) plus et terminalt *salvage*-led, der straffer degradation. Cellespænding U og indløbstemperatur T_{in} er de primære reguleringsvariable; det resulterende ikke-lineære program løses med gradientbaseret optimering (SLSQP/Trust-Constr) og single-shooting fremadafvikling.

Tre scenarier undersøger, hvor meget tidsmæssig reguleringsfrihed betyder, og hvad der driver optimum. I et ubegrænset 200-dages baseline-scenarium vælger optimizoren konstant potentiostatisk-isoterm (pot-iso) drift ved et exotermt arbejdspunkt ($U^* \approx 1.32\,\text{V}$, $T_{\text{in}}^* \approx 751\,^\circ\text{C}$); yderligere segmentvis frihed i U og T_{in} ændrer ikke den optimale trajektorie. Nickelmigration når brudgrænsen ved horisontens slutning, mens interconnect-oxidation forbliver sekundær. Når en fast brintleveringskontrakt og buffertank indføres, bliver lagerbevidst drift relevant: Optimizoren veksler mellem milde pot-iso-perioder og galvanostatiske intervaller styret af tankbeholdning og strafpriser, på bekostning af lavere gennemsnitlig nettoøkonomi. Et afsluttende 24-timers scenarium med cyklisk elpris og cyklisk tankbegrænsning flytter produktionen mod billige middagsintervaller hovedsageligt ved modulering af T_{in}, mens spændingen forbliver næsten konstant.

I-V-validering, sammenligning af degradationstrajektorier med publicerede referencesimulationer og gradienttjek understøtter brugen af værktøjsketten til kvalitative optimal-reguleringsstudier på ingeniørniveau. Hovedkonklusionen er, at optimal SOEC-drift er scenariosafhængig: Ubegrænset drift favoriserer stationær potentiostatisk-isoterm regulering ved exoterm spænding; en fast leveringskontrakt introducerer tankmediere galvanostatiske intervaller mod termoneutral drift; og en cyklisk elpris imødekommes hovedsageligt ved modulering af indløbstemperaturen, mens spændingen holdes nær den termoneutrale værdi.

1 Introduction

Most hydrogen produced today is derived from fossil fuels. This so-called grey or black hydrogen remains the cheapest route in direct production cost, but carries a large climate penalty through fossil carbon emissions [6]. Solid oxide electrolysis (SOE) offers an alternative: High-temperature electrolysis in a solid oxide electrolysis cell (SOEC) can convert steam into hydrogen using electricity with high efficiency [10]; when the electricity is low-carbon, the product qualifies as green hydrogen. The technology is not new, but capital and operating costs have kept it less competitive than fossil routes on open markets. Substantial effort is therefore directed at lowering the levelized cost of hydrogen through improved materials, scale-up, and smarter operation [4, 14].

A practical consequence is that many early green-hydrogen projects are structured around long-term delivery contracts between buyers and producers rather than spot sales alone. This setting matters for operation: The cell must meet an agreed hydrogen flow while electricity prices, inventory, and stack health evolve over time. Degradation is central to that trade-off. Prolonged high-temperature operation increases area-specific resistance, raises power demand for the same output, and eventually forces stack replacement [2, 3]. Recent work has started to couple degradation models with economic objectives and optimal control. It remains open how much temporal freedom in voltage and temperature actually changes the optimum, and how contractual delivery or time-varying prices reshape it.

1.1 Literature review

Preceding this thesis, degradation-aware modelling and economic optimization of SOEC operation have recently been studied mainly at stack or plant scale; the main results are summarized below.

Beyrami et al. [2] find that galvanostatic operation turns exothermic as resistance grows and potentiostatic operation stays nearer thermoneutral, whereas potentiostatic–galvanostatic (PG) operation limits area-specific resistance increase over long horizons, maintains stable efficiency, and emerges as the optimal operating mode. In follow-on work they report more than $50{,}000\,\mathrm{h}$ lifetime at $0.6\,\mathrm{A/cm^2}$ when optimizing against component failure criteria [3]. At plant scale, degradation accounts for roughly 8% of hydrogen production cost, with economically optimal current densities of 0.6-$0.7\,\mathrm{A/cm^2}$ and 4-5 year stack replacement [4].

Giridhar et al. [14] show that dynamic optimisation over extended horizons lowers levelized cost of hydrogen relative to single-objective strategies. Flexible schedules outperform fixed galvanostatic or potentiostatic operation, with replacement intervals of roughly 2-5 years depending on electricity price and mode. Cai et al. [8] maximize hydrogen output or minimize energy use under temperature constraints when pairing SOEC operation with intermittent renewables.

For model calibration, Rizvandi et al. [24] provide commercial-cell polarization data; Larrain [17] and Brylewski et al. [7] quantify interconnect oxidation kinetics; Skafte et al. [25] define electrothermally balanced reference conditions (figure 3.3, figure 3.5).

Several shortcomings remain. Most degradation-aware optimisation studies are formulated at stack or plant scale and compare a small set of fixed operating modes (galvanostatic, potentiostatic, or PG), rather than treating cell voltage and inlet temperature as freely co-optimised controls at repeat-unit resolution. Contractual hydrogen delivery, buffer storage, and time-varying power prices enter only indirectly, if at all. Spatial models in the literature are often two-dimensional, whereas tractable optimal-control studies typically rely on one-dimensional channel models such as that of Beyrami et al. [2]. Against this background, it is still unclear how much extra temporal freedom in U and T_{in} changes the optimum once degradation and external obligations are included together.

1.2 Introduction to the thesis

This thesis examines how optimal operating conditions for a solid oxide electrolysis cell (SOEC) change when cell voltage and inlet temperature are free to vary independently over time, a fixed hydrogen delivery contract applies, and electricity prices vary in time. Unlike Giridhar et al. [14], who optimize stack-scale schedules by switching between galvanostatic and potentiostatic modes for levelized hydrogen cost, the present formulation allows joint voltage–temperature control. It further resolves nickel migration and interconnect oxidation as slow states, each subject to a **critical degradation** limit: a preset maximum value of that degradation measure (layer thickness for nickel migration or interconnect oxidation) beyond which the repeat element is treated as failed and must be replaced. Buffer storage is added to meet offtake obligations. The work is organized around the following question:

> *How does physically modelled degradation influence the optimal control of an SOEC when electrochemistry and temperature of the cell are captured in a one-dimensional flow model? To what extent do external operating constraints and time-varying inputs reshape that optimum?*

The analysis targets system-scale operation, but the plant is represented by a simplified average stack repeat unit (SRU) plus a lumped heat exchanger (HEX). Within that SRU, a steady-state electrochemical and thermal model along the gas channel, largely following Beyrami et al. [2], is solved with a finite volume method and coupled to slow degradation states for nickel migration and interconnect oxidation. Preheat and heat recovery enter through the auxiliary sensible-heating term P_{heat} rather than a resolved HEX model; facility-level quantities follow by scaling the per-SRU result (chapter 2).

The primary controls are cell voltage U and inlet temperature T_{in}. The central question is whether the optimizer uses both degrees of freedom, or whether a restricted potentiostatic-isothermal mode remains optimal. Three scenarios are studied over horizons of days rather than multi-year stack lifetimes, reflecting real operation where future electricity prices and offtake cannot be known with certainty:

1. **Unrestricted baseline**: 200-day horizon without delivery or price constraints.
2. **Constant delivery contract**: Fixed hydrogen offtake with buffer storage.
3. **Daily cyclic operation**: 24-hour horizon with a time-varying power-price profile.

The governing physics and degradation models are presented in table 2.1 and chapter 2. The optimal-control formulation is developed in chapter 3 (equation (3.1)). The numerical results are reported in chapter 4 to section 4.3.

The discussion in chapter 5 to chapter 7 interprets how much control freedom is actually used, whether the implied lifetimes are plausible, and which modelling choices bound the conclusions. Together, the thesis provides a system-scale toolchain, built on an average SRU representation with lumped heat exchange, for comparing baseline, delivery-constrained, and price-driven operation under a common degradation-aware objective.

2 Physical Modelling

2.1 Modelling govering SOEC equations

In this chapter the aspects of solid oxide electrolysis cell's (SOEC's) will be introduced. Briefly visiting different scales of operation, before a more detailed description of how the mole flow is modelled, completing the section with energy balances that determines the behavior of temperature through the average cell.

The basics are presented with the assumption that the readers have a background in applied mathematics. Derivations throughout the chapter therefore assume familiarity with standard intermediate steps; lengthier or less routine derivations are given in the main text or in an appendix. Readers who wish to proceed directly to the governing equations will find them summarized in table 2.1.

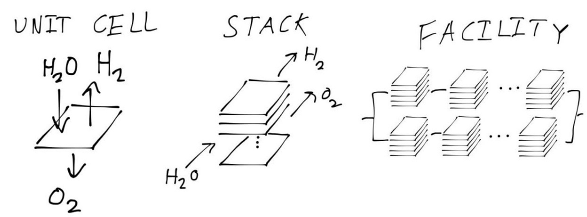


Figure 2.1: A sketch of the scales: The three different scales mentioned in this paper from smallest (left) to largest (right)

The three different scales of SOEC discussed in this paper are defined as follows. The first and smallest scale is the unit cell, where the focus is only on the dynamics occurring within a single SOEC cell.

Next scale is the stack. At this level, multiple unit cells are stacked together, which naturally introduces more complexity. Figure 2.1 is a quick visualization of the different scales.

Finally, the largest scale considered is at facility scale. At this level, multiple SOEC stacks are connected together; the facility operates much like an electrical circuit, with each stack behaving as its own component with distinct characteristics.

Although the application target is facility-scale operation, the plant is not modelled stack by stack. A **repeat element** is one repeating stack layer: a single electrochemical cell together with its fuel and air flow channels. Every stack is taken to be the same and every repeat element within every stack is identical. Instead, one **average stack repeat unit (SRU)**, together with a lumped heat exchanger (HEX), stands in for the full system (figure 2.1, left). **Lumped heat exchange** means the plant heat exchangers are not modelled in detail: Their net effect, raising inlet gases to T_{in} and recovering sensible heat from the outlet streams, is captured by the auxiliary term P_{heat} rather than a spatial heat-exchanger model. If the plant contains, for example, $5{,}000$ such repeat elements, a facility-scale quantity is obtained by multiplying the per-SRU result by $5{,}000$. This chapter develops that single-SRU formulation step by step, from channel-scale transport and electrochemistry to the energy balance that sets the temperature field.

2.1.1 Basics of SOEC

Solid oxide electrolysis couples steam splitting, ion transport through a dense electrolyte, and oxygen release at the opposing electrode. Figure 2.2 summarizes these processes at repeat-element scale [26].

The cell is composed of three main components: The **cathode**, the **electrolyte**, and the **anode**. At the cathode, water molecules are split into dihydrogen (H_2) and oxide ions (O^{--}). This process is only possible due to the presence of electrons provided by an external electric current: These electrons participate directly in the reaction at the cathode, facilitating the reduction of water.

The newly-formed oxide ions then migrate through the solid **electrolyte**, propelled by the electric field generated by the applied voltage, highlighting the central role of electrons and electrical energy in driving the overall process. Once the oxide ions reach the **anode**, they release their excess electrons and recombine to form dioxygen (O_2). The electrons released travel through the external circuit, closing the electrical loop.

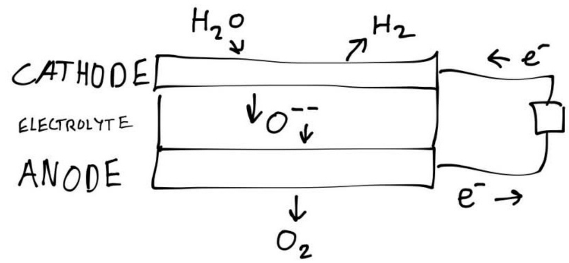


Figure 2.2: A sketch of a unit cell: The basic principles of an SOEC cell are illustrated. The cathode pulls apart the hydrogen and oxygen in the water. The oxygen is transferred through the electrolyte, then formed to dioxygen in the anode, all due to an applied current.

Summing up, the operation of the SOEC is fundamentally tied to the movement of electrons, both in the chemical reactions at the electrodes and in sustaining the ionic and electronic currents through the device. This is captured in the overall reaction, $2H_2O + 4e^- \to 2H_2 + 2O^{--} \to 2H_2 + O_2 + 4e^-$, where the electrons facilitate water splitting at the cathode and are recovered at the anode. For modeling purposes, and since the focus will generally not be on the internal mechanisms, the model simplifies to the equation (eq. 2.1).

$$H_2O \to H_2 + \frac{1}{2}O_2 \tag{2.1}$$

To clarify the terminology used in SOEC modeling: The upper side of the cell, where water is introduced and dihydrogen (H_2) is produced, is referred to as the **fuel side**. This name reflects the role of water as the “fuel,” acting as the reactant that is split during electrolysis. The opposite side—where dioxygen (O_2) is evolved, is known as the **air side**, as atmospheric air is typically supplied here. The designations “fuel side” and “air side” are standard throughout the SOEC literature [2, 22] and are used to distinguish these two main compartments of the cell.

The chemical flow in a cell

The equation in 2.1 treats the reaction as strictly one-way, but in reality, some produced hydrogen and oxygen can recombine to form water again, especially as the cell approaches equilibrium based on its electrical state. This reverse reaction is rare at the start but becomes more significant near equilibrium. Yielding equation 2.2.

$$2H_2O \underset{p_r}{\overset{p_l}{\leftrightharpoons}} 2H_2 + O_2 \tag{2.2}$$

Here, p_l stands for the transition probability that two hydrogen and one oxygen molecule will combine and turn back into water (moving from right state $\rightarrow$ left state), while p_r is the probability for splitting water into hydrogen and oxygen (left state $\rightarrow$ right state).

From these probabilities, the local forward and reverse reaction rates can be evaluated (assuming concentrations and probabilities are known everywhere). The net volumetric rate of change of H_2O is their difference; per unit cell volume it has units $\mathrm{mol}/(\mathrm{s}\cdot\mathrm{m}^3)$:

$$R_{H_2O} = p_r\mu^2_{H_2O} - p_l\mu^2_{H_2}\mu_{O_2}$$

Where μ_x is the concentration $\left(\frac{\mathrm{mol}}{\cdot\mathrm{m}^3}\right)$ of molecule x.

To obtain the production rate for the whole cell, this quantity is integrated over the thickness (height) of the cell. Which gives the total rate N_{c,H_2O} (later this will be treated as a local source, within the cell, but for now it is kept as a property for the whole cell):

$$N_{c,H_2O} = \int_0^d R_{H_2O}$$

This gives the average production rate per area of the cell, which is a useful parameter to work with. But realistically, calculating this from scratch could get a bit complicated. Thankfully, Faraday already did the heavy lifting here. By Faraday's law [26], the rate per cell area can be written as in equation 2.3:

$$N_{c,H_2O} = \frac{i}{2F} \tag{2.3}$$

Where i is the current, F is Faraday's constant, and the 2 comes from the number of electrons involved in the chemical equation.

For now, equation 2.3 is kept as it is, since the formulation is still at the global level; hence the current in the equation is the global current. Later, when making all this local, the current must be computed using the Nernst equation. More about that later. First, a form that describes the cell locally is introduced.

Behavior along the cell channels

To describe the cell locally, figure 2.3 is used. The fuel side is modeled as a channel (the upper section in 2.3), where the steam (H_2O) flows over the cathode. The width of the channel is w and the length is L, with the channel extending from the inlet at $z = 0$ to the outlet at $z = L$ (where L is the length of the cell). The gas flows through a constant cross-sectional area A_{int}.

For any control volume within this channel, mass conservation dictates that the accumulation of water over time equals the flow in plus the production of the cell reaction minus the flow out. Following the formulation by Primdahl and Mogensen [22] for a continuously stirred tank reactor (CSTR), the mass balance for the water molar fraction $x_{\mathrm{H_2O}}$ is:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = A_{\text{int}} N_{\text{in},\mathrm{H_2O}} - A_{\text{int}} N_{\text{out},\mathrm{H_2O}} + A_{\text{cell}} N_{c,\mathrm{H_2O}} \tag{2.4}$$

Where V is the volume of the gas in the reactor, N_i is the molar flux for species i, and A_{cell} is the active electrode area.

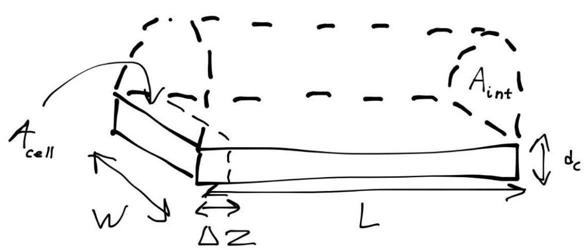


Figure 2.3: Schematic illustration of channel discretization for modeling local behavior along the fuel side of a SOEC. The channel is divided into N segments of length Δz, showing the flow of steam over the cathode.

To obtain the local behavior along the channel, discretization along the channel length is required, meaning the length is split into sections. This is also illustrated in figure 2.3. Hence $z_i - z_{i-1} = \Delta z$ for all $i = 1, \dots, N$, with $z_0 = 0$ and $z_N = L$, giving a sequence:

$$z_0 < z_1 < \cdots < z_N$$

Each section $z \in (z_{i-1}, z_i)$ can then be viewed as its own cell, where the inlet flow at z_{i-1} is the outlet flow of the previous interval. Letting $\Delta z \to 0$ and inserting the definition of $N_{c,\mathrm{H_2O}}$ (equation 2.3), and the PDE in equation 2.5 pops right out. (Appendix 1)

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{\text{int}}} N_{c,\mathrm{H_2O}} - \frac{\partial N_{\mathrm{H_2O}}}{\partial z} \tag{2.5}$$

Where $N_{\mathrm{H_2O}} = v_{\text{fuel}}\, \mu_{\mathrm{H_2O}}$ and $N_{c,\mathrm{H_2O}} = \nu_{\mathrm{H_2O}} \frac{i}{n_e F}$, with n_e the number of electrons in the reaction being 2.

By similar computations for H_2 and O_2, equation (2.6) and equation (2.7) will appear:

$$\frac{\partial \mu_{\mathrm{H_2}}}{\partial t} = \frac{w}{A_{\text{int}}} N_{c,\mathrm{H_2}} - \frac{\partial N_{\mathrm{H_2}}}{\partial z} \tag{2.6}$$

$$\frac{\partial \mu_{\mathrm{O_2}}}{\partial t} = \frac{w}{A_{\text{int}}} N_{c,\mathrm{O_2}} - \frac{\partial N_{\mathrm{O_2}}}{\partial z} \tag{2.7}$$

Where $N_{\mathrm{H_2}} = v_{\text{fuel}}\, \mu_{\mathrm{H_2}}$ and $N_{\mathrm{O_2}} = v_{\text{air}}\, \mu_{\mathrm{O_2}}$, with ν_i the stoichiometric coefficient for species i, μ_i the molar concentration of species i, and $v_{\text{fuel/air}}$ the corresponding velocity. And $N_{c,i} = \nu_i \frac{i}{n_e F}$.

From the chemical equation of the reaction:

$$2H_2O \leftrightharpoons 2H_2 + O_2$$

The stoichiometric coefficients are determined as $-\nu_{H_2O} = \nu_{H_2} = 2\nu_{O_2} = 1$.

This equation describes the local evolution of the species concentration. As noted earlier, the current i is no longer a global current. To determine it, the Nernst equation and more is required; the following subsection introduces the overpotential balance that supplies this relation.

Introducing the overpotentials

Not all of the applied cell voltage drives electrolysis directly. Part of it is consumed overcoming internal voltage losses, called **overpotentials**. As a simple picture, they can be viewed as resistances within the cell: Extra voltage must be supplied to push current through these losses before the intended production rate is reached. More precisely, the balance separates reversible (Nernst), kinetic (activation), and resistive (ohmic) contributions, each set by local gas composition, temperature, and operating conditions. Here degradation can enter through the latter two, raising the voltage required at fixed output (chapter 2).

From the equation stated in 2.5 only the current density i remains to be determined. The control variable is the voltage U, which equals the overpotential of the whole cell (η_{cell}) or cell voltage.

The cell voltage can be written as the sum of all overpotentials over the cell as in equation 2.8. Concentration overpotentials that appear when accounting for diffusion are neglected here, and the concentration is assumed to be the same everywhere on each side of the cell. (section 2.3)

$$\eta_{\text{cell}} = \eta_{\text{nernst}} + \eta_{\text{act}} + \eta_{\text{ohm}} \tag{2.8}$$

$$\eta_{\text{nernst}} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log\left(\frac{x_{H_2}\sqrt{x_{O_2}}}{x_{H_2O}\sqrt{x_0}}\right), \tag{2.9}$$

Where $x_0 = p_{\text{ref}}/p_{\text{air}} \approx 1$, ΔG is Gibbs free energy approximated by the polynomial $\Delta G = -0.0031\,T^2 - 49.119\,T + 244778$. [2]

$$\eta_{\text{ohm}} = iR_{\text{deg}} \tag{2.10}$$

Where R_{deg} is the resistance induced by degradation mechanisms.

$$\eta_{\text{act}} = \eta^a_{\text{act}} + \eta^c_{\text{act}} \tag{2.11}$$

Where η_{nernst} is the Nernst overpotential, representing the voltage generated by the electrochemical reaction, and η_{act} is the activation overpotential, which accounts for the energy barrier that must be overcome for the reaction to proceed, η_{ohm} is the Ohmic resistence, which accounts for resistance that is active due to degredative mechanisms. [2]

The Ohmic overpotential is essentially Ohm's law with an effective resistance; it is through this resistance that degradation enters the voltage balance. The Nernst overpotential describes the willingness of the reaction to proceed. When there is a lot of H_2O and very little H_2 and O_2, the Nernst overpotential is large in absolute value but negative. This drives the reaction away from the high concentration in H_2O and closer to equilibrium. Away from the Gibbs term and the other overpotentials, the *equilibrium* occurs where the fraction is 1. The Nernst overpotential can therefore be viewed as the *force* that pulls the concentrations towards equilibrium.

The activation overpotential is probably the most difficult part to compute. The governing equations are listed first; their meaning is explained afterwards.

The overpotentials for the anode and cathode arise by solving equation (2.12), also known as the Butler-Volmer equation.

$$i = i_0^\zeta \left(\exp\left[\alpha_\zeta \frac{n_e F}{RT}\eta_{\text{act}}^\zeta\right] - \exp\left[-(1-\alpha_\zeta)\frac{n_e F}{RT}\eta_{\text{act}}^\zeta\right] \right), \qquad \zeta \in \{a, f\}. \tag{2.12}$$

Where α_ζ is the symmetry factor for the corresponding side, and i_0^ζ is the exchange current for the respective side, with $\alpha_a = 0.65$ and $\alpha_c = 0.59$. [18]

The exchange current is computed as in equation (2.13).

$$i_0^\zeta = \lambda_\zeta \gamma_\zeta T \exp\left[-\frac{E_{\text{act}}^\zeta}{RT}\right] \prod_{i \in \text{Species}_\zeta} \left(\frac{x_i}{x_0}\right)^{m_i}, \qquad \zeta \in \{a, f\}. \tag{2.13}$$

Where Species$_a$ $= \{O_2\}$ and Species$_c$ $= \{H_2O, H_2\}$. $\lambda_a = 1$ (in principle it does not appear in the equation) and λ_c is also determined by degradation mechanisms, discussed later. γ_ζ is the pre-exponential factor for the respective side with values $\gamma_a = 1.515 \cdot 10^8$ and $\gamma_f = 7.666 \cdot 10^5$. E_{act}^ζ is the activation energy, with $E_{\text{act}}^a = 1.40 \cdot 10^5$ and $E_{\text{act}}^f = 1.05 \cdot 10^5$. And $m_{H_2O} = 0.33$, $m_{H_2} = -0.1$ and $m_{O_2} = 0.22$. [18]

This describes the electrical energy required to overcome the energy barrier for the reaction. Somewhat similar to how water does not necessarily freeze at 0 C°. The adequacy of Butler-Volmer kinetics for activation losses is debated in the literature [29]; for this project the fitted form of [18] is taken as a sufficient approximation.

Computing the current i

With the overpotentials introduced, it is clear that some of them depend on the current i, which is the missing piece in 2.3 needed to complete the coupled differential equations 2.5, 2.6 and 2.7.

The current is obtained by solving equation (2.8), with $\eta_{\text{cell}} = U$, and i defined as in equation (2.14)

$$i = \underset{i}{\text{Solve}}\left[U - \eta_{\text{nernst}} = \eta_{\text{ohm}}(i) + \eta_{\text{act}}(i)\right] \tag{2.14}$$

Using various simplifications this equation can be solved in different ways; for now it is solved numerically, using a root solver from the python library Scipy [28].

With the equations for mole transport in place, the temperature behavior is treated next, before all equations are summarized in a table.

2.1.2 Energy Balance

In order to determine the energy balances, the derivation starts at the thermal energy equation given by equation (2.15) [16]. A number of simplifications are then applied to obtain equation (2.16).

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T \tag{2.15}$$

where ρ is density, h specific enthalpy, λ_T thermal conductivity, h_i and N_i the specific enthalpy and flux of species i, and Q_T the volumetric heat source (e.g. electrochemical heating).

Starting with equation (2.15), which states how much energy a volume accumulates in enthalpy per unit volume (left-hand side), the right-hand side equals compression (the change in pressure) plus heat conduction (second-order differential term in temperature), enthalpy transport by species (H_2O, H_2, O_2), and the volumetric heat source Q_T.

The first simplification is to neglect the energy accumulation due to heat conduction ($\nabla \cdot (\lambda \nabla T) = 0$). Also assuming constant pressure, $h \approx c_p T$ and $\partial_t p = 0$, which yields [2]:

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + Q_T \tag{2.16}$$

Where $c_p = \sum_{e \in \mathcal{M}} c_{p,k} y_k$, with y_k being the mass fractions (Not the molar fraction), for further information look at Appendix 2.

It remains to define the source term, which follows from the power expression in equation (2.17) [2]. Power is the energy per unit time, hence the energy added to the cell.

$$Q_T = \frac{i\, w}{A_{\text{int}}} U \tag{2.17}$$

At steady state

At steady state ($\partial T / \partial t = 0$), with the along-channel species transport relations from equation (2.5)–equation (2.7) also at steady state, equation (2.16) simplifies further to equation (2.18).

$$0 = i \frac{w}{A_{\text{int}}} (U - U_{\text{th}}) - C_p \frac{\partial T}{\partial z} \tag{2.18}$$

Where $U_{\text{th}} = -\frac{\Delta h}{n_e F}$ with $\Delta h = h_{H_2O} - h_{H_2} - \frac{h_{O_2}}{2}$, (Appendix 2) [2].

With the energy balance in place, the channel-resolved electrochemical and thermal model is complete. In the full optimization problem this block is treated as **quasi-steady**: transport and temperature in the channel equilibrate much faster than degradation and the control horizon, so these steady-state equations are solved at each time step while the slow states evolve. The following chapter treats those degradation mechanisms (chapter 2).

Description	**Equation**	**Ref.**
Faraday's law	$N_{c,i} = \nu_i \frac{i}{2F}$	equation (2.3)
Molar concentration DEs	$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{\text{int}}} N_{c,i} - \frac{\partial N_i}{\partial z}$	Eqs. 2.5, 2.6, 2.7
Cell voltage	$\eta_{\text{cell}} = \eta_{\text{nernst}} + \eta_{\text{ohm}} + \eta_{\text{act}}^a + \eta_{\text{act}}^c$	equation (2.8)
Ohmic overpotential	$\eta_{\text{ohm}} = iR_{\text{deg}}$	equation (2.10)
Nernst overpotential	$\eta_{\text{nernst}} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log\left(\frac{x_{\text{H}_2}\sqrt{x_{\text{O}_2}}}{x_{\text{H}_2\text{O}}\sqrt{x_0}}\right)$	equation (2.9)
Butler–Volmer	$i = i_0^{a/f}\big(\exp[\alpha_{a/f}\frac{n_e F}{RT}\eta_{\text{act}}^{a/f}] - \exp[-(1-\alpha_{a/f})\frac{n_e F}{RT}\eta_{\text{act}}^{a/f}]\big)$	equation (2.12)
Exchange current	$i_0^{a/f} = \lambda_{a/f}\gamma_{a/f} T \exp\big[-\frac{E_{\text{act}}^{a/f}}{RT}\big] \prod_{i\in\mathcal{S}_{a/f}} (x_i/x_0)^{m_i}$	equation (2.13)
Gibbs free energy	$\Delta G \approx -0.0031T^2 - 49.119T + 244778$	equation (2.9)
Energy balance	$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i\in\mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + \frac{i\,w}{A_{\text{int}}} U$	Eqs. 2.16, 2.17

Table 2.1: Overview of model equations (Equation covering the development for ohmic resistance and degredation can be found in chapter 2).

2.2 Modelling degradation

This section examines the two components R_{ohm} that appear in equation (2.10) and λ_c, which appears in equation (2.13). The latter is more complicated, so the ohmic resistance is treated first. Below, an equation is set up that describes how degradation affects the ohmic resistance.

$$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left(\frac{E_{\text{act}}^{\text{ohm}}}{RT}\right) + R_{\text{contact}} + \sum_{d \in \mathcal{D}_{\text{ohm}}} R_d \tag{2.19}$$

Here, the first two terms are time-invariant and depend only on the present conditions of the cell. In contrast, the degradation terms, represented by the summation, evolve and increase as the cell ages. The first degradation mechanism considered is the resistance arising from nickel migration.

2.2.1 Nickel migration

Nickel migration describes the gradual recession of nickel in the fuel-side electrode away from the electrolyte interface. Initially the electrode is well infiltrated with nickel, over time the nickel-filled region shrinks and a gap of length $h_{\text{Ni-mi}}$ opens between the remaining nickel and the electrolyte. Because nickel provides the main electronic conduction paths in the cermet, this depleted zone increases the effective ohmic resistance, modelled by equation (2.20).

$$R_{\text{Ni-mi}} = \frac{h_{\text{Ni-mi}}}{\left(\frac{\varepsilon}{\tau}\sigma_{\text{elec}}\right)} \tag{2.20}$$

Where $h_{\text{Ni-mi}}$ is the thickness of nickel absence layer, which develops over time, σ_{elec} is the electrode conductivity, ϵ is the zirconia fraction and τ is the tortuosity of the zirconia fraction [2]

The sketch in figure 2.4 illustrates how the nickel migration develops over time, and the impact it has.

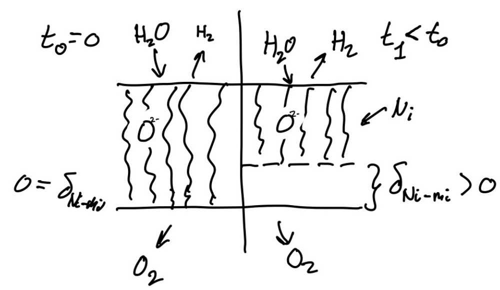


Figure 2.4: For a cell without degradation (left), nickel migration has not occurred and $d_{\text{Ni-mi}} = 0$ at the layer between the nickel-filled electrode and the electrolyte. Over time the nickel-filled part of the electrode creeps away from the electrolyte (right), giving rise to resistance induced by the raw electrode on the traveling species.

The conductivity of the electrode is given by equation (2.21). This conductivity is in reality the reduction in conductivity due to the creep of the nickel part of the electrode.

$$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,o}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right) \tag{2.21}$$

Javid Beyrami et al. [2] describe the development of nickel migration by the following differential equation, where c_1 and c_2 are negative and c_3 and c_4 are positive. For high values of current and temperature, migration increases; if either T or i is too small or too large, migration also increases.

$$\frac{\partial h_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_4$$

Where c_1, c_2, c_3 and c_4 are fitted to experimental data.

2.2.2 Interconnect Oxidation

Another important degradation mechanism to consider is interconnect oxidation. This process also contributes to the overall ohmic resistance, and is relatively straightforward to understand: It occurs when the metallic interconnect undergoes oxidation, in other words, when the metal effectively “rusts.”

The resistance resulting from interconnect oxidation is in structure much like that of nickel migration. Specifically, it is determined by the thickness of the oxidized layer divided by its (reduced) electrical conductivity, as shown in equation (2.22) [2], [17], [7].

$$R_{\text{IC-ox}} = \frac{h_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}} \tag{2.22}$$

Where the oxidized interconnect conductivity is given as a function of temperature in equation (2.23).

$$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp\left\{-\frac{E_{\text{el}}}{RT}\right\} \tag{2.23}$$

Beyrami et al. [2] describe the growth of the oxide layer by the following rate law.

$$\frac{dh^2_{\text{IC-ox}}}{dt} = k_{\text{mg}} \exp\left\{-\frac{E_{a,\text{ox}}}{RT}\right\} \tag{2.24}$$

The squared-thickness form in equation (2.24) reflects parabolic oxidation kinetics: Layer growth follows a square-root-of-time law, and the rate depends on local temperature alone through the Arrhenius term. Unlike nickel migration, interconnect oxidation is therefore not sensitive to current density. As $h_{\text{IC-ox}}$ increases, the contribution $R_{\text{IC-ox}}$ in equation (2.22) rises and adds to the degradation summation in the total ohmic resistance. The constants k_{mg} and $E_{a,\text{ox}}$ are fitted to experimental data [2], [7]. Together with nickel migration, this completes the ohmic degradation terms; the remaining mechanism acts on the cathode exchange current and will be discussed next.

2.2.3 Nickel Agglomeration

A further degradation mechanism affecting the cathode exchange current is nickel agglomeration. Unlike the ohmic contributions discussed above, it is typically modelled, not as ohmic resistance, but as a multiplier on λ_c, the normalized triple phase boundary [2]. Whether the underlying process is nickel agglomeration, or another phenomenon, is an ongoing debate, and the mechanistic details are therefore not pursued here. Because agglomeration develops on a relatively short timescale, λ_c is assumed to have reached its asymptotic value of approximately 0.55. [2]

Description	Equation	Ref.
Total ohmic resistance	$R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp\left\{\frac{E_{\text{act}}^{\text{ohm}}}{RT}\right\} + R_{\text{contact}} + \sum_{k\in\mathcal{D}_{\text{ohm}}} R_k$	
Nickel-migration resistance	$R_{\text{Ni-mi}} = \frac{d_{\text{Ni-mi}}}{(\frac{\varepsilon}{\tau}\sigma_{\text{elec}})}$	equation (2.20)
Electrode conductivity	$\sigma_{\text{elec}} = \frac{\sigma_{\text{elec,o}}}{T} \exp\left(-\frac{E_{a,\text{elec}}}{RT}\right)$	equation (2.21)
Ni-mi thickness rate	$\frac{\partial d_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_4$	[2]
Interconnect oxidation resistance	$R_{\text{IC-ox}} = \frac{d_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}}$	equation (2.22)
IC oxide conductivity	$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp\left\{-\frac{E_{\text{el}}}{RT}\right\}$	equation (2.23)
IC oxidation rate	$\frac{dd_{\text{IC-ox}}^2}{dt} = k_{\text{mg}} \exp\left\{-\frac{E_{a,\text{ox}}}{RT}\right\}$	equation (2.24)
Normalized triple phase boundary	$\lambda_c = 0.55$	[2]

Table 2.2: Overview of degradation mechanisms (ohmic contributions and slow-state dynamics). The base cell equations are summarized in table 2.1.

2.3 Assumptions

The average SRU model in the preceding sections couples molar transport, electrochemistry, and an energy balance along a one-dimensional channel. The resulting relations are collected in table 2.1; degradation is summarized in table 2.2. The closed formulation rests on the simplifications listed below. Each item states the assumption and why it is reasonable for the average SRU model; the modelled system boundary—what is resolved, lumped, and omitted—is set out in section 2.3.4. How the assumptions shape the optimization results will be discussed in section 5.4.

2.3.1 Geometry, flow, and thermodynamics

1. **Constant channel area:** The cross-sectional area A_{int} is uniform along the channel. Commercial repeat-element geometries use nearly constant channel dimensions over the flow length [24], so spatial variation in A_{int} is small relative to other uncertainties.

2. **One-dimensional flow:** Properties vary only in the axial direction z; width w and height d enter as geometric scalars. The channel length is much larger than its width, and repeat-unit models therefore resolve composition and temperature along z only [2, 22].

3. **Ideal gas:** Gas mixtures obey the ideal-gas law. Operating temperatures (700–$900\,^{\circ}$C) and near-atmospheric pressure give low compressibility, which is standard in channel-resolved SOEC models [2, 22].

4. **Negligible concentration overpotential:** Mass-transport limitations at the electrodes are not resolved; molar fractions on each electrode side are taken as uniform (equation (2.8)). The inlet is steam-rich and the fitted Butler-Volmer parameters describe the IV curve well in the operating window (figure 3.3) [18]; explicit concentration losses matter mainly at higher current or lower steam supply [19].

5. **Negligible heat conduction:** The conductive term $\nabla \cdot (\lambda_T \nabla T)$ is dropped in the energy balance (equation (2.16)). At the prescribed inlet flows, advection and electrochemical heating dominate the along-channel temperature rise in repeat-unit models [2, 16].

6. **Constant pressure:** $\partial_t p \approx 0$ and $h \approx c_p T$ in the enthalpy formulation. Gas velocities are low (large Mach number not reached), so pressure work and compressibility effects are secondary to enthalpy transport [16].

7. **Prescribed inlet flows:** Fuel and air volumetric flow rates are inputs to this model and are not optimized. The study targets optimal U and T_{in}; balance-of-plant flow scheduling is outside the scope of this thesis. Flows are set from commercial-cell data [24] and applied uniformly in the average SRU model (chapter 2, items 14-15).

2.3.2 Electrochemistry and degradation

8. **Butler-Volmer kinetics:** Anodic and fuel-side (cathode) activation overpotentials follow equation (2.12) with parameters from [18]. This form is standard for high-temperature solid-oxide cells and matches the experimental polarization used for validation (figure 3.3).

9. **Two dynamic ohmic degradation mechanisms:** Nickel migration and interconnect oxidation evolve via table 2.2; other ageing paths are not modelled dynamically. These two mechanisms dominate long-term ASR growth in the Beyrami framework and set the failure criteria used here [2, 3]. Feed-stream impurities are also known to accelerate degradation, but are not represented as separate states in this model; electrothermally balanced (AC:DC) operation can largely suppress such contamination-driven ageing [25], although the quantitative link to the nickel-migration and interconnect-oxidation laws adopted here is not yet established (section 5.6.3).

10. **Nickel agglomeration at steady state:** Because agglomeration develops on a short timescale relative to the optimization horizon, the cathode multiplier λ_c is fixed at ≈ 0.55 [2] rather than solved as an additional state.

11. **Spatially resolved degradation:** Slow states are stored per FVM cell ($\mathbf{d}_{\text{Ni-mi},i}, \mathbf{s}_{\text{IC-ox},i} \in \mathbb{R}^{N_{\text{cell}}}$); interconnect oxidation is advanced in squared-thickness form (equation (3.18)). Current and temperature vary along z, so migration and oxidation rates must be resolved spatially to capture along-channel profiles [2].

2.3.3 Quasi-steady coupling and time integration

12. **Quasi-steady SRU physics:** At each time step the FVM steady-state problem (molar fractions and temperature along z) is solved for the current controls and degradation state before the slow ODEs are updated. Thermal and mass-transport transients in the channel are fast (seconds to minutes) compared with the optimization horizon (days to months), so the repeat unit can be treated as quasi-steady while degradation evolves [2, 14].

13. **Implicit Euler in time:** Degradation states are advanced with the implicit Euler scheme on a uniform grid (equation (3.16)). The degradation ODEs are stiff at high temperature and current; implicit Euler gives stable marching on the coarse time grids used in the scenarios without restricting the step size to the fastest local rate.

2.3.4 Average SRU model

The motivating application is a full electrolysis plant, but the optimization model deliberately closes a smaller **system** so that channel-resolved physics remain tractable. Everything the optimizer sees is built from three ingredients: (i) one **average SRU** with fuel and air channels resolved along z; (ii) **lumped heat exchange**, with preheat to T_{in} and recovery from outlet enthalpy at $T_{\text{out}} - T_{\text{loss}}$ entering through P_{heat} in equation (3.2) while stack electrical demand is $P_{\text{cell}} = I_{\text{cell}}U$; and (iii) **uniform scaling** of per-SRU flows and powers to facility scale. Compressors, manifold routing, stack-to-stack thermal coupling, and heat-exchanger transients lie outside this boundary. Contractual hydrogen delivery and buffer storage enter at system scale in chapter 3, but the spatial plant model remains the SRU plus lumped HEX described above. Stating this boundary before the results is necessary for interpreting them: total power cost includes both P_{cell} and P_{heat}, and T_{in} is a meaningful control because preheat and recovery are lumped rather than resolved in detail.

14. **Average SRU formulation:** The governing equations describe one repeat element, a single layer with coupled fuel and air channels, resolved along z only (chapter 2). That channel is taken to represent stack-average behaviour; electrical series connection of multiple layers and spatial structure at stack or facility level are not modelled.

15. **Uniform SRUs across stacks:** All repeat elements are assumed to share the same geometry, inlet flows, and controls (U, T_{in}). Repeat-element maldistribution and stack-to-stack coupling are neglected; facility-scale hydrogen delivery is obtained by scaling per-SRU production [24].

16. **Lumped preheat and heat recovery:** As summarized above, sensible duty to reach T_{in} and a fixed recoverable outlet enthalpy ($T_{\text{out}} - T_{\text{loss}}$) enter through P_{heat} in equation (3.2). No heat-exchanger geometry, effectiveness, or transient is resolved; compressors and other auxiliary loads are likewise omitted.

3 Optimal Control and Solution Method

3.1 Optimal Control Problem

With the SRU model and degradation laws in place, operation can be simulated for any admissible inlet conditions and cell voltage within a reasonable range. What remains is to specify an economic objective and formulate the corresponding optimal control problem.

Following discussions with Dynelectro and DTU Energy, and a review of how the objective is formulated in most literature [14], [4], [8], the general structure adopted is to maximize integrated running revenue plus a terminal salvage term: revenue (R) is hydrogen sales income minus power cost, and salvage (Ψ) credits remaining useful stack life at the end of the horizon.

$$
\begin{aligned}
\max_{\mathbf{u}\in\mathcal{U}_{\text{ad}}} \int_0^{t_{\text{horizon}}} & R \,\mathrm{d}t + \Psi \\
\text{s.t.} \qquad & \frac{\partial n_i}{\partial t} \text{ satisfies table 2.1}, \quad \text{for } i \in \mathcal{S} \\
& \frac{\partial d_i}{\partial t} \text{ satisfies table 2.2}, \quad \text{for } i \in \mathcal{D} \\
& d_i \leq d_{i,\text{critical}}, \qquad \text{for } i \in \mathcal{D}
\end{aligned}
\tag{3.1}
$$

Where $\frac{\partial n_i}{\partial t}$ represents the multiphysics of the system, here stated as the flows where $\mathcal{S}$ represents the collection of species; $\frac{\partial d_i}{\partial t}$ represents the degradation of the system and $\mathcal{D}$ the collection of degradation mechanisms included in this thesis; and $d_{i,\text{critical}}$ is the ***critical degradation*** *limit for mechanism i (section 1.2), enforced by the last constraint.*

Here $\mathbf{u}(t) = [U(t),\, T_{\text{in}}(t)]^\top$ collects the control inputs. The set $\mathcal{U}_{\text{ad}}$ is the admissible control space: The constrained set of trajectories for which U and T_{in} remain within their physical bounds (e.g. $U \in [U_{\text{min}}, U_{\text{max}}]$ and $T_{\text{in}} \in [T_{\text{in,min}}, T_{\text{in,max}}]$; table 4.2) and satisfy any scenario-specific restrictions imposed on the controls (such as piecewise-constant segments or fixed voltage). **d** collects the degradation state variables;

$$
\mathbf{d} = \Big[d_i(t)\Big]_{i\in\mathcal{D}} .
$$

Viewing equation (3.1), the integrand $R(\mathbf{u}, \mathbf{d})$ is the **revenue rate**; maximizing its time integral corresponds to maximizing operating income over $[0, t_{\text{horizon}}]$. The terminal term $\Psi(\mathbf{d}(t_{\text{horizon}}))$ is the **salvage** contribution: an economic credit for the residual value of the SRU at the end of the horizon. Because prices and degradation cannot be forecast over the full stack lifetime, the objective optimizes only over $[0, t_{\text{horizon}}]$; Ψ then approximates the value of capacity not yet consumed: how far each spatially resolved degradation state $d_{i,k}$ still lies below its critical limit $d_{i,\text{critical}}$, scaled by repeat-element capital cost and averaged over all $2N_{\text{cell}}$ entries (equation (3.4)). In effect, it balances near-term hydrogen revenue against stack life that could still earn income beyond t_{horizon} (figure 3.1).

It is worth noting that minimizing degradation is, at its core, a part of minimizing production cost: When a cell degrades the resistance rises, leading to higher power for the same hydrogen output, and in the worst case; degradation leads to a malfunctioning cell, requiring replacement of the cell or, at facility scale: A stack. It should also be noted that using terminal degradation is a modeling choice; using the running degradation rate inside the integral would have been equally valid and may be simpler to implement.

3.1.1 Revenue

The revenue rate is taken in the explicit form shown in equation (3.2).

$$
\begin{aligned}
R &= c_{\mathrm{H_2}} \Delta \dot{n}_{\mathrm{H_2}} - c_P P \\
\Delta \frac{\partial n_{\mathrm{H_2}}}{\partial t} &= A_{\mathrm{int}} N_{\mathrm{out,H_2}} - \frac{\partial n_{\mathrm{in,H_2}}}{\partial t} \\
P &= P_{\mathrm{cell}} + P_{\mathrm{heat}}
\end{aligned}
\tag{3.2}
$$

Here $\frac{\partial n_{\mathrm{in,H_2}}}{\partial t}$ is specified as an input (part of $\mathbf{u}$), P is the total electrical power, P_{cell} the power supplied to the stack, and P_{heat} the auxiliary sensible heating power.

Thus, R is the value of net hydrogen production (outlet minus inlet, in appropriate units) minus the cost of power. All quantities in equation (3.2) follow from the SRU model summarized in table 2.1. The split between P_{cell} and P_{heat} reflects the average SRU model in chapter 2: Preheat and heat recovery are lumped, not resolved as a heat-exchanger model.

From the model data, the power of the cell is computed by $P_{\mathrm{cell}} = I_{\mathrm{cell}} U$ with:

$$
I_{\mathrm{cell}} \approx \frac{A_{\mathrm{cell}}}{N_{\mathrm{cell}}} \sum_{k=1}^{N_{\mathrm{cell}}} i_k .
$$

The heating power is the **sensible** auxiliary preheat duty: For each stream, the enthalpy difference between the **inlet** composition at the target inlet temperature T_{in} and the same composition at the recovered outlet temperature $T_{\mathrm{out}} - T_{\mathrm{loss}}$. Composition enthalpy of the electrochemical conversion is excluded from P_{heat} and is accounted for via P_{cell}. Positive P_{heat} means external preheat is required ($T_{\mathrm{in}} > T_{\mathrm{out}} - T_{\mathrm{loss}}$); negative values are recovery credits.

$$
P_{\mathrm{heat}} = \sum_{\zeta \in \{a,f\}} \dot{n}_\zeta \left(\sum_{i \in \mathcal{S}_\zeta} x_{i,\mathrm{in}}\, h_i(T_{\mathrm{in}}) - \sum_{i \in \mathcal{S}_\zeta} x_{i,\mathrm{in}}\, h_i(T_{\mathrm{out}} - T_{\mathrm{loss}}) \right)
\tag{3.3}
$$

Outlet heat is treated as recoverable toward the inlet with an absolute temperature loss T_{loss}. Here $\dot{n}_\zeta$ is the molar flow on side ζ and $x_{i,\mathrm{in}}$ are the inlet mole fractions.

With that, the running revenue or integrand is settled; the salvage term is discussed next.

3.1.2 Salvage

After operating up to time t_{horizon}, the running profit captures cash flow from hydrogen sales net of power cost. A salvage term is included so that the optimizer does not overuse the cell: It should retain enough residual value to justify its capital cost. The idea is to estimate, in a simple aggregate way, how close each spatially resolved degradation state $d_{i,k}$ still lies below its end-of-life limit $d_{i,\text{critical}}$ (equation (3.1)).

Equal weight is assigned to each mechanism at each FVM cell, leading to

$$\Psi = \frac{c_{\text{cell}}}{|\mathcal{D}|N_{\text{cell}}} \sum_{k=1}^{N_{\text{cell}}} \sum_{i \in \mathcal{D}} \left(1 - \frac{d_{i,k}}{d_{i,\text{critical}}}\right)^{\alpha_\Psi} \tag{3.4}$$

Here c_{cell} *scales the salvage credit to repeat-element capital cost;* $\mathcal{D} = \{\text{Ni-mi}, \text{IC-ox}\}$ *indexes the two dynamic degradation mechanisms,* N_{cell} *is the number of axial FVM cells (table 4.2);* $d_{i,k}$ *is the value of mechanism* i *in cell* k *at* $t = t_{\text{horizon}}$*,*

$d_{i,\text{critical}}$ *is the critical degradation limit for mechanism* i *(the same thresholds enforced in equation* (3.1) *and listed in table 4.2); and* α_Ψ *controls how sharply salvage declines as* $d_{i,k}$ *approaches* $d_{i,\text{critical}}$ *(figure 4.1).*

For each pair (i, k), the factor $1 - d_{i,k}/d_{i,\text{critical}}$ is the unused fraction of the margin to the critical limit at $t = t_{\text{horizon}}$. Averaging over all $2N_{\text{cell}}$ entries and scaling by c_{cell} yields a terminal value that increases when the cell stops farther from end-of-life.

Extra note on salvage

The ideal aim would be to extract the most value from a cell over its lifetime, which would correspond to optimizing the integral in equation (3.1) with $\Psi = 0$ and t_{horizon} a conditioned variable equal to the lifetime. To optimize over the lifetime is not practical, since degradation and prices cannot be predicted that far ahead. Instead, the objective optimizes value over a short future horizon while retaining potential revenue as illustrated in figure 3.1.

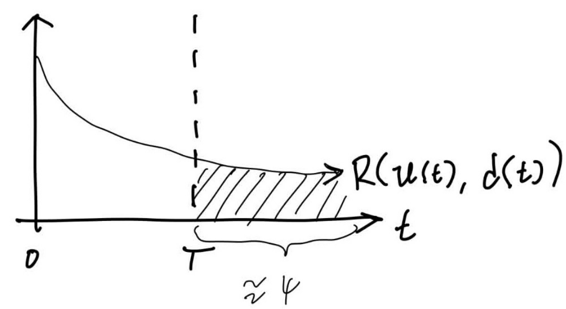


Figure 3.1: Sketch for the lifetime revenue, and the aim of a salvage term: The shaded area under the running revenue rate R is the part of the lifetime revenue that is not a direct part of the optimized horizon. This part of the lifetime revenue is hence the hardest part to incorporate. The aim is to have a model that approximates the maximal revenue left after the optimized horizon and thereby balances what is gained now against the potential future revenue.

The objective is essentially the area under the revenue-rate curve. The formulation used here approximates maximization of that area over the full lifetime, but only up to the finite horizon t_{horizon}.

3.1.3 H_2 storage and delivery agreement

Beyond the simple revenue model in equation (3.2), a contract-style extension adds buffer storage and a fixed hydrogen delivery obligation. The cell and degradation equations are unchanged; only the economic layer is extended. The contractual offtake is represented by a constant delivery rate $\dot{n}_{\mathrm{H_2},\text{agreement}}$. In practice, delivery may occur in batches; here it is approximated as a continuous flow over $[0, t_{\text{horizon}}]$. Shortfalls relative to production are met from inventory or, if the tank is empty, from spot purchases at rate $\dot{n}_{\text{purchase}}$, penalized in the revenue at $\kappa_{\text{buy}}\, c_{\mathrm{H_2}}$. The resulting revenue rate is given in equation (3.5).

$$R_{\text{tank}} = c_{\mathrm{H_2}}\, \dot{n}_{\mathrm{H_2},\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{\mathrm{H_2}}\, \dot{n}_{\text{purchase}}. \tag{3.5}$$

Inventory is modelled as a lumped storage tank. The net production surplus $\Delta\dot{n}$ is the difference between stack output and the agreement rate. When inventory is positive, the tank absorbs the full surplus or shortfall; spot purchases activate only when the tank is empty and production still falls short of delivery. This gives equation (3.6) and equation (3.7).

$$\Delta\dot{n} = \dot{n}_{\text{prod}} - \dot{n}_{\mathrm{H_2},\text{agreement}}$$

$$\dot{m}_{\text{tank}} = \Delta\dot{n} + \dot{n}_{\text{purchase}}, \qquad m_{\text{tank}}(t) \geq 0 \tag{3.6}$$

$$\dot{n}_{\text{purchase}} = \begin{cases} -\Delta\dot{n} & \text{if } m_{\text{tank}} = 0 \text{ and } \Delta\dot{n} < 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7}$$

Equivalently, over a control interval of length Δt with inventory m_k at the start of the interval, the discrete update used in the solver is

$$m_{k+1} = \max(0,\ m_k + \Delta\dot{n}\,\Delta t),$$

$$\dot{n}_{\text{purchase}}\,\Delta t = \max(0,\ -m_k - \Delta\dot{n}\,\Delta t).$$

At the end of the optimization horizon (end-of-horizon) the tank may still hold inventory. Two terminal treatments are used in the results. In the first, remaining hydrogen is valued as liquidation on the open market, giving the additional term in equation (3.8).

$$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\mathrm{H_2}}\, m_{\text{tank}}(t_{\text{horizon}}), \tag{3.8}$$

where $\kappa_{\text{sell}} \approx \frac{1}{9}$ when end-of-horizon stock must compete with *black* hydrogen. In the second, a cyclic inventory constraint replaces terminal liquidation:

$$m_{\text{tank}}(0) = m_{\text{tank}}(t_{\text{horizon}}), \tag{3.9}$$

which is natural when the operating window is meant to repeat periodically.

Under equation (3.9), many inventory trajectories can yield the same revenue. If $m_{\text{tank}}(0) = m_{\text{tank}}(t_{\text{horizon}}) = m_0$ is feasible, then so is any uniform upward shift $m_{\text{tank}}(t) + a$ with the same $a > 0$, because only changes in inventory, not the absolute level, enter the tank balance in equation (3.6). The cyclic constraint therefore does not by itself fix how much hydrogen should be held in reserve.

To break this tie, a **minimum-inventory penalty** is subtracted from the objective:

$$-\min_{t} m_{\text{tank}}(t).$$

When maximizing total return, a lower minimum inventory level is preferred: among trajectories with the same revenue, the optimizer selects the one that keeps the smallest tank level over $[0, t_{\text{horizon}}]$.

Full objective (tank mode)

When tank mode is active, the optimizer solves

$$\begin{aligned} \max_{u} \quad & \int_0^{t_{\text{horizon}}} R_{\text{tank}} \, \mathrm{d}t + \Psi(\mathbf{d}(t_{\text{horizon}})) + \Psi_{\text{tank}}(m_{\text{tank}}(t_{\text{horizon}})) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = 0 \end{aligned} \tag{3.10}$$

or

$$\begin{aligned} \max_{u} \quad & \int_0^{t_{\text{horizon}}} R_{\text{tank}} \, \mathrm{d}t + \Psi(\mathbf{d}(t_{\text{horizon}})) - \min_{t} m_{\text{tank}}(t) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = m_{\text{tank}}(t_{\text{horizon}}) \end{aligned} \tag{3.11}$$

instead of equation (3.1) with equation (3.2). Fixed delivery revenue decouples cash flow from instantaneous production: The optimizer minimizes power cost and spot purchases while building inventory when production is favourable, subject to the usual salvage trade-off on degradation.

An overview of all relevant equations for the optimal control problem is stated in table 3.1.

Description	Equation	Ref.
General objective function	$\int_0^{t_{\text{horizon}}} R, \mathrm{d}t + \Psi$	equation (3.1)
Revenue	$R = c_{\mathrm{H}_2} \Delta \dot{n}_{\mathrm{H}_2} - c_P P$	equation (3.2)
Salvage	$\Psi = \frac{c_{\text{cell}}}{2N_{\text{cell}}} \sum_{k=1}^{N_{\text{cell}}} \sum_{i \in \mathcal{D}} \left(1 - \frac{d_{i,k}}{d_{i,\text{critical}}}\right)^{\alpha_\Psi}$	equation (3.4)
Tank balance	$\dot{m}_{\text{tank}} = \Delta \dot{n} + \dot{n}_{\text{purchase}},\ m_{\text{tank}} \geq 0,\ \Delta \dot{n} = \dot{n}_{\text{prod}} - \dot{n}_{\mathrm{H}_2,\text{agreement}}$	equation (3.6)
Spot purchase	$\dot{n}_{\text{purchase}} = -\Delta \dot{n}$ if $m_{\text{tank}} = 0$ and $\Delta \dot{n} < 0$; else 0	equation (3.7)
Contract revenue	$R_{\text{tank}} = c_{\mathrm{H}_2} \dot{n}_{\mathrm{H}_2,\text{agreement}} - c_P P - \kappa_{\text{buy}}\, c_{\mathrm{H}_2}\, \dot{n}_{\text{purchase}}$	equation (3.5)
Terminal liquidation (alternative A)	$\Psi_{\text{tank}} = \kappa_{\text{sell}}\, c_{\mathrm{H}_2}\, m_{\text{tank}}(t_{\text{horizon}})$	equation (3.8)
Cyclic inventory constraint (alternative B)	$m_{\text{tank}}(0) = m_{\text{tank}}(t_{\text{horizon}})$	equation (3.9)
Minimum-inventory penalty (with alternative B)	$-\min_t m_{\text{tank}}(t)$	equation (3.11)
Cell Power	$P_{\text{cell}} = I_{\text{cell}} U$	
Heating Power	$P_{\text{heat}} = \sum_{\zeta \in \{a,f\}} \dot{n}_\zeta \left(\sum_{i \in \mathcal{S}_\zeta} x_{i,\text{in}} h_i\,(T_{\text{in}}) - \sum_{i \in \mathcal{S}_\zeta} x_{i,\text{in}} h_i\,(T_{\text{out}} - T_{\text{loss}}) \right)$	

Table 3.1: Overview of the optimal control formulation (objective, revenue, terminal value, and power bookkeeping). The underlying cell equations are summarized in table 2.1; degradation is treated in chapter 2. Optional tank economics (section 3.1.3) add contract revenue and inventory dynamics; end-of-horizon inventory is closed either by terminal liquidation (alternative A) or by a cyclic constraint with a minimum-inventory penalty (alternative B).

3.2 Building the numerical solver

This chapter presents the construction of numerical tools required to determine the optimal control $\mathbf{u}^*(t)$ for the average SRU model. The problem considered here is complex and necessitates a structured, step-by-step methodology. The central question addressed is: What tools and methods are needed to solve the optimal control problem outlined in chapter 3?

The overall strategy is as follows:

1. **Steady-state cell equations:**
   As the focus is on long-term behavior, it is assumed that the dynamics described by the differential equations in table 2.1 occur much faster than the overall optimization time-scale, treating them as instant. This allows for replacement with a set of steady-state equations, which are simpler to solve. The Finite Volume Method (FVM) is used to solve these equations numerically.

2. **Degradation dynamics:**
   With the steady-state SRU solver established, its outputs serve as algebraic inputs to the degradation model (table 2.2). This model captures SRU degradation over time through differential equations, which are solved numerically using the implicit Euler method.

3. **Objective and optimal control:**
   The optimal control objective (table 3.1) is addressed with **single shooting**: the control $\mathbf{u}(t)$ is replaced by a finite set of piecewise-constant values, and the degradation states and accumulated revenue are obtained by integrating the model forward in time from known initial conditions. Only those control values are passed to the optimizer as decision variables. In this approach, both the revenue and degradation equations are marched forward together on the same time grid.

Structuring the problem in these three parts (steady-state solution, degradation modeling, and control optimization) enables a systematic approach to the full optimal control challenge.

3.2.1 Finite volume method (FVM)

With the descriptive equations in place (table 2.1, chapter 2, and chapter 3), a numerical solution method is required. All equations can be written in the general form

$$\frac{\partial u}{\partial t} = Q_u - \frac{\partial F_u}{\partial z} \tag{3.12}$$

where Q_u denotes the source term and F_u the flow term; both can be computed directly from the state. For now only the steady-state solution is of interest, which simplifies the problem considerably and yields the following *boundary value problem* (BVP):

$$0 = Q_u - \frac{\partial F_u}{\partial z}$$

The *finite volume method* is used to solve this steady-state BVP [13].

As shown in figure 3.2, the domain is split into smaller subdomains coupled by flux from one volume into the next. Integration of the density u over a small volume Ω_k of volume V_k proceeds as follows.

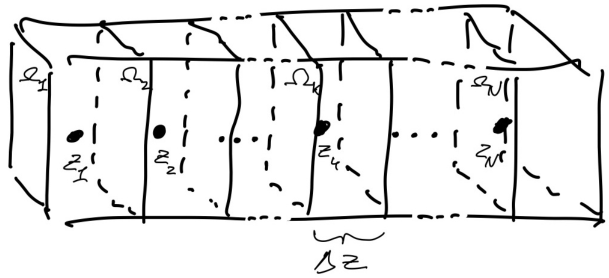


Figure 3.2: A sketch of the finite volume method. A large volume, corresponding to the air and gas channel, is split into smaller volumes. The greater volume is divided into N volumes, each represented by the midpoint, yielding a 1D system instead of 3D. Integration over the densities in each small volume gives the mass in that volume. The figure also introduces z_k (midpoint), Ω_k (small volume), and Δz (length of the volume).

$$\int_{\Omega_k} \frac{\partial u}{\partial t} = \int_{\Omega_k} Q_u \, dV - \int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV$$

The source term is approximated by a rectangle (midpoint rule): $\int_{\Omega_k} Q_u \approx Q_u(z_k) A_{\text{int}} \Delta z$. For the flux term, the fundamental theorem of calculus is used to evaluate the antiderivative in the z-direction, with the assumption that the concentration is constant perpendicular to z, giving:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \, dV = A_{\text{int}} F_u \Big|_{\partial \Omega_k} = A_{\text{int}} \Big(F_u(z_{k+\frac{1}{2}}) - F_u(z_{k-\frac{1}{2}}) \Big)$$

The boundary values $F_u(z_{k-\frac{1}{2}})$ are approximated by the value in the previous FVM cell, consistent with the flow direction along the SRU channel (in cell k there is no information from cell $k+1$, but there is from cell $k-1$). The integral is therefore approximated as:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \approx \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{\text{int}}$$

Combining the source and flux approximations gives the discrete equation equation (3.13).

$$\frac{\partial}{\partial t} \int_{\Omega_k} u \, dV = Q_u(z_k) A_{\text{int}} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{\text{int}} \tag{3.13}$$

or at steady state:

$$0 = Q_u(z_k) A_{\text{int}} \Delta z - \Big(F_u(z_k) - F_u(z_{k-1}) \Big) A_{\text{int}}$$

Applying boundary conditions

The problem has two boundary conditions. The first is a Dirichlet condition at the inlet, imposed by defining the first flux as:

$$F_u(z_{-1}) = F_u(z_{\text{inlet}})$$

For the Neumann condition at the outlet, no change is made to the equations, since the upstream flux in the FVM makes it possible to neglect a Neumann boundary term at the end.

FVM on specific equations

Applying the FVM to the governing equations yields the following discretization; see appendix 3 for derivations.

$$|\Omega_k|\frac{\partial \mu_i}{\partial t}(z_k) \approx wN_{c,i}\Delta z - A_{\text{int}}(N_i(z_k) - N_i(z_{k-1})), \tag{3.14}$$

And

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T A_{\text{int}}\Delta z - A_{\text{int}} \sum_{i\in\mathcal{S}} v_i c_{p,i}\Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big) \tag{3.15}$$

3.2.2 FVM model validation and Grid independence

IV-curve validation

After implementing all the equations given in table 2.1, the model was validated against current-voltage data (IV-curve) from experimental data [24]. The error is below 10^{-2} except at higher voltage/current (current ≈ 0.75, voltage ≈ 1.1), where the error is larger. Figure 3.3 compares the model with the data.

This level of agreement is sufficient for the intended operating range, as very high voltages are not relevant for typical operation. The main variable of interest is the change in moles across the cell, which exhibits similar trends; at high voltage or current, the error in molar change decreases further, since the molar fractions approach their extrema (0 or 1), making relative errors negligible.

Grid independence

Grid refinement was used to assess how much the states change when the mesh is refined, and thereby to estimate the resolution required for reliable simulations later. The steady-state solver was run while doubling the number of grid points each time. Figure 3.4 indicates that a relative error target of approximately 10^{-3} would require on the order of 200 FVM cells. The optimization runs in table 4.2 instead adopt $N_{\text{cell}} = 20$, for which the study indicates spatial error on the order of 10^{-2}, adequate for comparing control modes, as discussed in section 5.5.

The steady-state equations can nonetheless be solved to a rather fine precision. This leads to the next step: solving the ODEs using the outputs of the FVM.

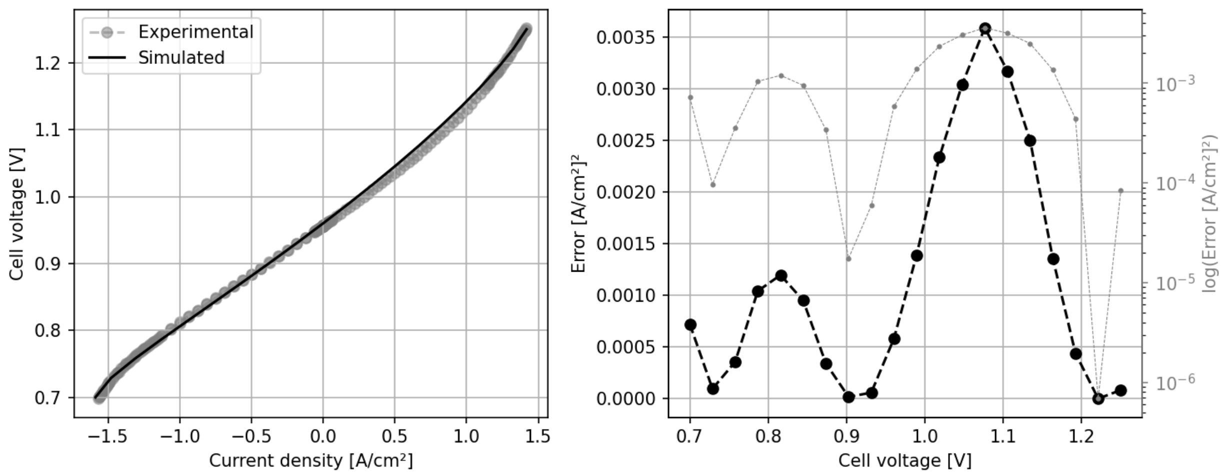


Figure 3.3: Model validation by comparing to an IV-curve from experimental data. The simulated current-voltage curve is compared to the experimental curve on the left. On the right, the error in current is shown as a function of voltage, on both logarithmic and linear scales. The experimental data is from figure 2 in [14].

3.2.3 Implicit Euler method (ODE solver)

As stated in the introduction to this chapter, the FVM gives the algebraic outputs needed to solve the differential equations that define degradation (table 2.2). Those ODEs are advanced with the implicit Euler method on a uniform time grid [15].

General form

Take the ODE: $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$. Discretize time on $t \in [0, T]$ into N_t steps with $\mathbf{x}_i = \mathbf{x}(t_i)$, $t_i = t_{i-1} + \Delta t_i$, $t_0 = 0$, and $t_{N_t} = T$. Given an initial state $\mathbf{x}_0$, the implicit Euler update is

$$\mathbf{x}_i = \mathbf{x}_{i-1} + \Delta t_i \, \mathbf{f}(t_i, \mathbf{x}_i), \qquad i = 1, \dots, N_t. \tag{3.16}$$

Because $\mathbf{x}_i$ appears on both sides, equation (3.16) is a fixed-point equation at each step. Damped fixed-point iteration is therefore a natural solver. The update is also Markovian: Step i depends only on $\mathbf{x}_{i-1}$ and the control on $[t_{i-1}, t_i)$, so the integration can proceed one time step at a time.

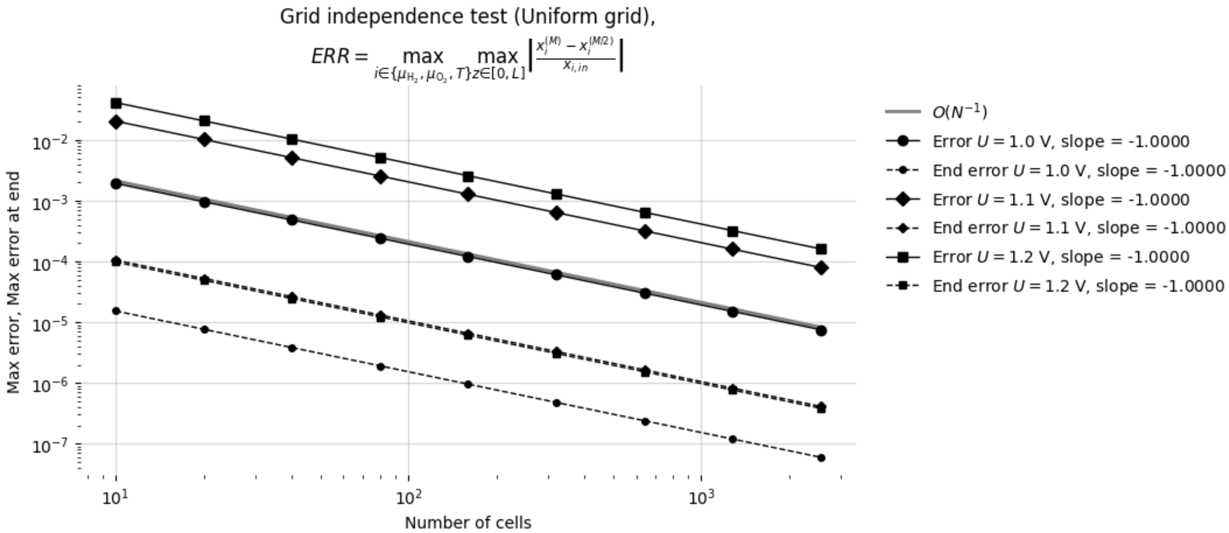


Figure 3.4: Grid-independence study of the steady-state FVM solver. The number of cells N is doubled repeatedly $(5, 10, \ldots)$ at fixed inlet conditions and three cell voltages ($U = 1.0$, 1.1, and 1.2 V). Solid markers show the maximum relative change in hydrogen and oxygen chemical potentials and temperature between successive refinements; dashed markers show the same quantity at the channel outlet. All curves fall approximately as $O(N^{-1})$ (grey reference). At $N \approx 200$ the error reaches $\sim 10^{-3}$; the optimization runs use $N_{\text{cell}} = 20$ (table 4.2), for which the error is $\sim 10^{-2}$.

Specific ODE setup

For this thesis the vector field $\mathbf{f}$ is expensive: Evaluating $\dot{\mathbf{d}}$ requires converged FVM fields (temperature and current density along the stack), which themselves depend on the degradation state. The degradation state advanced at each control interval is

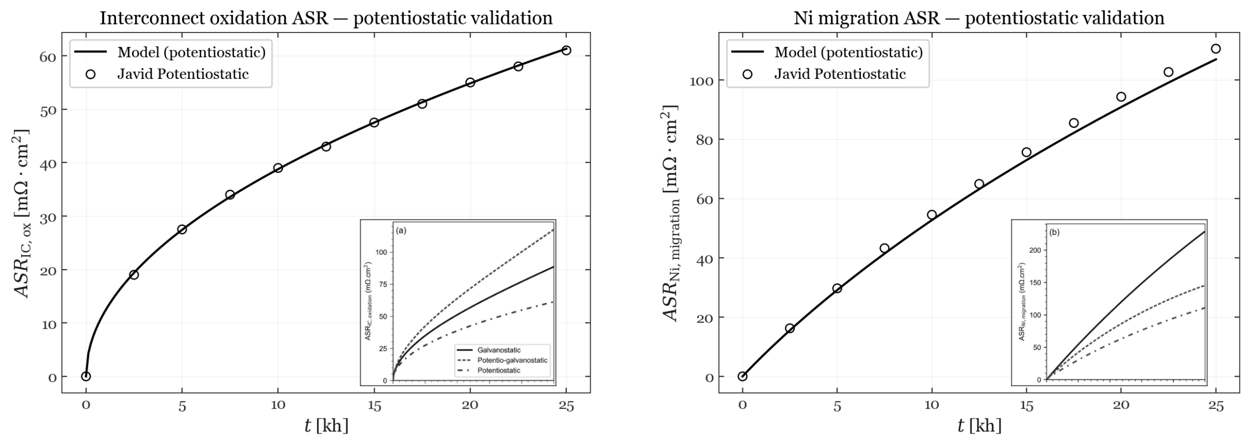
$$\mathbf{d}_i = \begin{bmatrix} \mathbf{d}_{\text{Ni}-\text{mi,i}} \\ \mathbf{s}_{\text{IC}-\text{ox,i}} \end{bmatrix}, \tag{3.17}$$

where $\mathbf{d}_{\text{Ni}-\text{mi,i}}, \mathbf{s}_{\text{IC}-\text{ox,i}} \in \mathbb{R}^{N_{\text{cell}}}$ *collect the per-cell slow states at time* t_i*, with* $\mathbf{s}_{\text{IC}-\text{ox,i}} = \mathbf{d}^2_{\text{IC}-\text{ox,i}}$*. Nickel migration stores layer thickness* $d_{\text{Ni}-\text{mi}}$ *directly; interconnect oxidation stores squared oxide thickness* $s_{\text{IC}-\text{ox}} = d^2_{\text{IC}-\text{ox}}$*, consistent with equation* (2.24) *and* $R_{\text{IC}-\text{ox}} = d_{\text{IC}-\text{ox}}/\sigma_{\text{IC}-\text{ox}}$ *with* $d_{\text{IC}-\text{ox}} = \sqrt{s_{\text{IC}-\text{ox}}}$.

With piecewise-constant controls $\mathbf{u}_i$ on $[t_{i-1}, t_i)$, implicit Euler gives

$$\begin{aligned} \mathbf{d}_{\text{Ni}-\text{mi,i}} &= \mathbf{d}_{\text{Ni}-\text{mi,i}-1} + \Delta t_i\, \dot{\mathbf{d}}_{\text{Ni}-\text{mi}}(\mathbf{d}_i, \mathbf{u}_i), \\ \mathbf{s}_{\text{IC}-\text{ox,i}} &= \mathbf{s}_{\text{IC}-\text{ox,i}-1} + \Delta t_i\, \dot{\mathbf{s}}_{\text{IC}-\text{ox}}(\mathbf{d}_i, \mathbf{u}_i), \end{aligned} \qquad i = 1, \ldots, N_t, \tag{3.18}$$

where $\mathbf{d}_i$ *stacks all per-cell degradation components at time* t_i*. The rates follow table 2.2:* $\dot{d}_{\text{Ni}-\text{mi,k}} = c_1 T_k + c_2 i_k + c_3 T_k i_k + c_4$ *depends on local temperature* T_k *and current density* i_k*, while* $\dot{s}_{\text{IC}-\text{ox,k}} = k_{\text{mg}} \exp\{-E_{a,\text{ox}}/(RT_k)\}$ *depends on* T_k *only.*



(a) Interconnect oxidation. At $t = 25\,\mathrm{kh}$: Model $61.3\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$, reference $61.0\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$. RMSE $= 0.26\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$.

(b) Nickel migration. At $t = 25\,\mathrm{kh}$: Model $106.9\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$, reference $110.5\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$. RMSE $= 2.45\,\mathrm{m}\Omega \cdot \mathrm{cm}^2$.

Figure 3.5: Potentiostatic validation of ASR for (a) interconnect oxidation and (b) nickel migration. Black line: Implicit-Euler simulation. Open markers: Digitized reference data from Beyrami et al (2024) [2].

Those FVM outputs are not explicit inputs to the ODE; they are obtained by running the steady-state FVM solver with control $\mathbf{u}_i$ and degradation state $\mathbf{d}_i$. Because $\mathbf{d}_i$ appears inside that solve, each time step is a coupled fixed-point problem: Guess $\mathbf{d}_i$, run FVM, evaluate the rates in equation (3.18), update $\mathbf{d}_i$, and iterate until convergence. Giving us the heavy computation of solving the steady state for each fixed-point iteration, leading to a large computational time.

In general, the problem can be viewed on a two-dimensional grid: Each spatial row (FVM cell) must be resolved repeatedly within a time step before advancing to the next column (time index).

3.2.4 Validation of the ODE solver

With the steady-state FVM validated in the previous section, the next step is to check that the coupled degradation integrator in equation (3.18) reproduces expected long-term behaviour.

The reference case follows the potentiostatic traces reported by Beyrami et al. [2] (figure 8): Fixed cell voltage $U = 1.285\,\mathrm{V}$, inlet temperature $T_{\text{in}} = 750\,^{\circ}\mathrm{C}$, and fuel composition $10\,\%$ H_2 / $90\,\%$ H_2O. At each time knot the FVM steady-state problem is solved, the degradation rates from table 2.2 are evaluated, and the states are updated with implicit Euler (equation (3.16)) until $t = 25\,\mathrm{kh}$. This produces a history of how the degradation mechanisms develop over time. From these quantities, the area-specific resistances (ASR) can be computed and compared with the traces from Beyrami et al. in figure 3.5.

The interconnect-oxidation agrees over the 25 thousand hours horizon. Nickel migration follows the same trend but sits slightly below the reference curve. Given the digitization uncertainty and the simplified inlet specification relative to the original study, this level of agreement is considered sufficient for the degradation model to be used in the optimal-control setup in section 3.2.5.

3.2.5 Solving the Optimal Control Problem

The optimal control problem (equation (3.1)) is solved using a nonlinear programming (NLP) solver. Following the direct single-shooting transcription [5], the control function $\mathbf{u}$ is parameterized by a finite vector of piecewise-constant values (as would also be the case in real applications); the state trajectory is then obtained by forward integration, so only $\mathbf{u}$ remains as a decision variable for the NLP.

$$\mathbf{u}^{(k)}(t) = \left\{ \mathbf{u}_i^{(k)} \quad t \in [t_i, t_{i+1}) \right.$$

The problem is split into N_t subproblems, each with constant control input. From equation (3.1) the objective φ can be rewritten as:

$$\begin{aligned} \varphi &= \int_0^{t_\text{horizon}} R(\mathbf{u}, \mathbf{d})\ \mathrm{d}t + \Psi(\mathbf{d}) \\ \Rightarrow \dot{\varphi} &= R(\mathbf{u}, \mathbf{d}) \end{aligned} \tag{3.19}$$

Thus φ is obtained by solving the auxiliary differential equation with initial condition $\varphi(0) = 0$ (numerical integration), concurrently with the state equations, and adding $\Psi(\mathbf{d})$ at the final time when the degradation at $t = t_\text{horizon}$ is known.

For the constraints in equation (3.1), which are equality constraints, they are used directly to compute degradation and molar flows, leaving only $\mathbf{u}$ as a decision variable for the optimizer. The NLP solver also requires the gradient $\nabla_\mathbf{u}\varphi$; adjoint gradient methods are used [23], [9]. Further details are in appendix 5; here it suffices that $\nabla_\mathbf{u}\varphi$ is available.

The full setup

Putting the pieces together, each control interval couples a steady-state FVM solve of the cell equations to an implicit-Euler update of degradation and to an accumulation of the running objective. Figure 3.6 sketches this single-shooting pipeline: Along the time axis the horizon is split into steps of length Δt, and at each step a piecewise-constant control $\mathbf{u}_i$ drives the spatial FVM solve; the resulting fields update $\mathbf{d}$ and contribute $R(\mathbf{u}, \mathbf{d})\,\Delta t$ to φ.

The same structure is stated as the discrete NLP

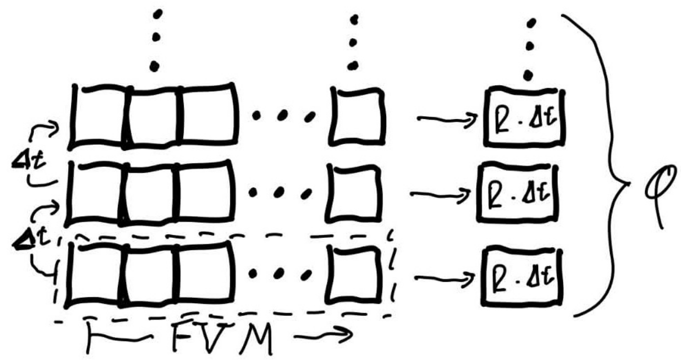


Figure 3.6: Overview of the single-shooting discretization. The horizon is divided into intervals of length Δt (vertical axis). At each interval a piecewise-constant control $\mathbf{u}_i$ drives a steady-state FVM solve of the cell equations (spatial cells along the stack, dashed box). The resulting fields and degradation state feed the running objective rate $R(\mathbf{u}, \mathbf{d})$, whose contribution $R\,\Delta t$ is accumulated forward to give the total objective φ.

$$
\begin{aligned}
&\max_{\mathbf{u}} \quad \varphi_N + \Psi(\mathbf{d}_N) \\
\text{s.t.} \quad & \varphi_{i+1} = \varphi_i + \dot{\varphi}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\
& \mathbf{d}_{i+1} = \mathbf{d}_i + \dot{\mathbf{d}}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\
& \varphi_1 = 0, \\
& \mathbf{d}_1 = \mathbf{d}_{\text{init}}.
\end{aligned}
\tag{3.20}
$$

with $\mathbf{d}$ obtained from the constraint equations rather than treated as free decision variables. A convex programming algorithm is used to find the optimal $\mathbf{u}$; the problem is generally convex. For this project the trust-constr method from SciPy is used [28], a trust-region method that employs the computed gradient of φ.

Solving the full setup

The NLP solver works in a simple loop [5]. It proposes a control trajectory $\mathbf{u}$, evaluates how good that trajectory is, computes how the objective would change if each control value were perturbed, and then updates $\mathbf{u}$. This repeats until convergence. Each iteration has two parts: a forward pass and a backward pass.

Forward pass Start from the known initial degradation $\mathbf{d}_{\text{init}}$. For each time interval $i = 1, \dots, N_t$, do the following:

1. Take the control $\mathbf{u}_i$ for that interval.
2. Solve the steady-state cell equations with FVM.
3. From the resulting fields, compute the revenue rate R and the degradation rates $\dot{\mathbf{d}}$.
4. Update $\varphi_{i+1} = \varphi_i + R\,\Delta t_i$ and $\mathbf{d}_{i+1} = \mathbf{d}_i + \dot{\mathbf{d}}\,\Delta t_i$.

After the last interval, add the terminal salvage term $\Psi(\mathbf{d}_{N_t})$. That gives the total objective $\varphi_{N_t} + \Psi(\mathbf{d}_{N_t})$ for the proposed $\mathbf{u}$. The degradation and revenue are never free variables in the NLP: they follow entirely from forward integration once $\mathbf{u}$ is fixed. This is the single-shooting idea: only $\mathbf{u}$ is optimized, and everything else is obtained by marching forward in time.

Adjoint gradient The SciPy trust-constr method [28] also needs $\nabla_{\mathbf{u}}\varphi$ at each iteration. Rather than finite differences, an adjoint sweep is used [23], [9]. After the forward pass, adjoint variables λ are propagated backward through the same stages. Each stage contributes a direct term (how R depends on $\mathbf{u}_i$) and an indirect term (how a change in $\mathbf{u}_i$ affects later stages through the integrated states). The full assembly is given in appendix 5; the partial derivatives of the cell model enter through the FVM Jacobian and the rate sensitivities in appendix 4.

NLP iteration With the objective and gradient in hand, trust-constr updates $\mathbf{u}$ and the loop starts again. In practice, one optimizer iteration means one forward trajectory and one backward adjoint solve.

For a deeper understanding of the direct single-shooting transcription and its relation to NLP, Section 5.1 of *Introduction to Model Based Optimization of Chemical Processes on Moving Horizons* [5] is recommended.

4 Results and Findings

With the full machinery in place, it is time to put it to work. The preceding chapters set up the model, the objective, and the solver; what is still missing before any results can be discussed is a list of the numerical settings used in the optimization runs. Those defaults are collected in table 4.2, table 4.3 and section 3.1.3. The first of the three tables states the settings used by default, unless anything else is stated.

Scen.	Description	Key parameter varied	Avg. revenue [c/d]	Avg. salv. loss [c/d]	Avg. net econ. [c/d]
1	Baseline (no delivery contract)	Pot-iso controls	8.65	0.45	8.20
2.1	Constant delivery, unlimited tank	$\kappa_{\text{sell}} = \frac{1}{9}$, pot-iso	6.18	0.17	6.01
2.2	Constant delivery, capped tank (13.5 g)	Tank cap; multi-phase schedule	6.15	0.17	5.98
3	24 h cycle with time-varying c_P	Cyclic tank; modulated T_{in}	6.24	0.19	6.05

Table 4.1: Overview of the result scenarios analysed in this chapter. Scenario 2 has two subcases (2.1 unlimited tank, 2.2 capped tank). Integrated revenue, salvage loss ($\Psi_0 - \Psi$), and net economy ($J - \Psi_0$) from section 4.1, section 4.2, and section 4.3 are divided by the scenario horizon (200 days for scenarios 1 and 2, 24 h for scenario 3). Daily averages are reported in euro cents per day (c/d).

The objective itself is unchanged from equation (3.1): Maximize integrated revenue plus terminal salvage over a finite horizon. The prices and critical degradation limits are the concrete numbers plugged into the formulation. Unless stated otherwise, the opimized scenario is setup, using the settings listed in table 4.2. Literature values for stack geometry and flows [24], end-of-life limits [3], cell cost [4], and energy prices [12] are indicated in the reference column; entries marked *chosen* are project defaults.

Several entries in table 4.2 are marked as chosen rather than taken from the literature. Brief rationale is given here: The voltage and inlet-temperature bounds keep the optimizer within a range of physically plausible operating conditions for the average SRU model. The number of FVM cells N_{cell} and the control segments N_t were set by the computational budget of the project: Finer discretizations were possible in principle, but the selected values keep each forward solve and outer iteration within tractable run times on the available hardware.

The 200-day horizon is chosen for practical reason. It is large enough that a non-constant control profile can show up within the optimization window, without needing to make too many Euler steps in the ODE solver, which requires more time to solve.

The parameter α_Ψ in equation (3.4) determines how sharply the terminal penalty rises as the cell approaches its end-of-life degradation threshold. In this work, a value of $\alpha_\Psi = 0.5$ is chosen: This means the penalty increases gently while the stack remains healthy, but becomes much steeper as the degradation nears the critical limit. With this choice, the optimizer remains sensitive to imminent failure, yet allows some flexibility for degradation earlier in the cell's lifetime.

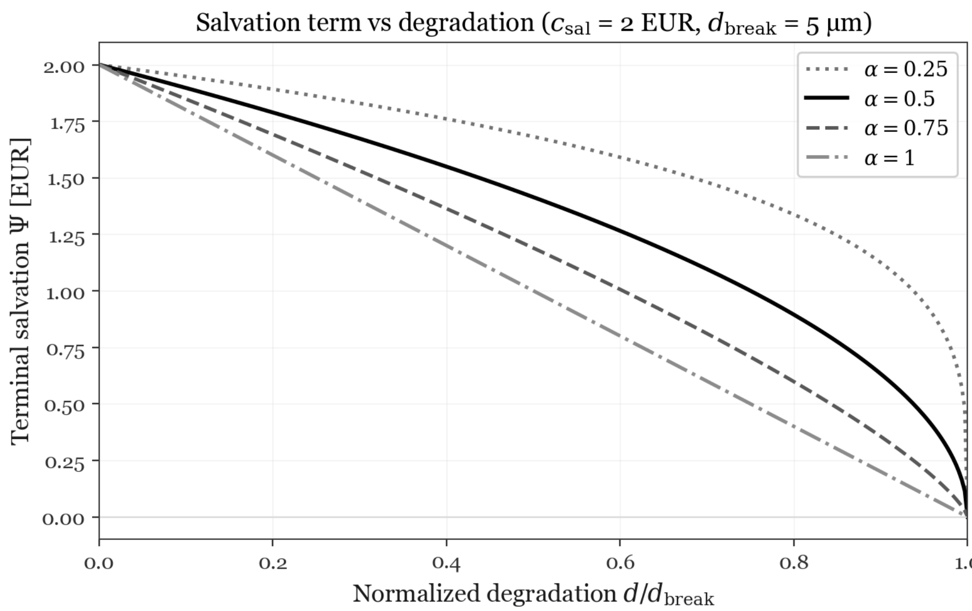


Figure 4.1: Effect of the salvage penalty sharpness parameter α_Ψ in equation (3.4) for a single degradation variable. For $\alpha_\Psi = 1$ (blue, linear) all degradation is penalized equally, while lower α_Ψ (red, green) creates a steeper penalty as end of life is approached, with near-zero penalty until close to the critical threshold. The value $\alpha_\Psi = 0.5$ (green) is chosen as a compromise.

This behavior is illustrated in figure 4.1, which plots the salvage penalty (for a single degradation parameter) for different values of α_Ψ. When $\alpha_\Psi = 1$, the penalty grows linearly with degradation. As α_Ψ approaches zero, the penalty remains negligible until the critical degradation limit is approached, after which the penalty jumps abruptly to its maximum. Selecting $\alpha_\Psi = 0.5$ strikes a balance: It penalizes all degradation to some degree, but especially emphasizes degradation close to failure, reflecting the greater importance of avoiding end-of-life conditions.

Throughout the chapter, three scenarios are optimized and shown here; afterward, a deeper analysis is carried out for sensitivity of relevant economic parameters for the scenarios where the analysis makes the most sense. An overview of the economic values given by the optimizers can be found in table 4.1.

4.1 Baseline optimization result

The first result is a baseline case, in which both controls are held constant over the full horizon: A single cell voltage U and a single inlet temperature T_{in}, broadcast to all $N_t = 5$ segments. The optimizer is run in potentiostatic-isothermal (pot-iso) mode, i.e. with fixed voltage and inlet temperature on every segment. This is the most restricted operating mode considered (U and T_{in} each collapse to one scalar decision variable). All settings follow table 4.2. The NLP returns:

$$U^* \approx 1.32\,\text{V}, \quad T_{\text{in}}^* \approx 751\,^\circ\text{C}$$

Setting	Baseline value	Unit	Reference
Horizon t_{horizon}	200	days	Chosen
Control segments N_t	5	—	Chosen
U bounds	$[0.8,\ 1.5]$	V	Chosen
T_{in} bounds	$[700,\ 900]$	$^{\circ}$C	Chosen
FVM cells N_{cell}	20	—	Chosen
Fuel / air flows	24 / 50	NL/h	[14]
Initial $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	0.75 / 1.5	μm	Values after 100 days burn-in
Critical $h_{\text{Ni-mi}}$ / $h_{\text{IC-ox}}$	5 / 10	μm	[18]
c_{H_2}	7.44	EUR/kg	[19]
c_P	0.079	EUR/kWh	[19]
c_{cell}	1000	EUR/m^2	[14], [1]
α_{Ψ}	0.5	—	Chosen
Optimization solver	SLSQP/Trust-Constr	—	—

Table 4.2: Numerical settings for the optimization results in this chapter (unless a subsection states otherwise).

Figure 4.2 shows the replayed trajectory at these controls. Mean nickel-migration thickness grows from $0.75\,\mu$m to the $5\,\mu$m critical degradation limit; interconnect oxidation from $1.5\,\mu$m to about $3.6\,\mu$m. Running revenue accumulates over the 200-day window while nickel migration reaches the enforced critical degradation limit at $t = t_{\text{horizon}}$.

The accumulated revenue is 17.30 EUR and the terminal salvage term is $\Psi \approx 1.01$ EUR. Because the degradation states are seeded from a 100-day burn-in (table 4.2), the salvage value at the horizon start is already $\Psi_0 \approx 1.91$ EUR rather than the pristine-cell value 2.00 EUR; the corresponding salvage loss before the optimization period is ≈ 0.09 EUR. Over the 200-day window the cell loses a further $\Psi_0 - \Psi \approx 0.90$ EUR of salvage value, giving a net economy $J - \Psi_0 \approx 16.40$ EUR on the $20\,\text{cm}^2$ cell.

The optimum operates exothermal rather than at thermoneutral conditions. The chosen voltage exceeds the thermoneutral value at the selected inlet temperature, so both reaction and ohmic heating raise the local temperature from inlet to outlet. At $t = t_{\text{horizon}}$, the outlet reaches about $780\,^{\circ}$C while the inlet remains near $751\,^{\circ}$C. Figure 4.3 illustrates these effects along the channel at $t = 0$, $t = t_{\text{horizon}}/2$, and $t = t_{\text{horizon}}$. By the end of the run, nickel migration emerges as the life-limiting degradation mechanism, with the nickel thickness within a few percent of the $5\,\mu$m cap across all positions z. Meanwhile interconnect oxidation remains well below its $10\,\mu$m cap and still shows a clear increase toward the warmer outlet. The almost uniform nickel profile at the critical degradation limit reflects two offsetting effects. The exothermal temperature field increases degradation toward the outlet, while the relatively even current-density distribution increases it toward the inlet. Together, these contributions produce a nearly uniform degradation pull along the cell. In contrast, interconnect oxidation continues to be more pronounced near the outlet, where temperatures are highest.

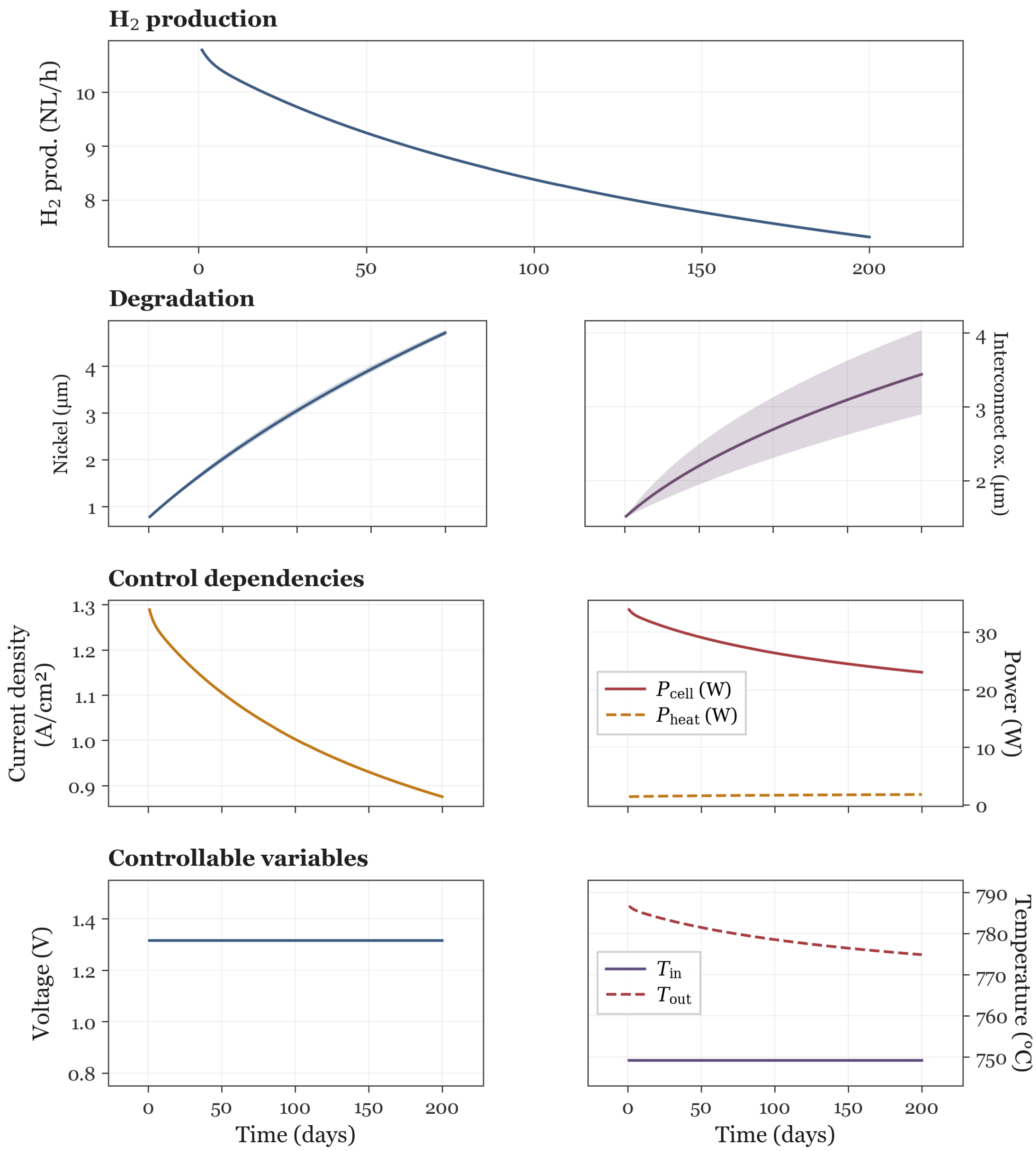


Figure 4.2: Baseline potentiostatic-isothermal optimum with no delivery contract ($N_t = 5$; table 4.2). Single controls $U^* \approx 1.32\,\mathrm{V}$ and $T_{\mathrm{in}}^* \approx 751\,^\circ\mathrm{C}$ are held over 200 days. Exothermal operation (U^* above the thermoneutral voltage at T_{in}^*) sustains positive P_{heat} and a rising temperature profile along the cell; $\dot{n}_{\mathrm{H_2,prod}}$ falls from ≈ 11.3 to $7.5\,\mathrm{NL/h}$ as nickel thickness approaches the critical degradation limit. Net economy $16.40\,\mathrm{EUR}$ on the $20\,\mathrm{cm}^2$ cell.

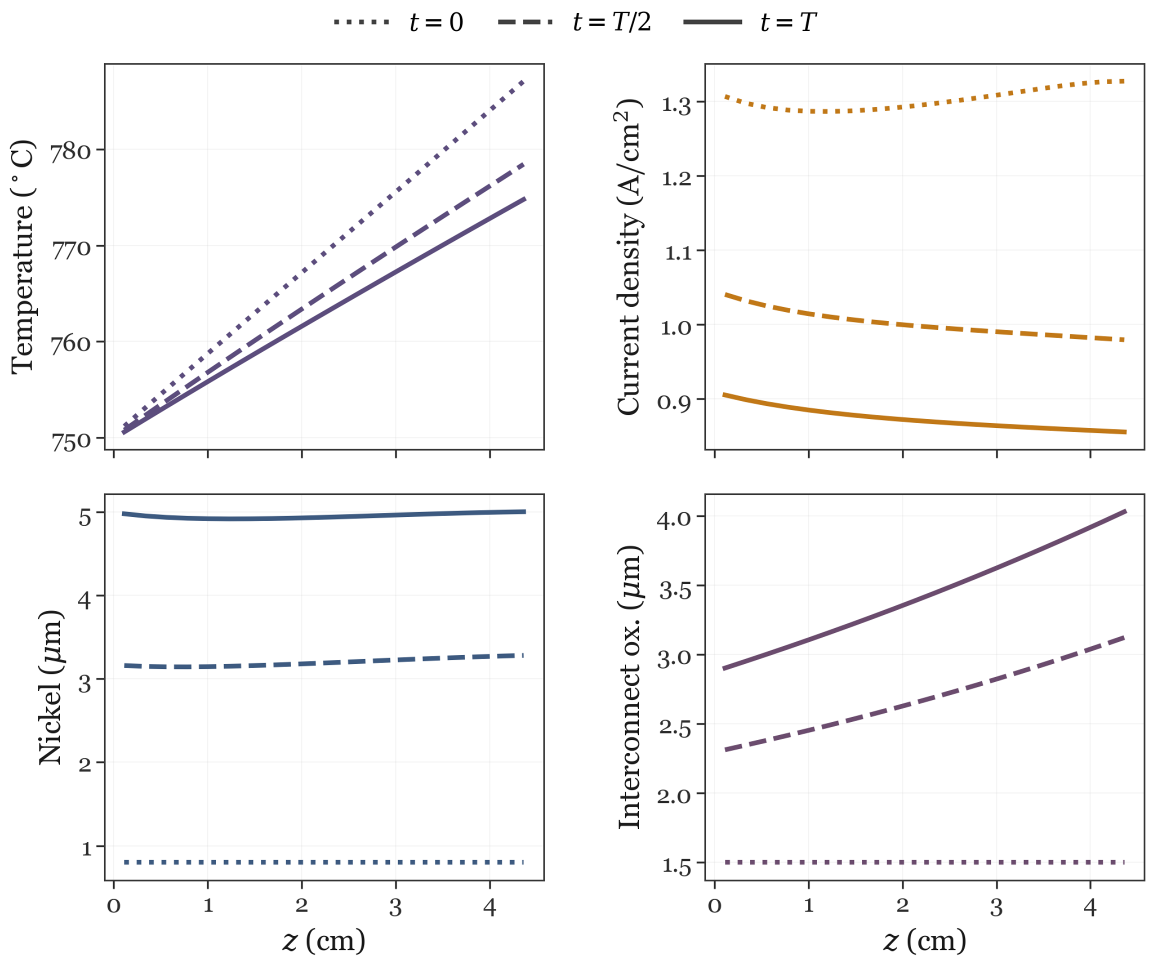


Figure 4.3: Spatial channel profiles at $t = 0$, $t = t_{\text{horizon}}/2$, and $t = t_{\text{horizon}}$ for the baseline pot-iso optimum (fine forward replay). Temperature rises along z under exothermal operation; nickel migration approaches the $5\,\mu$m critical degradation limit almost uniformly at $t = t_{\text{horizon}}$, while interconnect oxidation stays below its limit with stronger outlet weighting.

Setting	Value	Unit	Reference
Contractual delivery $\dot{n}_{\mathrm{H_2},\mathrm{agreement}}$	6.5	NL/h	Chosen
Initial tank inventory $m_{\mathrm{tank},0}$	0	kg	Chosen
Spot purchase penalty κ_{buy}	1.5	—	Chosen
Terminal liquidation credit κ_{sell}	$\frac{1}{9}$	—	Chosen
Tank inventory cap (case 1 / 2)	∞ 13.5 grams	Chosen / Update	
Updated variables	**Value**	**Unit**	**Reference**
Control segments N_t	20	—	Chosen

Table 4.3: Tank and delivery-agreement settings for the constant-delivery results (in addition to table 4.2). N_t is refined to 20 segments (10-day resolution) for the tank scenarios.

The next thing to do was to finally give the control more freedom and let the individual control segments take whatever value optimized the most, here a somewhat trivial thing happened. The control did not change and the potentiostatic-isothermal operation is the best. Even though it is a boring result, it makes perfect sense: The unrestricted baseline still favors steady potentiostatic-isothermal controls, with voltage and inlet temperature locked at the exothermal optimum that extracts revenue up to the nickel-migration critical degradation limit. So to make use of the dynamic flexibility of the model, the introduction of operation settings was introduced.

4.2 Constant delivery rate

The next result uses the storage-tank implementation from section 3.1.3 with a fixed contractual delivery rate. All other settings follow table 4.2; the additional tank parameters are listed in table 4.3.

The agreement rate $\dot{n}_{\mathrm{H_2},\mathrm{agreement}} = 6.5\,\mathrm{NL/h}$ per cell sits below the unconstrained baseline production at $t = 0$ (section 4.1), so surplus hydrogen can be stored rather than sold immediately. Scaled to a production setup of two stacks with about 150 cells each, the combined daily delivery is approximately $4.3\,\mathrm{kg\,H_2}$. The spot-purchase penalty $\kappa_{\mathrm{buy}} = 1.5$ fines under-delivery at 50% of the contract hydrogen price. The terminal liquidation credit $\kappa_{\mathrm{sell}} = \frac{1}{9}$ values leftover inventory at roughly 11% of the contract price, reflecting competition with lower-cost *black* hydrogen on an open market.

4.2.1 Optimum at low liquidation credit

With $\kappa_{\mathrm{sell}} = \frac{1}{9}$, terminal resale is so weak that the optimizer prefers to hoard surplus hydrogen rather than produce aggressively and liquidate at the optimization horizon. Figure 4.4 shows the optimum at this liquidation credit: A single pot-iso held for all 200 days:

$$U^* \approx 1.289\,\mathrm{V}, \qquad T^*_{\mathrm{in}} \approx 733.5\,^{\circ}\mathrm{C}.$$

Initial production ($\approx 8.1\,\mathrm{NL/h}$) exceeds the agreement rate, so the tank fills monotonically to a peak of $\approx 0.27\,\mathrm{kg}$ around day 170. As degradation lowers production below $6.5\,\mathrm{NL/h}$, the inventory buffer covers the shortfall; no spot purchases occur. Because unlimited storage is free in the model and κ_{sell} is small, the optimizer does not schedule galvanostatic intervals, but instead chooses constant gentle operation that minimizes salvage loss while primarily minimizing the added resistance induced by degradation, and uses the tank inventory to operate at lower production rates near the end.

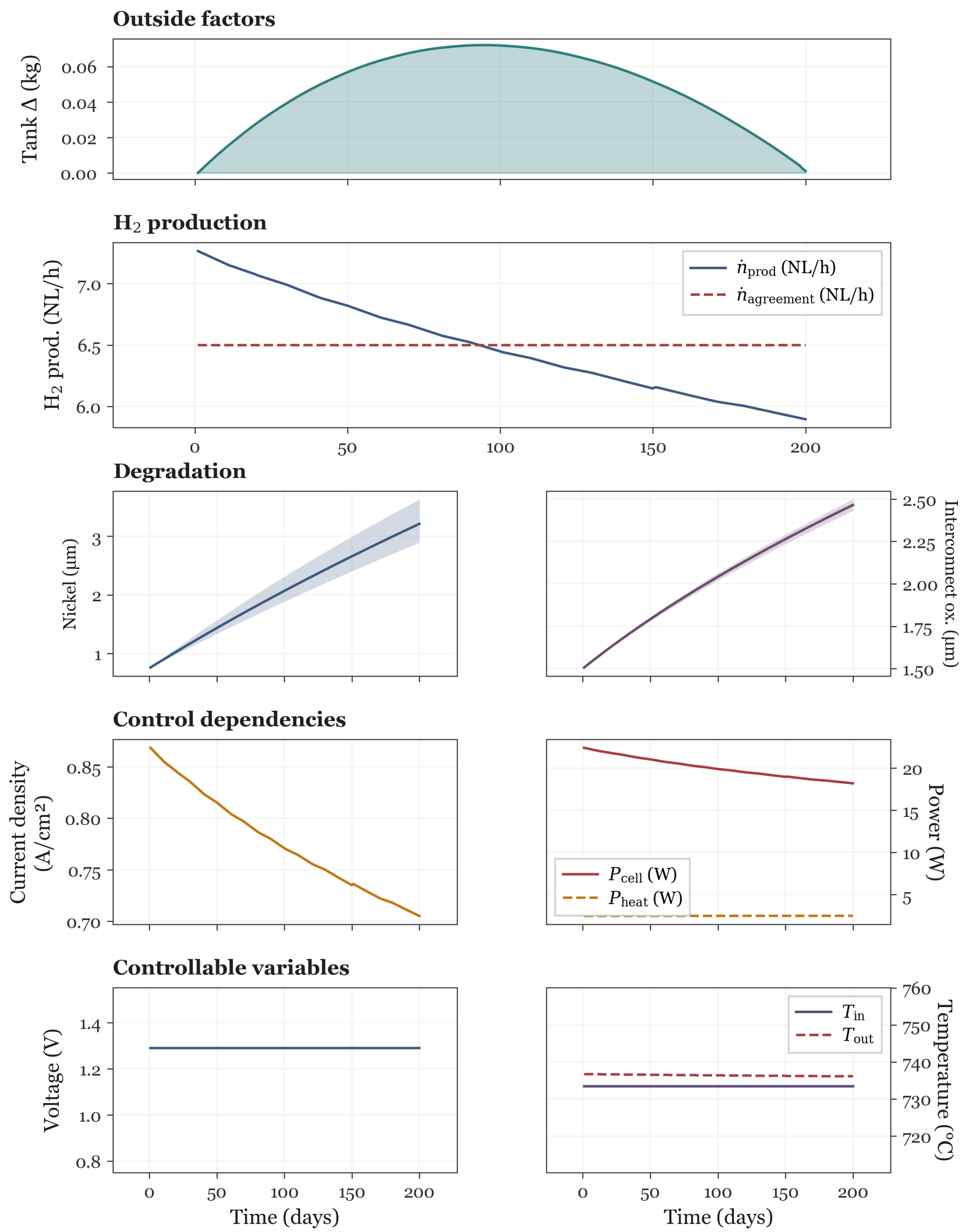


Figure 4.4: Constant-delivery optimum at $\kappa_{\text{sell}} = \frac{1}{9}$ and $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$ ($N_t = 20$; table 4.3). Single controls $U^* \approx 1.289\,\text{V}$, $T^*_{\text{in}} \approx 750\,^\circ\text{C}$; tank inventory peaks at $\approx 0.27\,\text{kg}$ (day 170) and remains near $\approx 0.26\,\text{kg}$ at the horizon. Running revenue $\approx 11.64\,\text{EUR}$, terminal salvage $\approx 1.50\,\text{EUR}$, net economy $\approx 13.36\,\text{EUR}$.

The accumulated revenue is ≈ 12.36 EUR and the terminal salvage lost is ≈ 0.34 EUR; the terminal liquidation credit contributes only marginally. The net economy is ≈ 12.02 EUR on the $20\,\mathrm{cm}^2$ cell over 200 days.

4.2.2 Limited tank reserve via scheduled mode change

The unconstrained optimum in figure 4.4 fills storage to roughly 72 grams, which is more than many practical tanks would hold. Figure 4.5 illustrates an alternative from the same contract and κ_{sell}, with an added maximum tank content of ≈ 13.5 grams. The schedule has three phases:

1. **Gentle pot-iso** (U, T_{in} slightly below the $\kappa_{\text{sell}} = \frac{1}{9}$ optimum) fills the tank more slowly and reduces early degradation stress.
2. **Galvanostatic window** (fixed U, stepped T_{in} tuned to track $\dot{n}_{\text{H}_2,\text{agreement}}$) holds delivery while inventory approaches the cap.
3. **Closing pot-iso** (return to U^*, hold T_{in} at the galvanostatic endpoint) lets production track degradation while inventory stays bounded.

Operationally it is preferable when inventory capacity is finite or carrying reserve hydrogen has off-model costs, even though the economic objective may differ slightly from figure 4.4. However, the important takeaway from this result is that the solver chooses galvanostatic and potentiostatic operation and adjusts by temperature.

The accumulated revenue is approximately 12.30 EUR, with a terminal salvage loss of about 0.34 EUR, resulting in a net revenue of roughly 11.96 EUR. Notably, the main penalty for deviating from the gentle potentiostatic-isothermal strategy appears as reduced revenue rather than a larger salvage loss. This reduction stems from increased resistance as degradation accelerates. Both strategies ultimately deliver the same amount of hydrogen and reach similar final cell states, operating at a constant thermoneutral voltage—where hydrogen production relative to power cost is maximized. However, during galvanostatic operation, the cell experiences faster degradation and higher resistance, which in turn increases the power cost per unit of hydrogen produced.

4.3 24-hour cycle with varying power price

The final scenario shortens the horizon to one day and replaces the flat power price with a time-varying profile. The cosine form in table 4.4 is a stylized stand-in for solar-driven day-night spreads: Electricity is cheapest around midday and most expensive around midnight, with roughly a $19\times$ ratio between the midnight maximum and the midday minimum. Together with the contractual delivery setup from section 4.2, this case tests whether the optimizer shifts production toward low-price periods while still meeting the agreement.

Terminal liquidation is disabled ($\kappa_{\text{sell}} = 0$), and the cyclic inventory constraint equation (3.9) is enforced so that $m_{\text{tank}}(0) = m_{\text{tank}}(t_{\text{horizon}})$. The day can therefore be studied as a repeating operating window without end-of-horizon sell-off. The updated settings are listed in table 4.4.

To help the optimizer converge to a sensible solution, continuity penalties have been added to the objective; see appendix 6 for more details. The weights are defined in table 4.4, and the reasoning behind the choices is discussed in section 5.3.2.

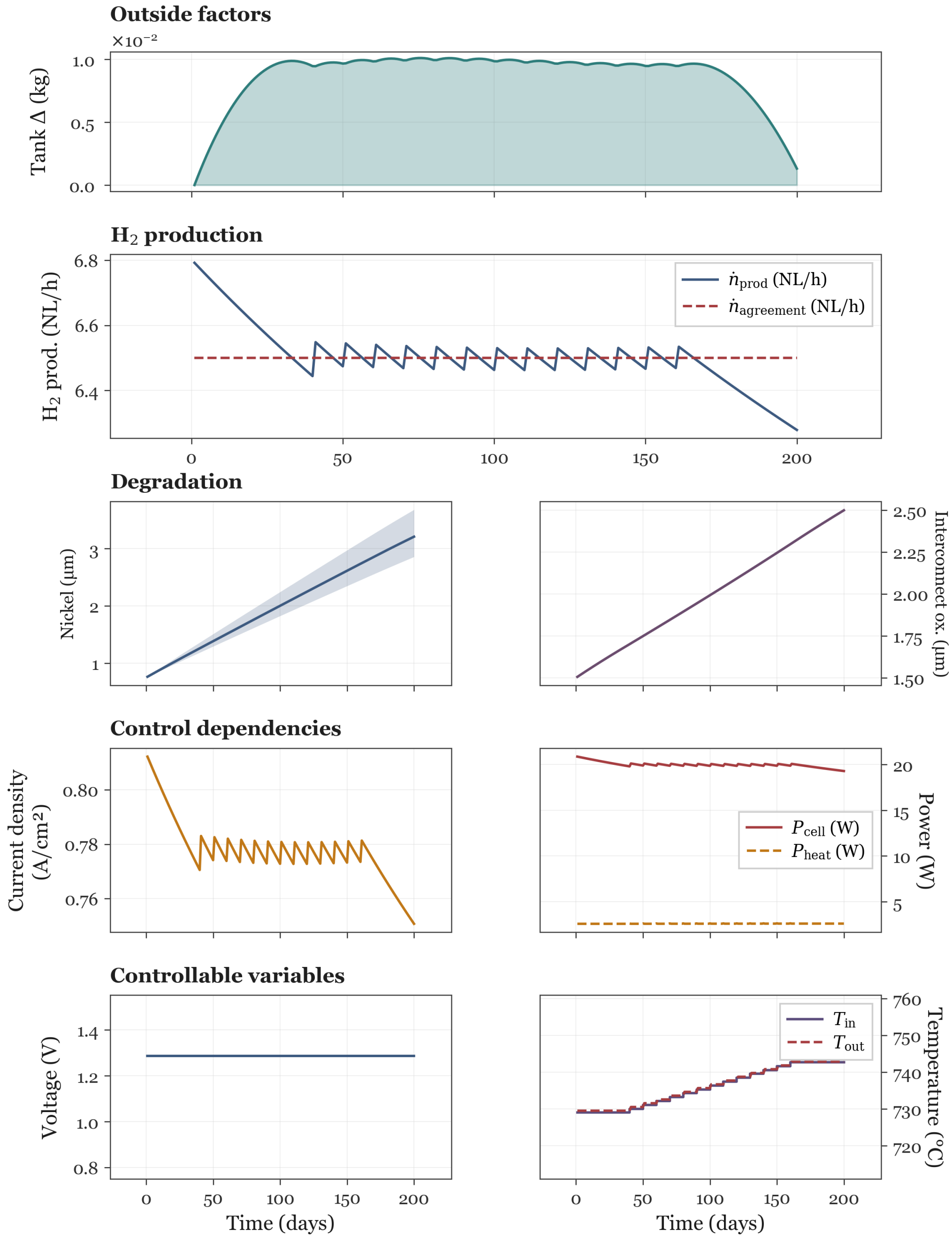


Figure 4.5: Limited capacity tank replay at $\kappa_{\text{sell}} = \frac{1}{9}$ and $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5$ NL/h (table 4.3). Reduced pot-iso fill (segments 0-3; $U \approx 1.289$ V, $T_{\text{in}} \approx 735\,^{\circ}$C), galvanostatic middle via stepped T_{in} at $U \approx 1.289$ V (segments 4-10, ramp to $\approx 748\,^{\circ}$C), and closing pot-iso hold at that same T_{in} (segments 11-19).

Setting	**Value**	**Unit**	**Reference**
Temperature continuity penalty w_T	10^{-5}	—	Chosen (appendix 6, section 5.3.2)
Voltage continuity penalty w_U	0.1	—	Chosen (appendix 6)
Updated variable	**Value**	**Unit**	**Reference**
Horizon t_{horizon}	24	h	Chosen
Control segments N_t	12	—	Chosen
Terminal liquidation κ_{sell}	0	—	Chosen
Initial tank inventory $m_{\text{tank},0}$	$m_{\text{tank},T}$	—	Periodic (3.1.3)
Power price $c_{P,k}$	$\bar{c}_P(1 - a\cos\theta_k)$, $\theta_k = 2\pi(h_k - 12\,\text{h})/(24\,\text{h})$	EUR/(W·s)	Chosen
Mean power price $\bar{c}_P$	0.079	EUR/kWh	[19]
Relative price swing a	0.9	—	Chosen

Table 4.4: Updated model settings for the 24-hour cyclic scenario. No terminal sale of hydrogen is allowed ($\kappa_{\text{sell}} = 0$), and the initial tank content is enforced as a periodic boundary. Twelve control segments are used for the day. The piecewise-constant power price on segment k follows a single cosine cycle over 24 h, with h_k the hour-of-day at the segment midpoint; the minimum $c_{P,k} = \bar{c}_P(1-a)$ occurs at midday ($h_k = 12\,\text{h}$) and the maximum $\bar{c}_P(1+a)$ at midnight, modelling lower prices when solar supply is high ($a = 0.9$ gives $\approx 19\times$ min/max spread). The continuity weights w_U and w_T enter the soft penalty appendix 6, including the last→first segment wrap consistent with the repeating day; with potentiostatic voltage only w_T is effective.

Using these settings, temperature remains the primary free control. The optimal trajectory is shown in figure 4.6. Production is highest when the power price is low and reduced when prices rise at night. The response is qualitatively as expected for load shifting: The optimizer modulates inlet temperature rather than exploiting both voltage and temperature.

The economic decomposition of the optimized run is as follows: Integrated running revenue over the 24,h window is $\Phi \approx 0.062\,\text{EUR}$ and the terminal salvage term is $\Psi \approx 1.91\,\text{EUR}$, giving a total objective $J \approx 1.97\,\text{EUR}$. Because the degradation states are seeded from the same 100-day burn-in as the long-horizon cases, the salvage value at the start of the day is already $\Psi_0 \approx 1.91\,\text{EUR}$; the additional salvage loss over the first day is therefore only $\approx 0.002\,\text{EUR}$, so the net economy from operation is effectively $\approx 0.060\,\text{EUR}$ for the $20\,\text{cm}^2$ cell.

This outcome is primarily shaped by two modelling choices: The cyclic tank constraint, which ensures no terminal credit for inventory and enforces production/storage balance, and the use of continuity penalties. For $w_T = 10^{-5}$, the optimizer finds an intermediate solution with a daily cyclic temperature swing and load-shifting behavior. As w_T is increased ($w_T \gg 0$), the solution approaches potentiostatic-isothermal operation with nearly flat T_{in}; as $w_T \to 0$, the solution becomes increasingly piecewise-constant, allowing sharper temperature changes between control segments, leading to a cell that is only turned on during low power price hours.

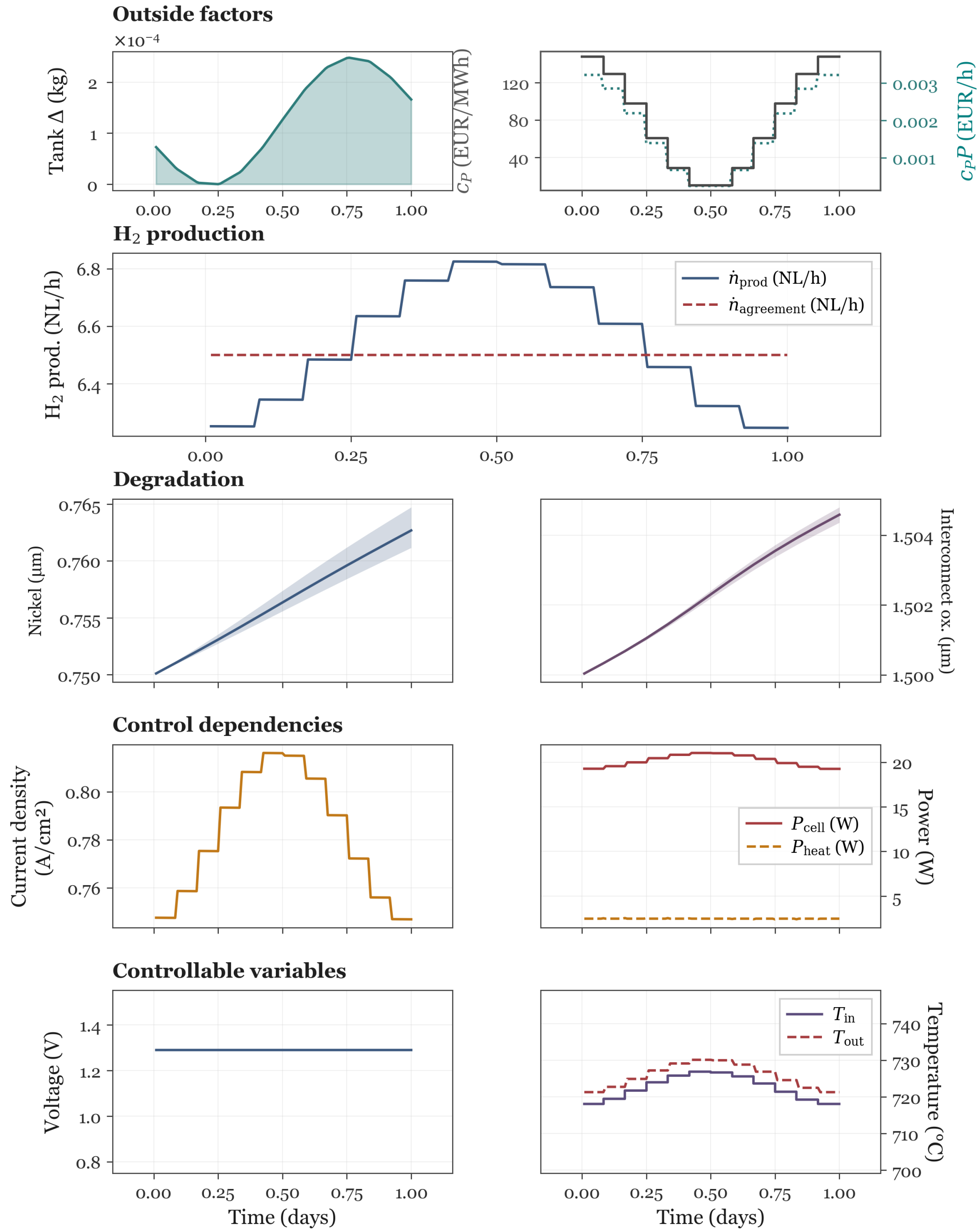


Figure 4.6: 24-hour cyclic optimum under a daily cyclic power price ($N_t = 12$; table 4.4). Fixed $U \approx 1.289\,\text{V}$ ($w_U = 0.1$); T_{in} is modulated ($w_T = 10^{-5}$) to shift production toward low-price midday segments (c_P and $c_P P$ in **Outside factors**). Peak $\dot{n}_{\text{H}_2,\text{prod}} \approx 6.8\,\text{NL/h}$ versus $\approx 6.2\,\text{NL/h}$ overnight; the agreement rate $6.5\,\text{NL/h}$ is met on average via cyclic tank use ($\kappa_{\text{sell}} = 0$). Running revenue $\Phi \approx 0.062\,\text{EUR}$, terminal salvage $\Psi \approx 1.91\,\text{EUR}$.

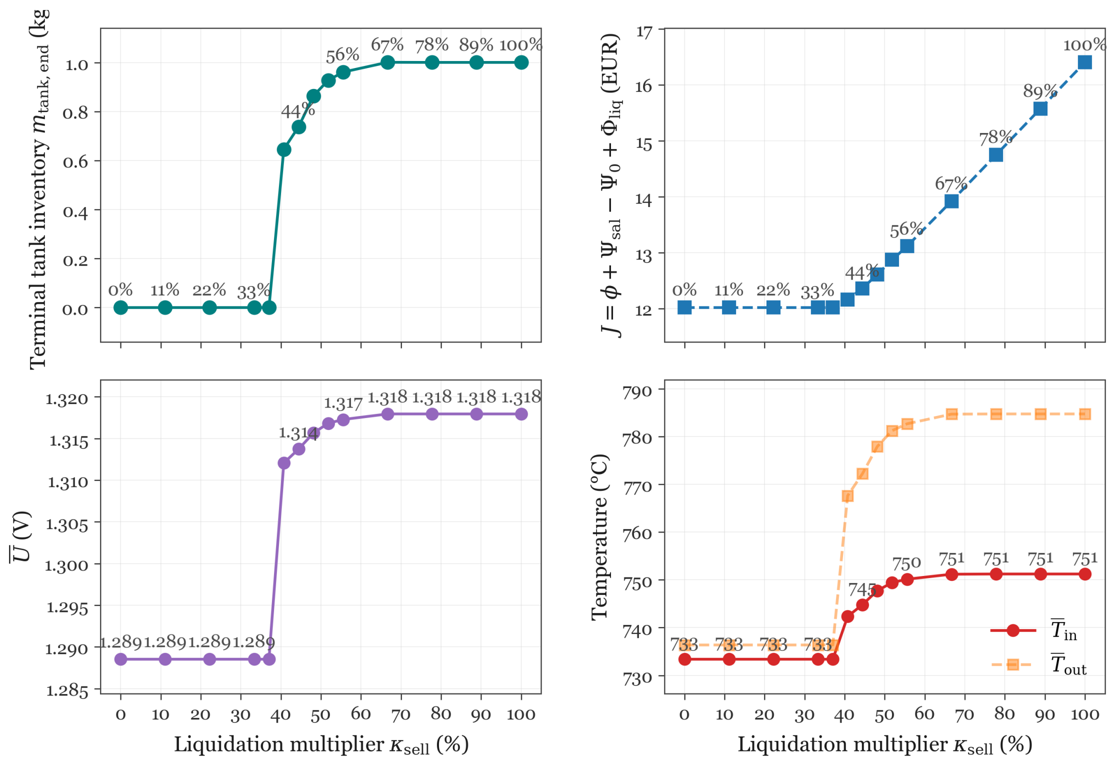


Figure 4.7: Terminal liquidation credit κ_{sell} sensitivity for the constant-delivery tank scenario ($\dot{n}_{H_2,\text{agreement}} = 6.5\,\text{NL/h}$, pot-iso controls, no tank cap). Top panels: Terminal tank inventory and net objective $J = \phi + \Psi_{\text{sal}} - \Psi_0 + \Phi_{\text{liq}}$; bottom panels: Mean voltage and inlet/outlet temperature. For $\kappa_{\text{sell}} \lesssim 37\%$ the optimizer leaves the tank empty and holds moderate U and T_{in}. Above $\approx 40\%$ it hoards surplus hydrogen and drives the cell toward the nickel critical degradation limit.

4.4 Sensitivity analysis

4.4.1 Liquidation sensitivity

The constant-delivery result at $\kappa_{\text{sell}} = \frac{1}{9}$ leaves negligible terminal inventory, whereas the unconstrained baseline favors higher current and temperature–behaviour that corresponds formally to $\kappa_{\text{sell}} = 1$. A sweep over the terminal liquidation credit was therefore carried out to identify when the optimizer prefers to drive the cell toward the nickel critical degradation limit rather than meet delivery through gentler operation. The outcome is shown in figure 4.7. For clarity, every point in the figure was obtained with potentiostatic-isothermal controls (U and T_{in} constant over the horizon), which is the mode the solver selects when no tank-cap constraint is imposed.

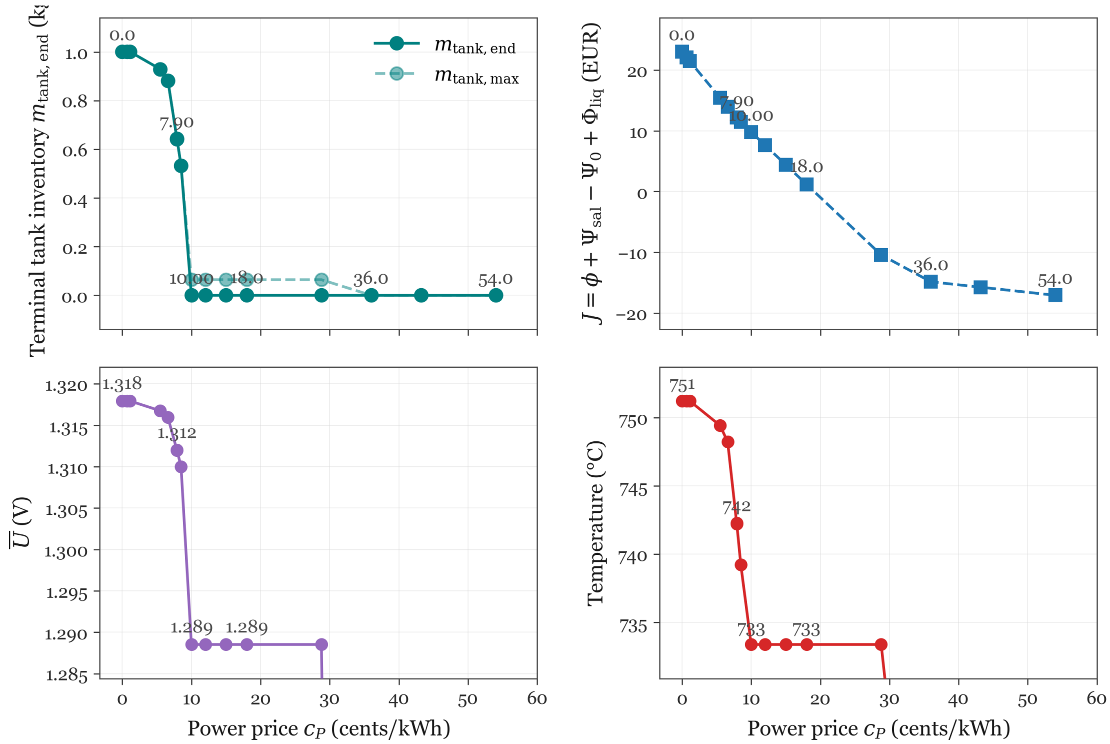


Figure 4.8: Power price c_P sensitivity at fixed $\kappa_{\text{sell}} = 40.74\%$ and $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$ (pot-iso controls). Top panels: Terminal and peak tank inventory and net objective J; bottom panels: Mean voltage and inlet temperature. At the baseline price ($\approx 7.9\,\text{cent/kWh}$, starred) the optimizer hoards $\approx 0.53\,\text{kg}$ by running the cell hard. As c_P rises, inventory falls and J decreases while production is kept alive down to $\approx 29\,\text{cent/kWh}$. Above $\approx 36\,\text{cent/kWh}$ electrolysis shuts down and contract hydrogen is purchased at spot.

Figure 4.7 exhibits a clear transition: Beyond a threshold, the marginal cost of additional production exceeds the value of terminal resale, and the optimizer drains the tank instead of filling it. This transition lies between $\kappa_{\text{sell}} \approx 37\%$ and $\kappa_{\text{sell}} \approx 40\%$, corresponding to an effective resale price of roughly 2.92-$3.16\,\text{cent/kWh}$. Above the breakpoint the optimum moves rapidly toward the critical degradation limit. Immediately below it, a narrow band remains in which modest over-production is still marginally worthwhile without triggering operation at the critical degradation limit. In that region, rising resistance makes further production only weakly attractive relative to power cost, and the salvage term contributes to the trade-off: The liquidation credit is not yet high enough to justify the extra degradation.

4.4.2 Power price sensitivity

Following this sweep, $\kappa_{\text{sell}} = 40\%$ was selected for the remaining one-parameter studies. The power price c_P was varied next. Because terminal liquidation acts as a post-horizon hydrogen price, lowering κ_{sell} is qualitatively similar to raising c_P; the c_P sweep therefore mirrors the liquidation response in reverse. Figure 4.8 confirms this expectation.

Two distinct regimes appear. In the first, the optimum shifts from operation at the critical degradation limit toward an empty terminal tank while still maintaining production to satisfy the delivery obligation; electrolysis remains cheaper than spot purchase even as inventory is run down. In the second, at sufficiently high c_P, production is abandoned entirely: The cell idles at minimum temperature and zero current, and all contractual hydrogen is bought on the spot market.

4.4.3 salvage price sensitivity

The sensitivity to the salvage price c_{sal} was examined at the same reference point ($\kappa_{\text{sell}} \approx 40\%$), where terminal inventory also begins to rise. Figure 4.9 shows the result.

The c_{sal} response, again, separates into two steps. At low salvage prices the optimum is dominated by resistance growth: Degradation raises ohmic losses and the terminal salvage term is secondary. As c_{sal} increases, salvage loss weighs more heavily until additional production is no longer justified relative to the value destroyed in the stack. At the highest prices examined, spot purchase becomes cheaper than operating the cell.

4.4.4 Delivery rate sensitivity

Finally, the contractual delivery rate $\dot{n}_{\text{H}_2,\text{agreement}}$ was varied to test which control adjusts when terminal liquidation forces the tank empty. Results in section 4.3 and section 4.2 suggest that, under potentiostatic operation, temperature alone should modulate production while voltage stays fixed. Figure 4.10 confirms this pattern at $\kappa_{\text{sell}} = 0$.

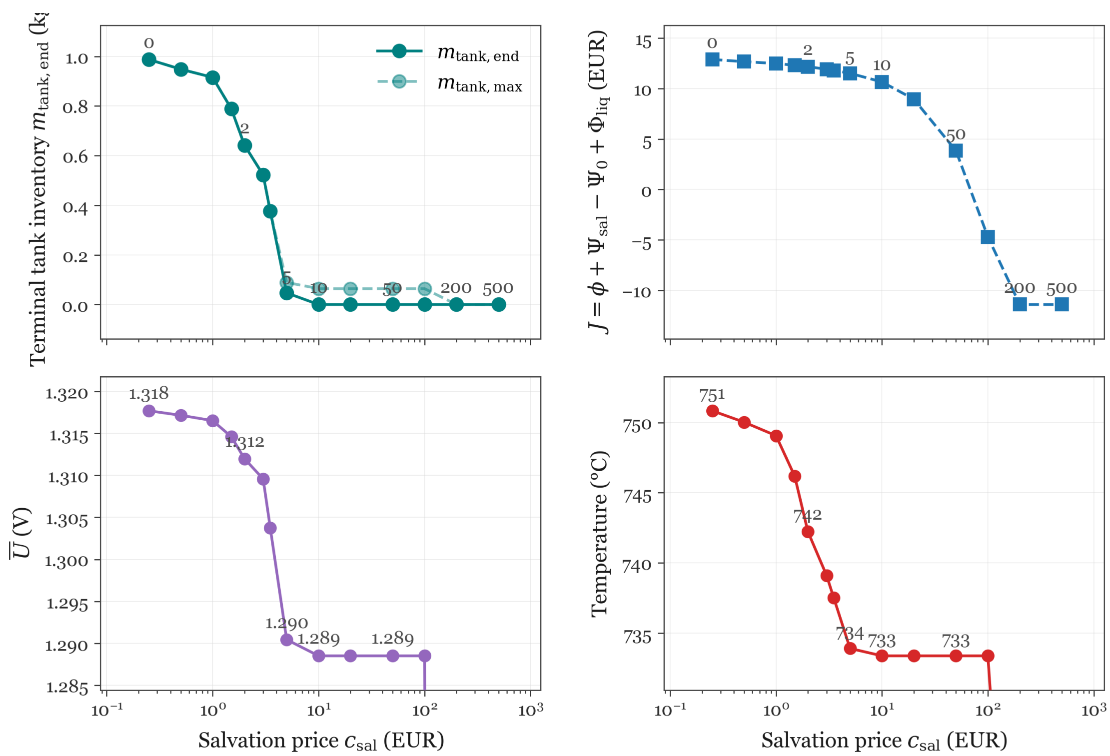


Figure 4.9: salvage price c_{sal} sensitivity at fixed $\kappa_{\text{sell}} = 40.74\%$ and $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$ (pot-iso controls). Top panels: Terminal and peak tank inventory and net objective J; bottom panels: Mean voltage and inlet temperature. At the baseline price (2 EUR, starred) the optimizer hoards $\approx 0.64\,\text{kg}$ by running the cell hard. As c_{sal} rises, inventory falls and J decreases; above $\approx 10\,\text{EUR}$ the optimum reverts to gentle pot-iso operation with negligible terminal inventory.

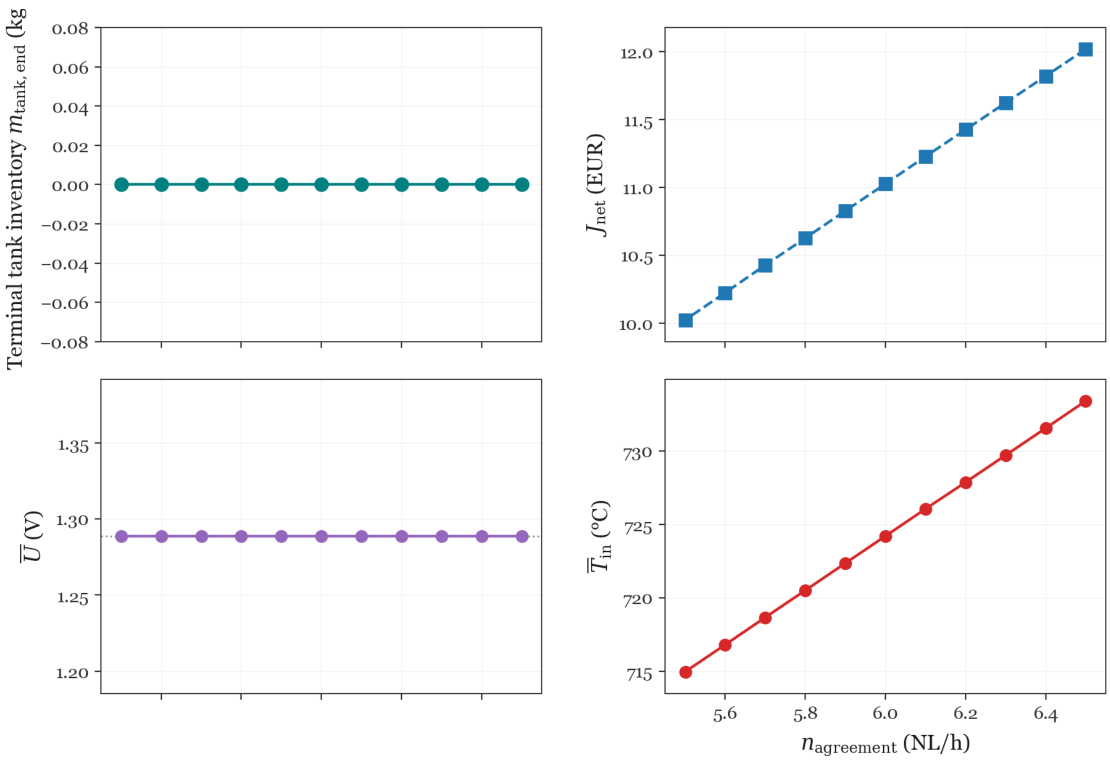


Figure 4.10: Contractual delivery rate sensitivity at $\kappa_{\text{sell}} = 0$ and pot-iso controls ($N_t = 20$). U is fixed to the 6.5 NL/h optimum (≈ 1.289 V); T_{in} is reduced for lower agreement rates so that $m_{\text{tank,end}} = 0$. Top panels: Terminal tank inventory and net objective J_{net}; bottom panels: Mean voltage and inlet temperature. J_{net} rises almost linearly from ≈ 10.0 to ≈ 12.0 EUR as $\dot{n}_{\text{H}_2\text{,agreement}}$ increases from 5.5 to 6.5 NL/h. Voltage stays flat while $\overline{T}_{\text{in}}$ falls from ≈ 734 to ≈ 715 °C.

5 Discussion

The results from chapter 4 show that the optimizer generally favors potentiostatic-isothermal operation, adjusting temperature as needed to meet the delivery agreement or shift production to lower-cost periods. In the unconstrained baseline (section 4.1), exothermal operation is optimal but limited by the salvage term and the nickel critical degradation limit. Otherwise, delivery contracts and price profiles dominate, with the salvage term having less impact. The following sections interpret these findings, discuss parameter choices, explain modelling assumptions, and outline limitations and future directions.

5.1 Interpretation of optimal operating strategies

All considered, the results indicate that extra control freedom is used only when contractual delivery or time-varying prices force a departure from potentiostatic-isothermal operation. The subsections below follow that logic.

5.1.1 Baseline without delivery obligations

The first question is how much of the available control freedom is actually used. For the unrestricted baseline (section 4.1), the answer is very little: Without a delivery floor, maximizing $\int R\,\mathrm{d}t + \Psi$ favours steady efficient production, and the optimal mode remains potentiostatic-isothermal. The additional decision variables are largely redundant in this scenario.

The baseline optimum is nevertheless exothermal: $U^* \approx 1.32\,\mathrm{V}$ exceeds the thermoneutral voltage at $T_{\mathrm{in}}^* \approx 751\,^\circ\mathrm{C}$, so heat release raises the temperature along the channel (roughly 751 to $780\,^\circ\mathrm{C}$ at $t = t_{\mathrm{horizon}}$; figure 4.3). Revenue is maximized up to the nickel-migration critical degradation limit at $5\,\mu\mathrm{m}$; interconnect oxidation stays near $3.6\,\mu\mathrm{m}$ and is not life-limiting. Because the chosen voltage is effectively pinned at this exothermal operating point, U is also constant across the other long-horizon cases considered. A repeat study with voltage fixed potentiostatic and only one remaining degree of freedom, either piecewise galvanostatic current with inferred temperature, or piecewise isothermal T_{in} with inferred current, would likely reproduce the same qualitative conclusions while reducing the decision space.

5.1.2 Effect of a fixed delivery contract

A second question concerns how the control variables are parametrized within each segment. In the present setup, both U and T_{in} are held constant over each control interval, simply because the NLP formulation is structured this way (section 3.2.5). In the fixed-delivery scenario (section 4.2), the production tends to fluctuate around the delivery agreement; in those periods, it would actually be more natural to hold the current constant (operate galvanostatically) instead of voltage or temperature. A natural next step would be to give the optimizer the freedom to decide, for each interval, whether to use potentiostatic, galvanostatic, or isothermal operation.

Even without this implementation, the current results already reveal when galvanostatic operation becomes relevant, and show how tank inventory enables or enforces these switches between operating modes.

5.1.3 Temperature jumps and practical implementation

Piecewise-constant optimal controls can imply large instantaneous changes in T_{in} between segments. Such jumps are difficult to implement on hardware and are not charged directly in the base formulation. In the 24-hour run, a squared continuity soft penalty was added retrospectively (appendix 6). A more physical alternative would be to include the auxiliary preheat power required to move from one inlet set-point to the next within a segment duration, or to model thermal cycling as an additional degradation driver.

5.2 Implied lifetime and proximity to critical degradation limits

Across the long-horizon runs, current densities are relatively high and the modelled degradation states approach the chosen critical degradation limits within 200 days (table 4.2). In the baseline replay (section 4.1), mean nickel-migration thickness reaches the $5\,\mu$m limit at $t = t_{\text{horizon}}$ while interconnect oxidation remains well below its $10\,\mu$m limit. The window is far shorter than stack lifetimes reported in the SOEC literature. Beyrami et al. predict lifetimes above $50{,}000\,\text{h}$ at $0.6\,\text{A/cm}^2$ when component failure criteria are enforced [3], and later embed stack lifetimes of 4-5 years into plant-scale economic optimization [4]. Giridhar et al. obtain replacement schedules on the order of 2 to 5 years depending on electricity price and operating mode [14].

The gap is partly expected. This thesis maximizes revenue plus salvage over a fixed 200-day horizon rather than full lifetime, and the delivery scenarios include galvanostatic periods that accelerate the modelled degradation mechanisms. It nevertheless suggests that one or more modelling choices may be too permissive. The fitted nickel-migration rate law (section 2.2.1) is a simple empirical relation and may overestimate the nickel-migration rate at the currents seen in optimization.

5.2.1 Revenue-Salvage balance in the objective

A complementary explanation for the aggressive control involves the way the scalar objective is formulated. In the baseline case (section 4.1), the cell still reaches its critical degradation limit, yet the salvage term is not driven to zero. This is a consequence of decisions made early in the project: Specifically, the choice of Ψ was intended to provide a simple trade-off between maximum and mean degradation. As a result, the balance between ongoing revenue and the terminal salvage term may not be fully aligned. If either c_{cell} or α_Ψ in equation (3.4) underestimates the true cost of approaching end-of-life, the optimizer may select aggressive (exothermal) temperature and current profiles that literature-based lifetime studies would typically avoid.

In this baseline, monotonic heating along the channel results in nearly uniform nickel-migration growth (figure 4.3), so failure occurs almost uniformly rather than at a single hotspot—but the $5\,\mu$m cap is still eventually reached at $t = t_{\text{horizon}}$. Meanwhile, ohmic losses increase linearly with resistance from nickel migration and interconnect oxidation, but the degradation rates themselves tend to slow as the states evolve. This means that a small increase in temperature can generate disproportionately more revenue for a given increment in resistance. Since the salvage term is bounded, it does not increase sharply enough throughout state space to fully offset this trend, unless α_Ψ is set extremely high.

One alternative is to replace the hard critical degradation limits, used in this thesis, with a failure probability that scales revenue at each instant. A schematic form is given in equation (5.1).

$$\int_0^{t_{\text{horizon}}} \left(1 - P_{\text{fail}}(U, T, I, \mathbf{x})\right) R \,\mathrm{d}t + \mathbb{E}[\tau_{\text{life}} \mid \mathbf{x}_T]\, \mathbb{E}[R \mid \mathbf{x}], \tag{5.1}$$

where P_{fail} is the probability that the cell reaches a failure criterion given the controls and degradation state $\mathbf{x}$, and $\mathbb{E}[\tau_{\text{life}} \mid \mathbf{x}_0]$ is the expected remaining lifetime from the initial state.

Rather than stopping the simulation at a hard cap, revenue would be discounted by the instantaneous risk of failure, and the second term would reward expected production over life. In principle, P_{fail} could rise sharply once degradation states pass critical thresholds, giving a stronger, and nonlinear, penalty for operating too aggressively than the linear resistance increase combined with the bounded salvage term. Such a formulation would also extend naturally to fully resolved stack- and plant-scale objectives beyond the present average-SRU abstraction.

Such a model would require much more data than is currently available. However, an approximate model might be feasible with a comprehensive literature review on the probability of failure for each cell component, as this information is likely partially known. From there, it could be possible to construct a probabilistic model, though this would require substantial additional work. With such a model, it would then be possible to consider whether to optimize for expected revenue or for a different objective, such as a lower quantile of performance, depending on the desired risk tolerance.

Another, simpler approach to this problem would be to keep the current structure, but use a salvage term that multiplies the maximum degradation by the mean degradation. This would yield a squared salvage term that simultaneously weighs both mean and maximum degradation.

5.3 Justification of project-specific settings

The results chapter (chapter 4) marks several inputs as *chosen* in table 4.2, table 4.3, and table 4.4. Settings for the horizon, control segments, finite-volume mesh, operating bounds, and α_Ψ are already motivated in chapter 4 and the grid-independence study (section 3.2.2), which covers the baseline case (table 4.2). The subsections below focus on the delivery and cyclic scenarios.

5.3.1 Delivery setup

Table 4.3 lists both delivery scenarios: Case 1 uses an uncapped tank (∞), which isolates the contract and penalty structure from inventory-cap effects; case 2 imposes a maximum inventory of $13.5\,\mathrm{g}$, chosen as a practical reserve level after the unconstrained optimum in figure 4.4 fills storage to roughly $72\,\mathrm{g}$.

The contractual delivery rate and the multipliers κ_{buy} and κ_{sell} are strongly coupled to the optimal control. Exploratory runs in section 4.2 guide the reported values. If $\dot{n}_{\text{H}_2,\text{agreement}}$ is raised above the unconstrained production rate at the baseline optimum (section 4.1), the solver eventually drives the cell to the critical degradation limits, because sustaining that throughput is too aggressive. If the agreement rate is set below the baseline terminal production rate ($\approx 7.5\,\text{NL/h}$ at $t = t_{\text{horizon}}$ in section 4.1), the solution depends mainly on κ_{sell}: With $\kappa_{\text{sell}} = 1$, the optimum coincides with the unrestricted baseline (including terminal tank liquidation); with $\kappa_{\text{sell}} = 0$, inlet temperature is reduced so that $m_{\text{tank}}(t_{\text{horizon}}) = 0$. Intermediate values of κ_{sell} interpolate between these extremes. The interpolation is shown in figure 4.7, where it is clear that a transition point exists near $\kappa_{\text{sell}} \approx 33\%$: Below this value the optimizer hoards hydrogen in the tank (figure 4.4); above it, production aligns with the unrestricted baseline (section 4.1). Figure 4.5 shows how the capped case forces a three-phase schedule that respects the $13.5\,\text{g}$ limit while still meeting the contract.

Together, the entries in table 4.3 were selected because they expose informative tank behaviour at $\dot{n}_{\text{H}_2,\text{agreement}} = 6.5\,\text{NL/h}$: Hoarding at low κ_{sell} with unlimited storage (figure 4.4) and a practical capped schedule with galvanostatic tracking (figure 4.5).

5.3.2 Cyclic setup

For the 24-hour case (table 4.4), the horizon and N_t were scaled to one day with twelve two-hour segments. Terminal liquidation is set to $\kappa_{\text{sell}} = 0$ so that the periodic inventory constraint from section 3.1.3 governs end-of-day inventory rather than a sell-off term.

The continuity penalty (appendix 6) was added because exploratory NLP runs without it often converged to degenerate solutions with the same objective value but large, unphysical segment-to-segment jumps in T_{in}. The reported weight $w_T = 10^{-5}$ keeps that term small enough for load shifting to remain worthwhile. With the inlet bounds $[700,\ 900]\,^{\circ}\text{C}$ (200 K window), a single $10\,\text{K}$ step (5% of that window) contributes $w_T(\Delta T_{\text{in}})^2 \approx 0.001\,\text{EUR}$ to P_{cont}, i.e. $\approx 20\%$ of the mean running revenue per two-hour segment ($\approx 0.0051\,\text{EUR}$). For the reported optimum, the total penalty is $P_{\text{cont}} \approx 3.3 \times 10^{-4}\,\text{EUR}$ ($\approx 0.5\%$ of the daily Φ), and typical adjacent steps ($|\Delta T_{\text{in}}| \approx 1.9$–$2.3\,\text{K}$) cost $\lesssim 1\%$ of mean segment revenue each. The cosine power-price profile provides a simple, interpretable day-night swing.

5.4 Influence of modelling assumptions

The assumptions collected in section 2.3 are introduced to keep the average SRU model tractable for gradient-based optimal control. They are not neutral with respect to the results in section 4.1-section 4.3: The simplifications below directly shape which operating modes appear optimal and how aggressively the optimizer can run the plant.

5.4.1 Average SRU model and scale

Items 14–16 (section 2.3.4, section 2.3) mean the model works with a single averaged channel, so the optimized voltage (U) and inlet temperature (T_{in}) are the same everywhere in the system. This is different from what you would see if you looked at a more detailed model where conditions can vary in different parts of the stack [22, 24]. The model also simplifies things by scaling up delivery and using a single, lumped value for heating power (P_{heat}), which skips details like the way gas is routed through manifolds or heat is lost in exchangers. This makes the model more likely to favor temperature changes as a control strategy in some cases (section 4.2 and section 4.3). However, this simplification by itself is not what would produce odd or unrealistic results.

5.4.2 Electrochemical and thermal simplifications

Negligible concentration overpotential and Butler-Volmer kinetics (section 2.3, items 4, 8) keep the current-voltage relation smooth in the parameter range fitted by [18]. When explicit concentration terms are retained instead, mass-transport limitations add a current-dependent voltage loss: Mao et al. report that concentration overpotential from porous-electrode transport *cannot be neglected under high operating currents*, and that if steam supply falls below the electrochemical consumption rate the polarization curve warps upward [19]. Parametric SOEC models show the same sensitivity to steam fraction, porosity, and tortuosity [20]. Gas-phase conversion can likewise introduce flow-dependent overpotentials when compositions are non-uniform [22]. The present model omits these contributions, so the optimizer can raise voltage and current without that additional mass-transport penalty, consistent with the exothermal baseline at $U^* \approx 1.32\,\mathrm{V}$ (section 4.1, figure 3.3).

Dropping axial heat conduction (section 2.3, item 5) makes the along-channel temperature rise depend mainly on advection and local ohmic/activation heat release. Stack-scale models with resolved thermal fields instead report non-uniform temperature distributions and outlet hot spots that can amplify stress risks [24]. That distinction matters for the argument that exothermal operation homogenises nickel-migration growth (figure 4.3, section 5.2): Conductive spreading would redistribute heat and could shift both the temperature profile and the preferred operating point away from the present exothermal optimum [25].

5.4.3 Degradation scope and time integration

Restricting dynamic degradation to nickel migration and interconnect oxidation, while fixing nickel agglomeration through $\lambda_c \approx 0.55$ (section 2.3, items 9–10), narrows which failure modes the optimizer anticipates. The baseline replay reaches the nickel cap at $t = t_{\text{horizon}}$ while interconnect oxidation stays secondary (section 4.1), so conclusions about exothermal preference and uniform migration are conditional on that mechanism ranking. If interconnect oxidation were dominant or agglomeration evolved on the optimization horizon, thermoneutral or lower-current strategies might look preferable (chapter 7).

Spatially resolved degradation states (item 11) are necessary for the observed along-channel profiles, but the fitted nickel-migration rate and hard critical degradation limits remain empirical choices. Together with implicit Euler time stepping (item 13), they set how quickly the optimizer can approach end-of-life within the 200-day window (section 5.2). A slower or threshold-sharper degradation law would likely reduce current density and temper the exothermal optimum without changing the broader scheduling logic seen in the delivery and cyclic cases.

5.5 Model and numerical limitations

The results above are useful for qualitative comparison of operating modes, but several modelling and numerical choices bound how far the trajectories should be extrapolated. How the physical assumptions in section 2.3 feed into the optimal strategies is discussed in section 5.4; the subsections below focus on overall scope, discretization, and solver effects.

5.5.1 Physical scope and mechanism coverage

The average SRU model is a single one-dimensional channel with quasi-steady electrochemistry and temperature (section 2.3, table 2.1, chapter 2). Flow rates, pressure, and geometry are fixed; only U and T_{in} are optimized. That formulation is appropriate for asking how the stack-average repeat unit responds to price and delivery incentives, but it does not resolve stack-to-stack thermal coupling, manifold maldistribution, heat-exchanger transients, or plant-level auxiliary loads beyond the lumped P_{heat} bookkeeping (see section 5.4).

Degradation is limited to nickel migration and interconnect oxidation (table 2.2). Other ageing paths, cathode delamination, chromium poisoning, seal leakagem are not modelled dynamically. Nickel agglomeration enters only as a fixed multiplier $\lambda_c \approx 0.55$ on the exchange current (chapter 2), so any further cathode loss at high current is not captured. Together with the fitted nickel-migration rate law (section 2.2.1), this narrow mechanism set likely contributes to the short implied lifetimes discussed in section 5.2.

5.5.2 Spatial and temporal discretization

Spatial resolution and the scenario-dependent values of N_t, t_{horizon}, and the continuity weights are discussed in section 5.3. In brief, grid-independence analysis (figure 3.4) indicates that the budget-limited mesh in table 4.2 accepts spatial error on the order of 10^{-2}, which is adequate for comparing control modes but not for quantitative lifetime forecasting.

5.5.3 Nonlinear programming and solution quality

The optimal control problem is transcribed as a single-shooting NLP solved with SciPy's trust-region method and adjoint gradients (section 3.2.5, appendix 5). Gradient checks support the implementation, but the problem is not guaranteed to be convex in practice. The delivery scenario, with tank limit (section 4.2), returned more distinct local minima with equal objective value (without considering continuity penalty), which shows that reported trajectories need not be unique without extra constraints such as minimum inventory, continuity cost, or an operating-reserve preference.

Hard critical degradation limits stop the forward integration when thresholds are reached. That keeps the solver well posed but introduces a non-smooth boundary that can interact with the bounded salvage term (equation (3.4)). Finally, the rolling horizon t_{horizon} is finite and scenario-dependent (200 days versus 24 hours), so terminal salvage can dominate short windows even when running revenue drives the qualitative scheduling pattern (section 4.3).

5.6 Future Work

The limitations above point to several natural extensions of the present framework. The topics below were considered during the project and are listed here as concrete directions for follow-up work, without repeating the conclusions already drawn in this chapter.

5.6.1 Modelling and degradation

- Extend the degradation state beyond nickel migration and interconnect oxidation (table 2.2), or calibrate the fitted rate laws (section 2.2.1) against long-duration experiments at the current densities seen in optimization (section 4.1).

- Replace the fixed agglomeration multiplier λ_c with a dynamic state when operating far from the calibration point (chapter 2, section 2.3).
- Include auxiliary preheat power or a thermal-cycling degradation term so that segment-to-segment temperature changes carry an explicit cost (section 5.1, appendix 6).
- Move from the average SRU abstraction—with lumped heat exchange—to fully resolved stack- or plant-scale models with shared thermal and gas-management constraints (section 5.5; cf. [24]).

5.6.2 Objectives and economics

- Implement the risk-aware objective sketched in equation (5.1), or compare the present $\int R\,\mathrm{d}t + \Psi$ form directly with LCOH- and efficiency-based objectives from the literature (section 5.2; [4, 8, 14]).
- Break objective degeneracies in the delivery case with a minimum-inventory penalty, reserve constraint, or preference for smoother tank utilisation (section 5.5, section 4.2).
- Test sensitivity to κ_{buy}, κ_{sell}, α_{Ψ}, and alternative terminal treatments for tank inventory (section 4.2, section 4.3, figure 4.7).

5.6.3 AC:DC optimization

The short-horizon and periodic-control ideas below connect to electrothermally balanced operation and dynamic SOEC response discussed by Skafte et al. [25].

- Develop an objective function specifically designed for very short time-horizon optimizations, in which the cell's mechanical and electrochemical responses cannot be assumed to be instantaneous (section 4.3). This would require accounting for the finite dynamic response of the cell, capturing any lag between input changes and their physical effects.
- Extend the degradation modelling framework to include not only irreversible but also reversible degradation mechanisms (chapter 2). By doing so, it becomes possible to study phenomena where the cell's performance may partially recover under certain operating conditions, enabling a more nuanced optimization of lifetime and efficiency.
- Formulate a single-period control trajectory that is completely unconstrained in shape, allowing the optimization procedure to discover non-standard or unexpected control profiles that might yield improved performance (section 3.2.5). This unconstrained formulation could help identify new strategies that would be otherwise overlooked under more restrictive parameterizations.
- Establish realistic constraints for the periodic control function in order to ensure that the resulting schedules are physically feasible and implementable in practice (appendix 6, section 5.3.2). These constraints could include bounds on ramp rates, limits on the magnitude of control variables, or conditions that enforce smooth transitions between periods, reflecting actual system limitations and safety requirements.

6 Use of AI

Generative AI tools were used throughout this thesis. Aside from occasional questions to ChatGPT [21] for clarification, the main tool used, was Cursor [1], an AI-assisted code editor.

The tools were used first for code debugging and script generation. The core optimal-control implementation was written manually, with autocomplete and agent assistance from Cursor; subsequent debugging was also carried out with Cursor. Many of the bash submission scripts for optimization sweeps on the DTU HPC cluster [11], together with plotting scripts and gradient test scripts, were generated by Cursor and checked manually before use.

Cursor was also used as a writing aid for the thesis. It helped revise and tighten the wording of individual sections already drafted by the author, (smoothing transitions, clarifying technical claims, and aligning terminology across chapters) as well as check grammar and draft supplementary prose in the mathematically heavy appendices (appendix 1, appendix 5); all equations and derivations were done manually, with the generated text serving as supporting exposition.

General discussion

The use of generative AI in academic work is a pressing matter, and familiarity with such tools is increasingly expected in graduate projects [27]. In that sense, competent use of today's AI assistants can be seen as part of completing a modern Master's thesis. Here, the tools accelerated both coding and writing substantially. Without them, the project would likely be less complete, in part because a hand fracture slowed typing during a central period of the work. Coding agents helped keep development moving in that interval.

That period also illustrates the limits of the approach: Cursor was, at times, relied on more heavily than was ideal, which introduced defects that required intensive manual debugging later. Even so, the time spent correcting those issues was still less than writing and debugging the same code one-handed would have taken.

In summary, all modelling choices, physical interpretation, optimization results, and conclusions remain the author's responsibility. Generated code and text were reviewed, tested where applicable (section 3.2.2, appendix 7), and revised before inclusion. Generative AI sped up routine tasks but did not replace the scientific judgment underlying this thesis.

7 Conclusion

Through an analysis of the optimal control of cell voltage U and inlet temperature T_{in} for an average stack repeat unit (SRU) with a lumped heat exchanger (HEX), using a one-dimensional channel model with physically based degradation and a terminal salvage term (chapter 2, chapter 4 to section 4.3), a clear pattern emerges in the optimal schedules.

Across the studied cases, that pattern is potentiostatic operation: The optimizer holds voltage constant and uses inlet temperature as the main scheduling lever (section 4.1, section 4.2, section 4.3). Depending on the constraints, voltage is set to either thermoneutral operation, where the cell neither requires nor releases net heat, or exothermal operation, where ohmic heating raises the temperature along the gas channel. In the unrestricted baseline, when nothing prevents the optimizer from driving the cell toward its critical degradation limit while still selling the hydrogen produced, the preferred point is the more aggressive exothermal one ($U^* \approx 1.32\,\text{V}$): It spreads nickel migration more uniformly along the channel rather than concentrating aging near the inlet, and nickel migration reaches the $5\,\mu\text{m}$ critical limit within the 200-day horizon while interconnect oxidation stays below its cap (figure 4.3, section 4.1, section 5.2). That willingness to spend the stack for short-horizon profit is a strong indication that the present salvage term, based on a cell-area price in equation (3.4), underestimates the true economic cost of approaching end-of-life (section 5.2).

Under a fixed delivery contract, where not all produced hydrogen can necessarily be sold, and with unlimited buffer storage, the optimum instead stays near thermoneutral ($U^* \approx 1.29\,\text{V}$) in a single potentiostatic–isothermal hold: Surplus hydrogen fills the tank early and covers later shortfalls as degradation lowers production, so no galvanostatic intervals are needed (figure 4.4, section 4.2). When tank capacity is capped, the schedule splits into three phases: A gentle pot-iso fill, a galvanostatic middle with stepped T_{in} to track the agreement rate, and a closing pot-iso hold. Voltage still remains near thermoneutral (figure 4.5). Over a 24-hour cycle with a time-varying power price, voltage stays fixed at that same thermoneutral point and load shifting is done entirely through T_{in}, raising production at midday when electricity is cheap and lowering it overnight (figure 4.6, section 4.3). In both constrained settings, delivery and price obligations reshape the schedule more than the salvage term.

In answer to the question posed in section 1.2:

> *Physically modelled degradation does reshape the optimum when critical degradation limits bind and nickel migration is the critical mechanism: The preferred mode is then exothermal potentiostatic operation that homogenises aging, through nickel migration, along the channel. Through the salvage term alone, however, degradation has only a weak effect on the schedule, and only in a narrow window of hydrogen and power prices. External operating constraints dominate the schedule instead: A fixed delivery obligation forces galvanostatic intervals (constant current) near thermoneutral voltage, with temperature adjusted as needed, while cyclic power prices mainly shift production in time through inlet temperature, with voltage held fixed.*

Relative to prior degradation-aware SOEC studies reviewed in section 1.1, most of which compare a few fixed modes (galvanostatic, potentiostatic, or switched) at stack or plant scale [2–4, 14], this work treats voltage and temperature as freely co-optimised controls on a spatially resolved SRU, includes nickel migration and interconnect oxidation as slow states with critical limits, and embeds contractual offtake, buffer storage, and time-varying prices in the same objective. The main qualitative addition is that freely varying both U and T_{in} over time rarely pays off. Thermoneutral potentiostatic operation is the preferred fixed mode, and it is worth adjusting inlet temperature to meet demand or shift load while keeping voltage at that thermoneutral point. When inventory limits force a change of mode, the capped-tank schedule mixes potentiostatic and galvanostatic intervals, in agreement with Beyrami et al. [2], who also find that potentiostatic–galvanostatic operation is optimal.

Zooming out, early green-hydrogen projects are often built around long-term delivery contracts rather than spot sales alone, so the operator must meet agreed offtake while electricity prices and stack health evolve. Within the limits of an average-SRU model and the present economic objective, degradation-aware optimal control still points toward simple, near-potentiostatic schedules; market and contractual constraints dominate the day-to-day plan. Stronger valuation of remaining cell life, and eventually stack- or plant-scale thermal coupling, would be needed before the same toolchain can support quantitative lifetime forecasts rather than comparative operating strategies.

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Appendix

1 Deriving PDEs for species balances

Following [22], eqs. [A-3]–[A-5]:

$$\frac{PV}{RT}\frac{\partial x_{\mathrm{H_2O}}}{\partial t} = \underbrace{A_{\text{int}} N_{\text{in},\mathrm{H_2O}}}_{\text{Inlet}} + \underbrace{A_{\text{cell}} N_{a,\mathrm{H_2O}}}_{\text{Cell contribution}} - \underbrace{A_{\text{int}} N_{\text{out},\mathrm{H_2O}}}_{\text{Outlet}}$$

$$N_{\text{in/out},\mathrm{H_2O}} = v_{\text{fuel}}\, \mu_{\text{in/out},\mathrm{H_2O}}$$

$$N_{a,\mathrm{H_2O}} = \nu_{\mathrm{H_2O}} \frac{i}{2F}$$

The equations are also required in terms of the concentration $\mu_{\mathrm{H_2O}}$, defined by:

$$\mu_{\mathrm{H_2O}} = \frac{n_{\mathrm{H_2O}}}{V}$$

Substituting these relations and the ideal gas law, then discretizing, gives

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{A_{\text{cell}}}{V} \nu_{\mathrm{H_2O}} \frac{i}{2F} + \frac{A_{\text{int}}}{V} v_{\text{fuel}} (\mu_{\mathrm{H_2O}}(z - \Delta z) - \mu_{\mathrm{H_2O}}(z))$$

With $A_{\text{cell}} = w\Delta z$ and $V = \Delta z A_{\text{int}}$, isolating the time derivative yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w\Delta z}{A_{\text{int}}\Delta z} \nu_{\mathrm{H_2O}} \frac{i}{2F} + \frac{A_{\text{int}}}{A_{\text{int}}\Delta z} v_{\text{fuel}} (\mu_{\mathrm{H_2O}}(z - \Delta z) - \mu_{\mathrm{H_2O}}(z))$$

$$= \frac{w}{A_{\text{int}}} \nu_{\mathrm{H_2O}} \frac{i}{2F} + v_{\text{fuel}} \frac{(\mu_{\mathrm{H_2O}}(z - \Delta z) - \mu_{\mathrm{H_2O}}(z))}{\Delta z}$$

Letting $\Delta z \to 0$ yields

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{\text{int}}} \nu_{\mathrm{H_2O}} \frac{i}{2F} - v_{\text{fuel}} \frac{\partial \mu_{\mathrm{H_2O}}}{\partial z}$$

or, since $N_{\mathrm{H_2O}} = v_{\text{fuel}} \mu_{\mathrm{H_2O}}$,

$$\frac{\partial \mu_{\mathrm{H_2O}}}{\partial t} = \frac{w}{A_{\text{int}}} \nu_{\mathrm{H_2O}} \frac{i}{2F} - \frac{\partial N_{\mathrm{H_2O}}}{\partial z}$$

The same steps for H_2 and O_2 give

$$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{\text{int}}} \nu_i \frac{i}{2F} - \frac{\partial N_i}{\partial z}, \qquad \text{for } i \in \{H_2O, H_2, O_2\}$$

with $\nu_i = \left(1, -1, -\frac{1}{2}\right)^\top$.

1.1 Unit check (H_2O)

A unit check on the H_2O equation gives

$$\frac{\partial \mu_{H_2O}}{\partial t} = \frac{w}{A_{\text{int}}} \frac{i}{2F} - v_{\text{fuel}} \frac{\partial \mu_{H_2O}}{\partial z}$$

Left side units: - $\frac{\partial \mu_{H_2O}}{\partial t}$: $[mol \cdot m^{-3} \cdot s^{-1}]$

Right side units: - $\nu_{H_2O} \frac{w}{A_{\text{int}}} \frac{i}{2F}$: - ν_{H_2O} is dimensionless. - $\frac{w}{A_{\text{int}}}$: $[m]/[m^2] = [m^{-1}]$ - $\frac{i}{2F}$: $[A \cdot m^{-2}]/[C \cdot mol^{-1}]$

$$[m^{-1}] \cdot [A\, m^{-2}]/[C\, mol^{-1}] = [A\, m^{-3}]/[C\, mol^{-1}]$$

Next, since $1\, A = 1\, C/s$:

$$= [C\, s^{-1}\, m^{-3}]/[C\, mol^{-1}] = [mol\, s^{-1}\, m^{-3}]$$

So the total units are $[mol\, m^{-3}\, s^{-1}]$

- $v_{\text{fuel}} \frac{\partial \mu_{H_2O}}{\partial z}$: $[m \cdot s^{-1}] \cdot \frac{[mol \cdot m^{-3}]}{[m]} = [mol \cdot m^{-3} \cdot s^{-1}]$ (since $\partial \mu / \partial z$ has units concentration per length, i.e. $[mol \cdot m^{-3}]/[m] = [mol \cdot m^{-4}]$).

Both terms on the right match the left-hand side, $[mol \cdot m^{-3} \cdot s^{-1}]$.

2 Deriving PDEs for energy balances

Starting from the thermal energy equation with $\mathcal{S} = \{\mathrm{H_2O}, \mathrm{O_2}, \mathrm{H_2}\}$:

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T$$

With $\frac{\partial p}{\partial t} = \frac{\partial p}{\partial z} = 0$ and $h \approx c_p T$, this becomes

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (c_p T N_i) + Q_T$$

Neglecting thermal conductivity,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \nabla \cdot (T N_i) + Q_T$$

In one dimension,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \frac{\partial T N_i}{\partial z} + Q_T$$

With $N_i = v_i \mu_i$,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + Q_T$$

2.1 Source term

The volumetric source follows from the cell power,

$$P = i A_{\text{cell}} U,$$

which is the current times the cell voltage. Dividing by the volume gives Q_T:

$$Q_T = \frac{P}{V} = \frac{A_{\text{cell}}}{V} i U$$

With $A_{\text{cell}} = w dz$ and $V = A_{\text{int}} dz$,

$$Q_T = \frac{w}{A_{\text{int}}} iU$$

The full equation is therefore

$$\rho c_p \frac{\partial T}{\partial t} = -\sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + \frac{w}{A_{\text{int}}} iU$$

2.2 Steady-state limit

Expanding the advective term and setting $\partial_t T = 0$ gives

$$0 = -\sum_{i \in \mathcal{S}} v_i c_{p,i} \left(\frac{\partial T}{\partial z} \mu_i + T \frac{\partial \mu_i}{\partial z} \right) + \frac{w}{A_{\text{int}}} iU$$

Substituting $h_i = c_{p,i} T$ and the steady-state species balance $v_i \frac{\partial \mu_i}{\partial z} = \frac{w}{A_{\text{int}}} \nu_i \frac{i}{n_e F}$ yields

$$0 = -\sum_{i \in \mathcal{S}} \left(v_i c_{p,i} \frac{\partial T}{\partial z} \mu_i + \frac{w}{A_{\text{int}}} \frac{i}{n_e F} \nu_i h_i \right) + \frac{w}{A_{\text{int}}} iU$$

Factoring out the common terms,

$$0 = -\frac{\partial T}{\partial z} \sum_{i \in \mathcal{S}} v_i c_{p,i} \mu_i - i \frac{w}{A_{\text{int}}} \sum_{i \in \mathcal{S}} \nu_i h_i + i \frac{w}{A_{\text{int}}} U$$

With

$$\sum_{i \in S} v_i c_{p,i} \mu_i = C_p, \quad U_{\text{th}} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i h_i$$

this gives

$$0 = i \frac{w}{A_{\text{int}}} (U - U_{\text{th}}) - C_p \frac{\partial T}{\partial z}$$

3 Computing the finite-volume discretization

3.1 Species balances

From equation (2.5), equation (2.6) and equation (2.7), integration over a cell volume gives

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \int_{\Omega_k} \frac{w}{A_{\text{int}}} N_{c,i}\, dV - \int_{\Omega_k} \frac{\partial N_i}{\partial z}\, dV$$

The time derivative can be taken outside the integral, and integrating the concentration over a volume gives the moles in that volume:

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t}\, dV = \partial_t \int_{\Omega_k} \mu_i\, dV = \frac{\partial n_i}{\partial t} = \dot{n}_i$$

Using the approximation in section 3.2.1 with $V_k = A_{\text{int}} \Delta z$,

$$\int_{\Omega_k} \frac{w}{A_{\text{int}}} N_{c,i}\, dV \approx \frac{w}{A_{\text{int}}} N_{c,i}(z_k) V_k = w N_{c,i}(z_k) \Delta z$$

For the flux term, integration over z is performed first, then over the cross-sectional area:

$$\begin{aligned}
\int_{A_{\text{int}}} \int_{z_{k-\frac{1}{2}}}^{z_{k+\frac{1}{2}}} \frac{\partial N_i}{\partial z}\, dz\, dA &\approx \int_{A_{\text{int}}} \Big(N_i(z_k) - N_i(z_{k-1}) \Big)\, dA \\
&= \Big(N_i(z_k) - N_i(z_{k-1}) \Big) A_{\text{int}}
\end{aligned}$$

Each cell therefore satisfies

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) = \dot{n}_i(z_k) \approx w N_{c,i} \Delta z - A_{\text{int}}(N_i(z_k) - N_i(z_{k-1}))$$

3.2 Energy balance

The steady-state form is discretized first, then the transient term.

3.2.1 Steady state

Starting from equation (2.18):

$$0 = i\frac{w}{A_{\text{int}}}(U - U_{\text{th}}) - C_p\frac{\partial T}{\partial z}$$

The same procedure as for the species balances gives

$$0 = \int_{\Omega_k} i\frac{w}{A_{\text{int}}}(U - U_{\text{th}}) - \int_{\Omega_k} C_p\frac{\partial T}{\partial z}$$

As before,

$$\int_{\Omega_k} C_p\frac{\partial T}{\partial z}\,dV \approx C_p(T(z_k) - T(z_{k-1}))A_{\text{int}}$$

Collecting terms,

$$0 \approx iw(U - U_{\text{th}})\Delta z - C_p(T(z_k) - T(z_{k-1}))A_{\text{int}}$$

3.2.2 Transient term

For the time derivative,

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} = \int_{\Omega_k} Q_T - v\sum_{i\in\mathcal{S}}\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z}$$

On the left-hand side, ρ integrated over the cell volume is the cell mass m_k, so

$$\int_{\Omega_k} \rho c_p\frac{\partial T}{\partial t} \approx m_k c_p\frac{\partial T}{\partial t}(z_k)$$

The source term follows the same pattern as above, $\int_{\Omega_k} Q_T = Q_T(z_k)A_{\text{int}}\Delta z$. For the advective flux, one species is shown explicitly; the others are analogous:

$$\int_{\Omega_k} c_{p,i}\frac{\partial T\mu_i}{\partial z} \approx c_{p,i}\Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big)A_{\text{int}}$$

The discrete energy balance is therefore

$$m_k c_p\frac{\partial T}{\partial t}(z_k) \approx Q_T(z_k)A_{\text{int}}\Delta z - \sum_{i\in\mathcal{S}} v_i c_{p,i}\Big(T(z_k)\mu_i(z_k) - T(z_{k-1})\mu_i(z_{k-1})\Big)A_{\text{int}}$$

4 Relevant gradients for the numerical solver

4.1 Molar balance

$$\text{Jacobian}_{\mu}[x] = \begin{bmatrix} \frac{1}{\mu^f} & -\frac{1}{\mu^f} & 0 \\ -\frac{1}{\mu^f} & \frac{1}{\mu^f} & 0 \\ 0 & 0 & \frac{\mu_{\text{in},N_2}}{(\mu_{\text{in},N_2}+\mu_{O_2})} \end{bmatrix}$$

Then

$$\nabla_{\mu} f(x) = \left(\text{Jacobian}_{\mu}[x]\right)^{\top} \nabla_x f$$

4.1.1 Nernst Equation

Gradient for molar fractions: ∇_x

$$\nabla_x \eta_{\text{nernst}} = \frac{RT}{2F} \begin{bmatrix} -\frac{1}{x_{H_2O}} \\ \frac{1}{x_{H_2}} \\ \frac{1}{2x_{O_2}} \end{bmatrix}$$

Gradient for temperature: ∇_T

$$\nabla_T \eta_{\text{nernst}} = \frac{\nabla_T(\Delta G)}{2F} + \frac{R}{2F} \log \left\{ \frac{x_{H_2}\sqrt{x_{O_2}}}{x_{H_2O}\sqrt{x_0}} \right\}$$

$$\nabla_T(\Delta G) = -0.0062T + -49.119$$

4.1.2 Activation overpotential

Gradient for molar fractions: ∇_x

$$\nabla_x(\text{Butler-Volmer}_{\zeta}) = \nabla_x(i_0^{\zeta})\Big(\exp\left[\alpha_{\zeta}\tfrac{n_e F}{RT}\eta_{\text{act}}^{\zeta}\right] - \exp\left[-(1-\alpha_{\zeta})\tfrac{n_e F}{RT}\eta_{\text{act}}^{\zeta}\right]\Big)$$

$$\nabla_x(i_0^a) = \gamma_a T \exp\left\{-\frac{E_{\text{act}}^a}{RT}\right\} \begin{bmatrix} 0 \\ 0 \\ m_{O_2}\left(\frac{x_{O_2}}{x_0}\right)^{m_{O_2}-1} \end{bmatrix}$$

$$\nabla_x(i_0^f) = \lambda_f \gamma_f T \exp\left\{-\frac{E_{\text{act}}^f}{RT}\right\} \begin{bmatrix} m_{H_2O}\left(\frac{x_{H_2O}}{x_0}\right)^{m_{H_2O}-1}\left(\frac{x_{H_2}}{x_0}\right)^{m_{H_2}} \\ m_{H_2}\left(\frac{x_{H_2O}}{x_0}\right)^{m_{H_2O}}\left(\frac{x_{H_2}}{x_0}\right)^{m_{H_2}-1} \\ 0 \end{bmatrix}$$

Gradient for temperature: ∇_T

$$\nabla_T(\text{Butler-Volmer}_\zeta) = \nabla_T(i_0^\zeta)\Big(\exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right] - \exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right]\Big)$$
$$+\, i_0^\zeta\Big(-\alpha_\zeta \tfrac{n_e F}{RT^2}\eta_{\text{act}}^\zeta \exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right] - (1-\alpha_\zeta)\tfrac{n_e F}{RT^2}\eta_{\text{act}}^\zeta \exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right]\Big)$$

$$\nabla_T(i_0^\zeta) = \left(1 + \frac{E_{\text{act}}^\zeta}{RT}\right)\lambda_\zeta\gamma_\zeta \exp\left\{-\frac{E_{\text{act}}^\zeta}{RT}\right\}\prod_{i\in\mathcal{S}}\left(\frac{x_i}{x_0}\right)^{m_i} = \frac{i_0^\zeta}{T}\left(1 + \frac{E_{\text{act}}^\zeta}{RT}\right)$$

Gradient for activation overpotential: $\nabla_{\eta_{\text{act}}^\zeta}$

$$\nabla_{\eta_{\text{act}}^\zeta}(\text{Butler-Volmer}_\zeta) = i_0^\zeta\Big(\alpha_\zeta \tfrac{n_e F}{RT}\exp\left[\alpha_\zeta \tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right] + (1-\alpha_\zeta)\tfrac{n_e F}{RT}\exp\left[-(1-\alpha_\zeta)\tfrac{n_e F}{RT}\eta_{\text{act}}^\zeta\right]\Big)$$

4.2 Degradation

4.2.1 Ohmic degradation

Gradient for molar fractions: ∇_x

$$\nabla_x R_{\text{ohm}} = \mathbf{0}$$

Gradient for temperature: ∇_T

$$\nabla_T R_{\text{ohm}} = \left(1 - \frac{E_{a,\text{ohm}}}{RT}\right)\frac{1}{B_{\text{ohm}}}\exp\left\{\frac{E_{a,\text{ohm}}}{RT}\right\} + \sum_{d\in\mathcal{D}_{\text{ohm}}}\nabla_T R_d$$

$$\nabla_T R_{\text{Ni-mi}} = d_{\text{Ni-mi}}\frac{T - E_{\text{act,elec}}/R}{\sigma_{\text{elec,0}} T\exp\{-\frac{E_{\text{act,elec}}}{RT}\}}$$

4.3 Temperature equations

4.3.1 Specific heat

Gradient for molar fractions: ∇_μ

$$\nabla_\mu C_p = \begin{bmatrix} c_{p,H_2O} \\ c_{p,H_2} \\ c_{p,O_2} \end{bmatrix}$$

Gradient for temperature: ∇_T

$$\nabla_T C_p = \sum_{i\in\mathcal{S}}\mu_i \nabla_T c_{p,i}$$

4.3.2 Thermaneutral voltage

Gradient for molar fractions: ∇_μ

$$\nabla_\mu U_{\text{th}} = \mathbf{0}$$

Gradient for temperature: ∇_T

$$\nabla_T U_{\text{th}} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i \nabla_T h_i$$

5 NLP Gradients

This appendix collects the gradients used in the NLP formulation of equation (3.1). The discrete chain rule for the stage model is stated first, followed by the partial derivatives that enter it and the linear systems for sensitivities of the spatial FVM model and the slow trajectory in time.

5.1 Discrete chain rule for the stage model

Between shooting stages, the integrated objective contribution ϕ and the degradation variables d satisfy a discrete sensitivity chain. For a scalar stage index i and a control component u_k,

$$\frac{\mathrm{d}}{\mathrm{d}u_k}\phi_{i+1} = \nabla_{u_i}\dot{\phi}\cdot\nabla_{u_k}u_i + \nabla_{d_i}\dot{\phi}\,\frac{\mathrm{d}}{\mathrm{d}u_k}d_i + \frac{\mathrm{d}}{\mathrm{d}u_k}\phi_i,$$

with the analogous relation for the degradation state,

$$\frac{\mathrm{d}}{\mathrm{d}u_k}d_{i+1} = (\nabla_{u_i}\dot{d}_i)\,\nabla_{u_k}u_i + (\nabla_{d_i}\dot{d}_i)\,\frac{\mathrm{d}}{\mathrm{d}u_k}d_i + \cdots,$$

where the trailing terms collect any additional dependence of the stage map on u_k (for example through implicit stage equations).

The partial gradients of the instantaneous rates are

$$\begin{aligned}
\nabla_u\dot{\phi} &= c_{\mathrm{H_2}}\nabla_u\frac{\partial n_{\mathrm{H_2}}}{\partial t} - c_P(I + u\nabla_u I),\\
\nabla_d\dot{\phi} &= c_{\mathrm{H_2}}\nabla_d\frac{\partial n_{\mathrm{H_2}}}{\partial t} - c_P u\nabla_d I - \left[c_{i,\text{utilization}}\frac{1}{d_{i,\text{critical}}}\right]_{i\in\mathcal{D}_{\text{fail}}},\\
\nabla_\phi\dot{\phi} &= 0,\\
\nabla_u\dot{d} &= c_1\nabla_u T + c_2\nabla_u i + c_3(\nabla_u i\,T + i\nabla_u T),\\
\nabla_d\dot{d} &= c_1\nabla_d T + c_2\nabla_d i + c_3(\nabla_d i\,T + i\nabla_d T),\\
\nabla_\phi\dot{d} &= 0.
\end{aligned}$$

Here the nickel-migration rate is $\dot{d} = c_1T + c_2i + c_3iT + c_4$ as in table 2.2; the constant c_4 drops out of the sensitivities.

Note that $\nabla_\zeta I = \frac{A_{\text{cell}}}{N_{\text{cell}}}\sum_{j=1}^{N_{\text{cell}}}\nabla_\zeta i_j$ for $\zeta\in\{u, d, \phi, \ldots\}$.

The remaining ingredients are the gradients of the current density with respect to cell voltage and local degradation state; these are given below. Further electrochemical partials are listed in appendix 4.

5.2 Gradient with respect to cell voltage: $\nabla_u i$

The current is expressed in terms of fuel- and air-side activation branches in the same notation as appendix 4. With scalars $\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)$ and $\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)$ denoting $\partial i/\partial\eta^f_{\text{act}}$ and $\partial i/\partial\eta^a_{\text{act}}$ at the operating point,

$$\nabla_u i = \frac{1}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

5.3 Gradient with respect to degradation: $\nabla_d i$

Similarly, with $\nabla_d R_{\text{ohm}}$ the partial of the ohmic resistance with respect to the local degradation variable,

$$\nabla_d i = -i\,\frac{\nabla_d R_{\text{ohm}}}{R_{\text{ohm}} + \dfrac{1}{\nabla_{\eta^f_{\text{act}}}(\text{Butler-Volmer}_f)} + \dfrac{1}{\nabla_{\eta^a_{\text{act}}}(\text{Butler-Volmer}_a)}}.$$

5.4 Sensitivity of the spatial FVM discretization

At each time level the converged spatial state satisfies a residual of flux form. With cell index k along the stack, let $Q(u_k)$ denote the cellwise source residual and $F(u_k)$ the numerical flux between neighbours. The source term is scaled by the cell thickness Δz and the flux difference by the interface area A_{int}, so the semi-discrete steady relation is

$$0 = \Delta z\, Q\left(u_k\right) - A_{\text{int}}\left(F\left(u_k\right) - F\left(u_{k-1}\right)\right).$$

Differentiating with respect to a parameter p (a control coefficient, a boundary value, etc.) and writing $\partial u_k/\partial p$ for the sensitivity of the converged state gives the familiar block recursion

$$\begin{aligned}\left(\Delta z\,\frac{\partial Q}{\partial u_k}\left(u_k\right) - A_{\text{int}}\frac{\partial F}{\partial u_k}\left(u_k\right)\right)\frac{\partial u_k}{\partial p} + A_{\text{int}}\frac{\partial F}{\partial u_{k-1}}\left(u_{k-1}\right)\frac{\partial u_{k-1}}{\partial p}\\ = A_{\text{int}}\left(\frac{\partial F}{\partial p}\left(u_k\right) - \frac{\partial F}{\partial p}\left(u_{k-1}\right)\right) - \Delta z\,\frac{\partial Q}{\partial p}\left(u_k\right).\end{aligned}$$

Stacking all cells, this is the linear system

$$J_{\text{fvm}}\frac{\partial \mathbf{u}}{\partial p} = b^{(p)}_{\text{fvm}},$$

with the block-lower-bidiagonal Jacobian and right-hand side

$$J_{\text{fvm}} = \begin{bmatrix} \Delta_1 & & & \\ A_{\text{int}} \dfrac{\partial F}{\partial u_1}(u_1) & \Delta_2 & & \\ & \ddots & \ddots & \\ & & A_{\text{int}} \dfrac{\partial F}{\partial u_{N-1}}(u_{N-1}) & \Delta_N \end{bmatrix},$$

$$\text{with } \Delta_k = \Delta z \, \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k),$$

$$b_{\text{fvm}}^{(p)} = \begin{bmatrix} A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_1) - \frac{\partial F}{\partial p}(u_{\text{in}}) \right) - \Delta z \, \frac{\partial Q}{\partial p}(u_1) \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_2) - \frac{\partial F}{\partial p}(u_1) \right) - \Delta z \, \frac{\partial Q}{\partial p}(u_2) \\ \vdots \\ A_{\text{int}} \left(\frac{\partial F}{\partial p}(u_N) - \frac{\partial F}{\partial p}(u_{N-1}) \right) - \Delta z \, \frac{\partial Q}{\partial p}(u_N) \end{bmatrix}$$

The same block structure matches the flux assembly used in appendix 3.

5.5 Sensitivity of the ODE system

For this section the problem is written in general form:

$$\begin{aligned} \max_u \Big\{ \text{Objective} = & \int_0^{t_{\text{horizon}}} \Phi(X, d, u)\, dt + \Psi(X, d, u) \Big\} \\ \text{s.t.} \qquad 0 &= R_{\text{fvm}}(X, d, u) \\ \dot{d} &= f(X, d, u) \end{aligned}$$

The method of adjoint gradients [23], [9] is used, with the following discretization ($t_0 = 0 < t_1 \cdots t_i \cdots t_N = T$):

$$X(t_i) \approx X_i, \quad d(t_i) \approx d_i, \quad u(t_i) \approx u_i.$$

$$\begin{aligned} R_{\text{fvm},i} &= R_{\text{fvm}}(X_{i+1}, d_{i+1}, u_i), \\ D_i &= d_{i+1} - f(X_{i+1}, d_{i+1}, u_i)\Delta t - d_i \\ \Phi_i &= \Phi(X_{i+1}, d_{i+1}, u_i)\Delta t \approx \int_{t_i}^{t_{i+1}} \Phi(X, d, u) dt \end{aligned}$$

The combined residual for the FVM constraints and the ODE defect is then:

$$R_i = \begin{bmatrix} D_i \\ R_{\text{fvm},i} \end{bmatrix}$$

The state variables are combined as

$$w_i = \begin{bmatrix} X_i \\ d_i \end{bmatrix}$$

The Jacobian of the residual with respect to the internal and control variables is

$$\frac{\partial R_i}{\partial w_i} = \begin{bmatrix} \mathbf{0} & \frac{\partial D_i}{\partial d_i} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \frac{\partial R_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial D_i}{\partial X_{i+1}} & \frac{\partial D_i}{\partial d_{i+1}} \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial R_i}{\partial u_i} = \begin{bmatrix} \frac{\partial D_i}{\partial u_i} \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i} \end{bmatrix}$$

$$\frac{\partial \Phi_i}{\partial w_{i+1}} = \begin{bmatrix} \frac{\partial \Phi_i}{\partial X_{i+1}} \\ \frac{\partial \Phi_i}{\partial d_{i+1}} \end{bmatrix} \quad \frac{\partial \Phi_i}{\partial u_i}$$

For implementation these blocks are spelled out in more detail:

$$\frac{\partial R_i}{\partial w_i} = \begin{bmatrix} \mathbf{0} & -\mathbf{I} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$
$$\frac{\partial R_i}{\partial w_{i+1}} = \begin{bmatrix} -\Delta t \frac{\partial f}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \mathbf{I} - \Delta t \frac{\partial f}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix}$$
$$\frac{\partial R_i}{\partial u_i} = \begin{bmatrix} -\Delta t \frac{\partial f}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix}$$

The blocks $\partial R_{\text{fvm},i}/\partial X_{i+1}$ coincide with J_{fvm} from the spatial sensitivity section (same flux assembly). At each time level:

$$\frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} = J_{\text{fvm}} \quad \frac{\partial R_{\text{fvm},i}}{\partial u_i} = \begin{bmatrix} | & | \\ b_{\text{fvm}}^{(u_1^{(U)})} & b_{\text{fvm}}^{(u_i^{(T_0)})} \\ | & | \end{bmatrix} \quad \frac{\partial R_{\text{fvm},i}}{\partial d_i} = b_{\text{fvm}}^{(d)}$$

With these blocks, the objective gradient with respect to u_i is

$$\nabla_{u_i} \text{Objective} = \frac{\partial \Phi_i}{\partial u_i} + \left(\frac{\partial R_i}{\partial u_i}\right)^\top \lambda_{i+1} \quad \text{for} \quad i = 1, \dots, N$$

The adjoint variables $\Lambda = (\lambda_k)_k$ satisfy

$$\left(\frac{\partial R_{N-1}}{\partial w_N}\right)^\top \lambda_N = -\frac{\partial \Phi_{N-1}}{\partial w_N} - \frac{\partial \Psi}{\partial w_N},$$
$$\left(\frac{\partial R_{i-1}}{\partial w_i}\right)^\top \lambda_i + \left(\frac{\partial R_i}{\partial w_i}\right)^\top \lambda_{i+1} = -\frac{\partial \Phi_{i-1}}{\partial w_i}, \qquad i = 1, \dots, N-1.$$

Equivalently, one large block system can be assembled for Λ.

$$\begin{bmatrix} \frac{\partial R_0}{\partial w_1} & & & & \\ \frac{\partial R_1}{\partial w_1} & \frac{\partial R_1}{\partial w_2} & & & \\ & \ddots & \ddots & & \\ & & & \frac{\partial R_{N-2}}{\partial w_{N-1}} & \\ & & & \frac{\partial R_{N-1}}{\partial w_{N-1}} & \frac{\partial R_{N-1}}{\partial w_N} \end{bmatrix}^\top \Lambda = - \begin{bmatrix} \frac{\partial \Phi_0}{\partial w_1} \\ \frac{\partial \Phi_1}{\partial w_2} \\ \vdots \\ \frac{\partial \Phi_{N-2}}{\partial w_{N-1}} \\ \frac{\partial \Phi_{N-1}}{\partial w_N} + \frac{\partial \Psi}{\partial w_N} \end{bmatrix}$$

6 Squared continuity penalty

Piecewise-constant controls on N_t segments can introduce jumps between adjacent intervals. When both voltage and inlet temperature are optimized segment-wise, the optimizer may exploit those jumps unless they are penalized. The implementation adds an optional soft penalty to the running objective, proportional to the sum of squared differences between neighbouring segment values.

For segment index $k = 0, \dots, N_t - 2$, let U_k and $T_{\text{in},k}$ denote the piecewise-constant controls on segment k. With non-negative weights w_U and w_T (CLI: `--continuity-weight-u`, `--continuity-weight-T`), the penalty is

$$P_{\text{cont}} = w_U \sum_{k=0}^{N_t-2} (U_{k+1} - U_k)^2 + w_T \sum_{k=0}^{N_t-2} (T_{\text{in},k+1} - T_{\text{in},k})^2.$$

It is added to the running economic cost before integration over the horizon. Setting $w_U = w_T = 0$ disables the term (the default in table 4.2).

In **potentiostatic** operation ($U_k \equiv U$), voltage is a single decision variable rather than a length-N_t profile, so adjacent voltage differences vanish and the w_U term contributes nothing. Only the inlet-temperature sum is active in that mode.

The penalty is smooth and cheap to evaluate; its gradient with respect to the control knots is implemented analytically in the NLP (`code/src/scripts/nlp/continuity.py`). Saved run metadata reports P_{cont}, P_U, and $P_{T_{\text{in}}}$ as the two sums in appendix 6.

7 Gradient verification plots

The NLP solver uses adjoint gradients assembled from the partial derivatives in appendix 4 and appendix 5. Before running the optimization cases in chapter 4–section 4.3, analytical derivatives were checked against central finite-difference (FD) approximations. Standalone scripts in `code/src/standalone/` generate the figures below; additional component-level checks exist in `code/figures/testers/gradients_test/` but are omitted here.

7.1 End-to-end NLP objective

Figure A7.1 sweeps uniform cell voltage and inlet temperature on a single-shooting problem with $N_t = 100$ time segments ($\Delta t = 4320\,\mathrm{s}$), exercising the full adjoint chain over a long horizon. The adjoint totals $\sum_j \partial f/\partial U_j$ and $\sum_j \partial f/\partial T_{\mathrm{in},j}$ match the FD curves over the operating range.

7.2 Spatial FVM Jacobian

The steady-state cell residual $r(\mu_{H_2}, \mu_{O_2}, T)$ must be differentiated when assembling the block-bidiagonal system in appendix 5. Figure A7.2 compares each Jacobian entry to a secant FD estimate; the curves agree on the test grid.

7.3 Electrochemical coupling

Current density enters both the degradation rates and the economic objective. Figure A7.3 and figure A7.4 verify $\nabla_U i$ and $\nabla_d i$ from appendix 5. Figure A7.5 checks the fuel-side Butler–Volmer branch and its partials w.r.t. temperature, composition, and activation overpotential.

7.4 Instantaneous economic rate

Figure A7.6 checks partial derivatives of the running economic rate $\dot{\phi}$ w.r.t. outlet chemical potentials and the stack-outlet temperature that feed the stage objective in appendix 5.

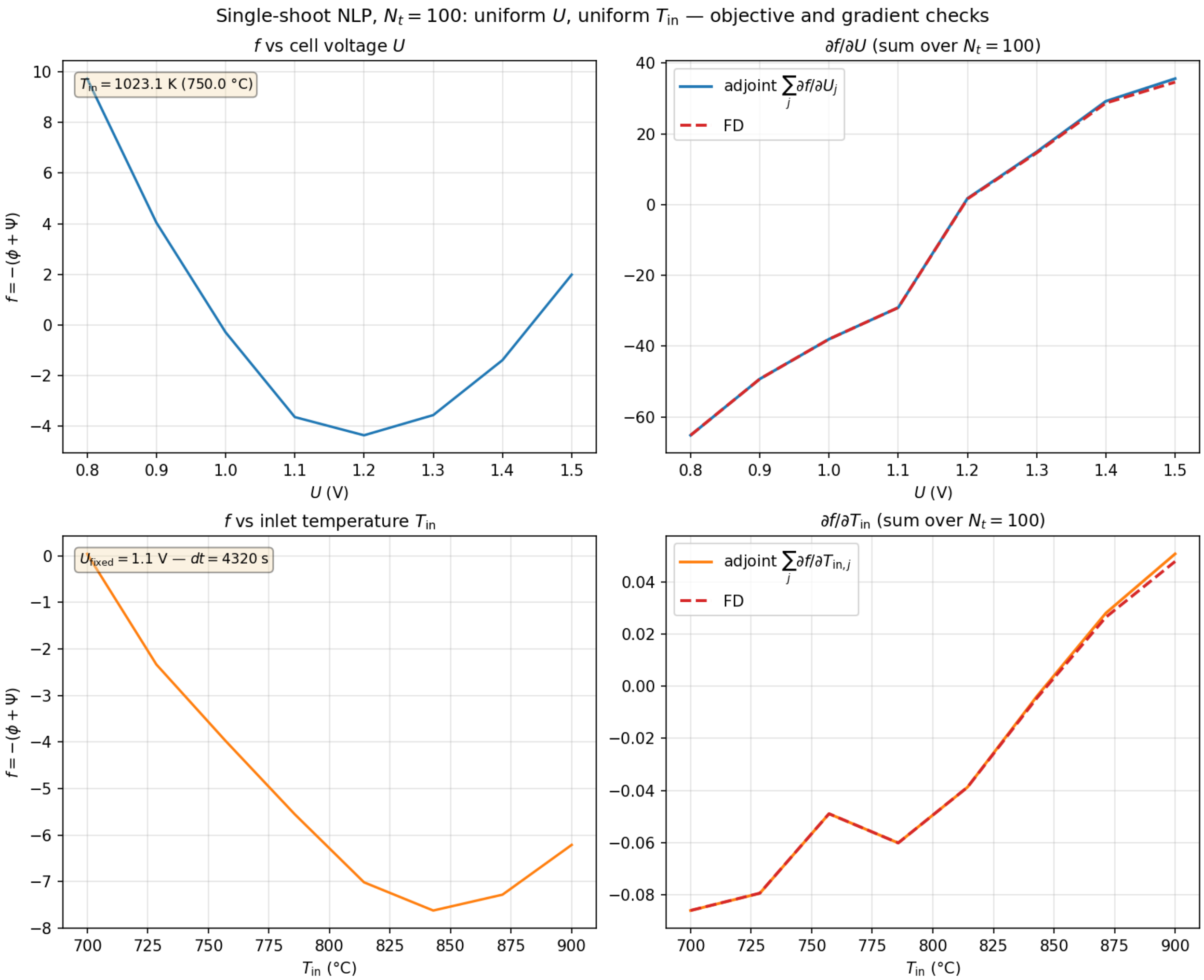

Figure A7.1: Single-shooting NLP gradient check at $N_t = 100$: Objective $f = -(\phi + \psi)$ and its derivatives w.r.t. uniform cell voltage U (top) and inlet temperature T_{in} (bottom). Solid lines: Adjoint; dashed: Central FD with all segment controls perturbed together.

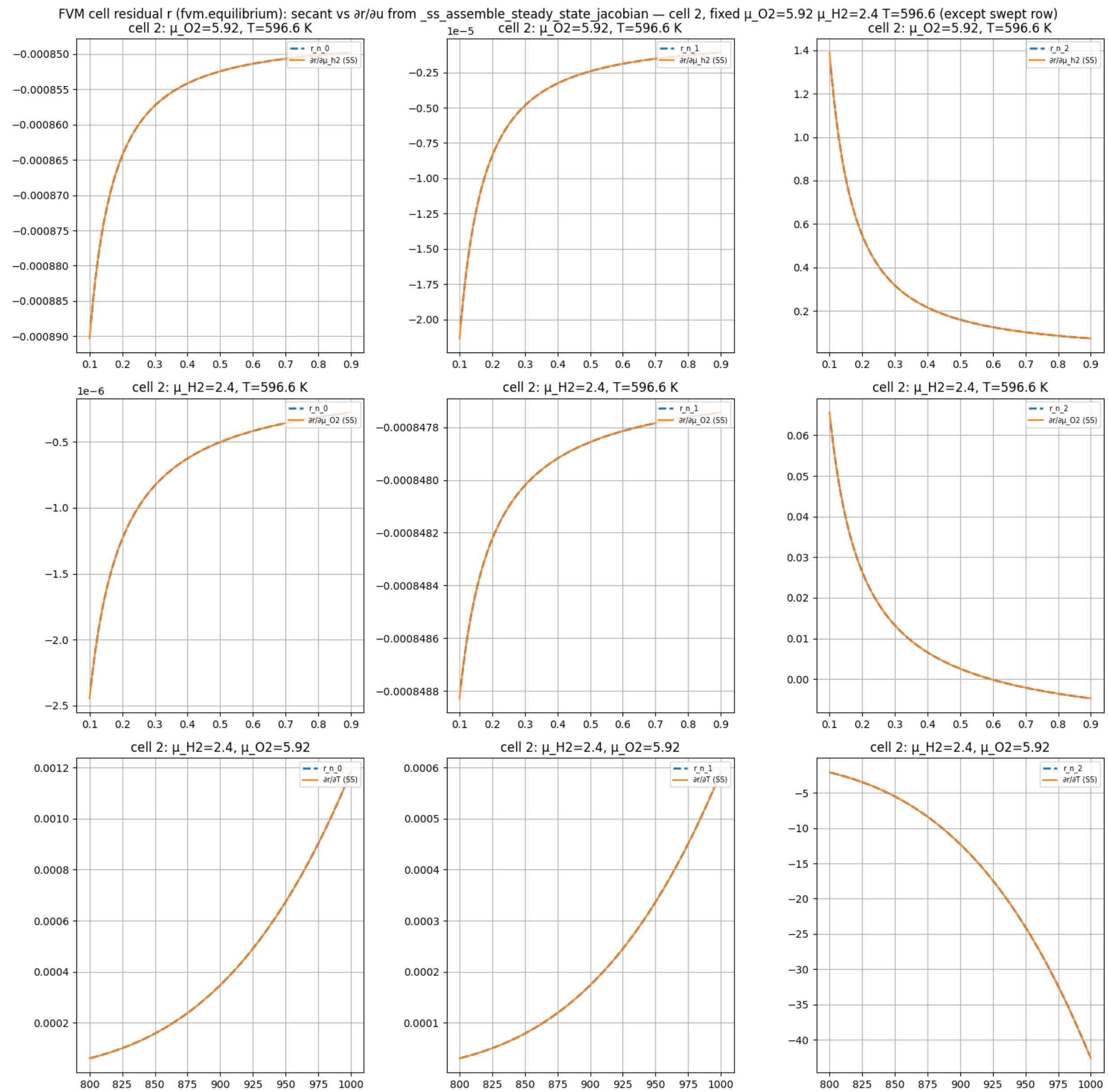

Figure A7.2: Steady-state FVM residual Jacobian vs. secant FD for the three residual components r_0, r_1, r_2 and inputs μ_{H_2}, μ_{O_2}, and T. Dashed: Numerical secant; solid: Analytical steady-state Jacobian.

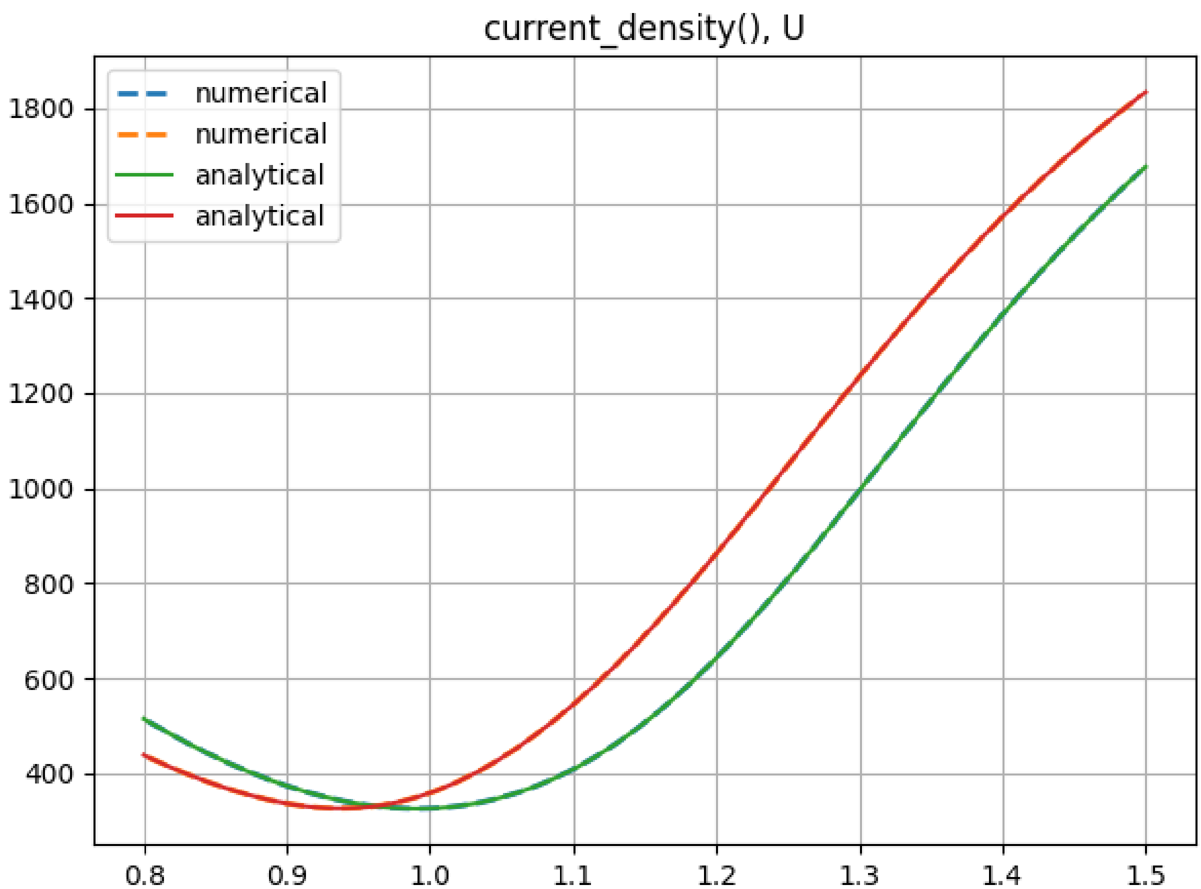

Figure A7.3: Analytical vs. numerical gradient of cell current density w.r.t. cell voltage U.

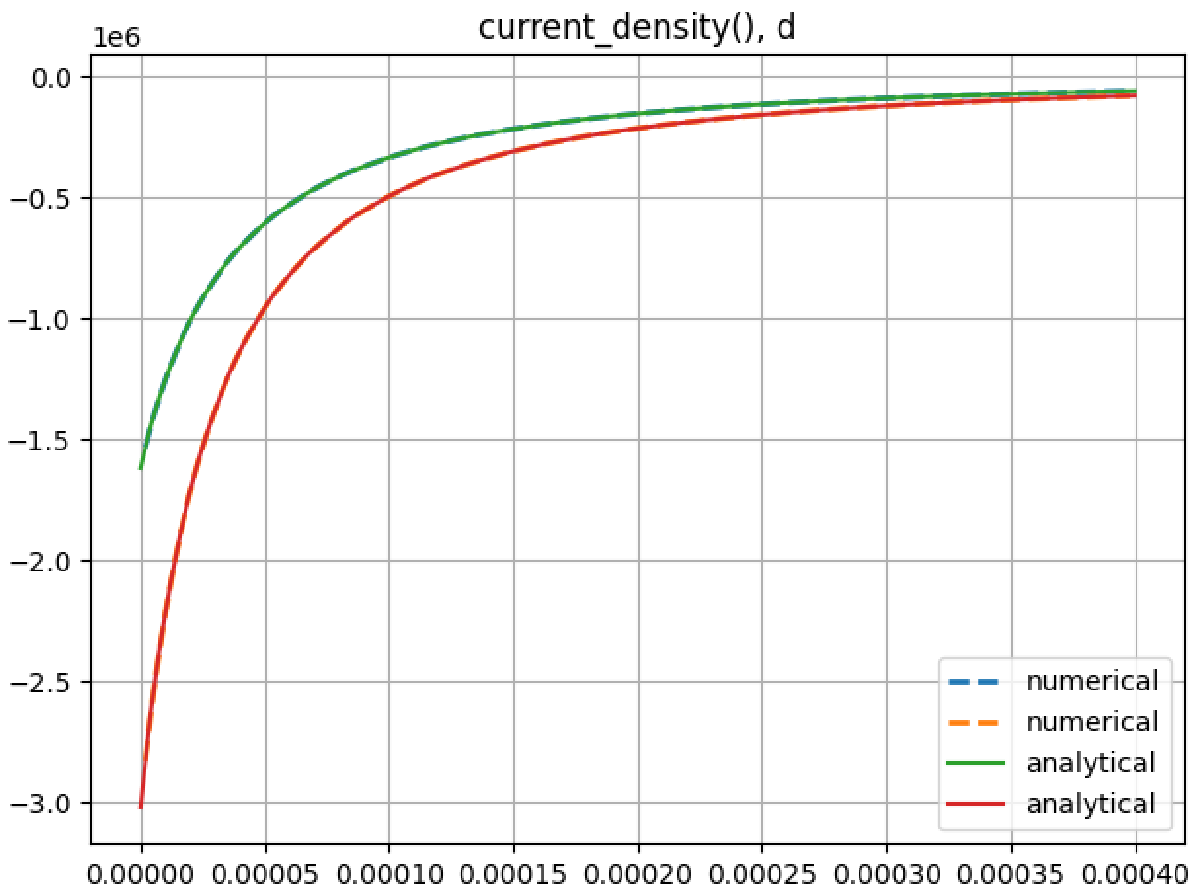

Figure A7.4: Analytical vs. numerical gradient of cell current density w.r.t. local degradation state d.

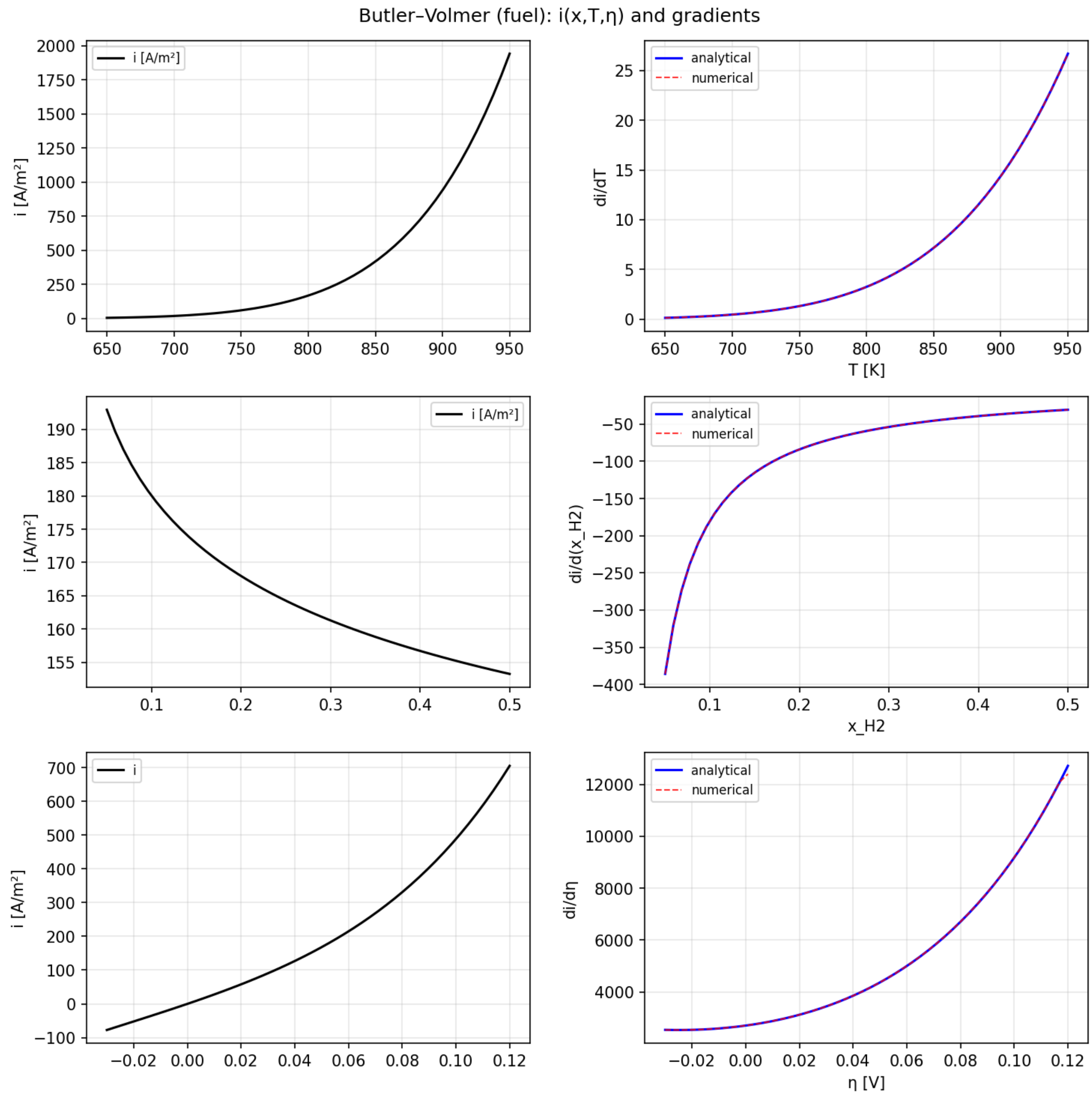

Figure A7.5: Fuel-side Butler–Volmer current $i(x, T, \eta)$ and gradients w.r.t. T, x_{H_2}, and activation overpotential η. Left column: i; right column: Analytical (solid) vs. numerical (dashed) partial derivatives.

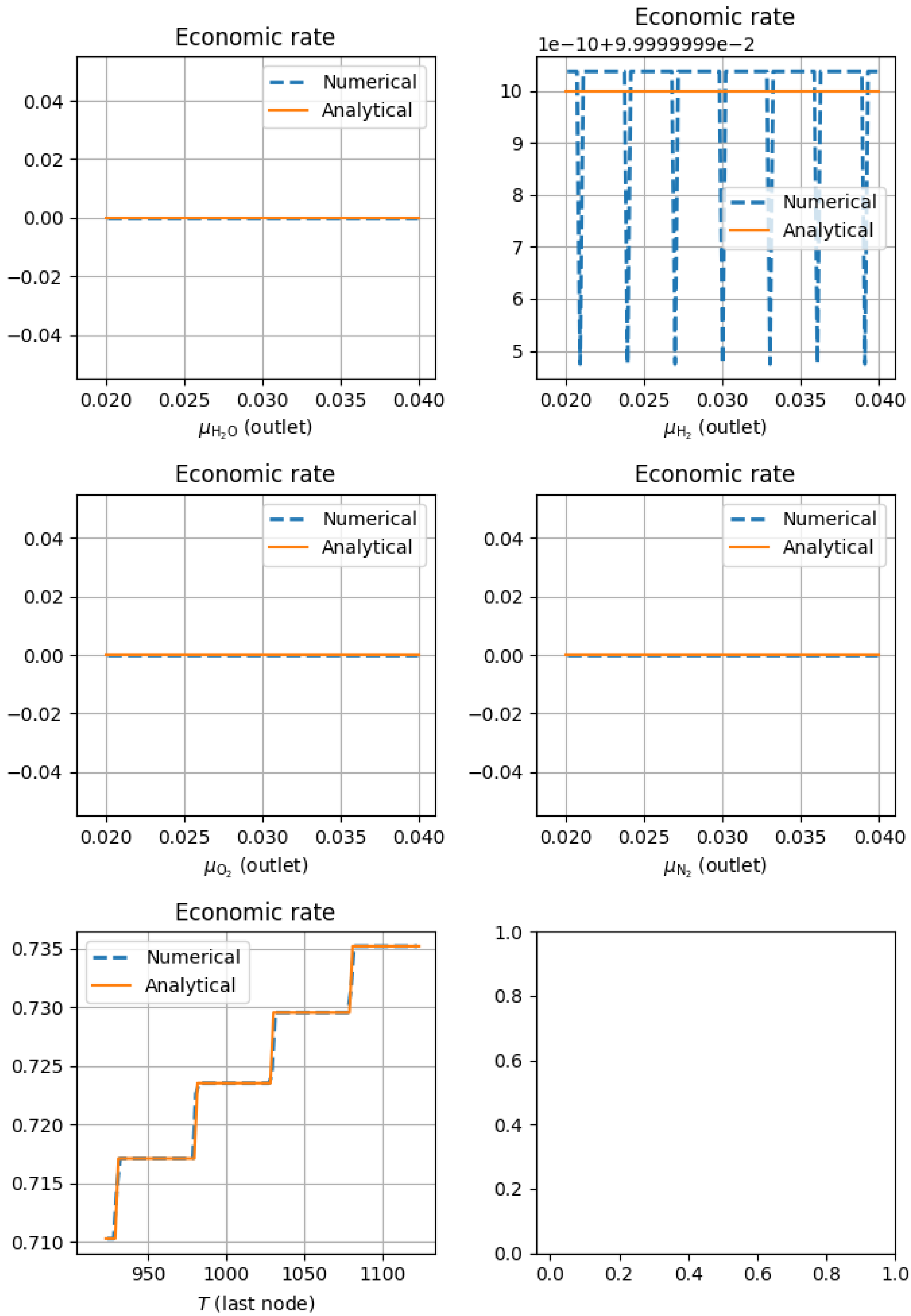

Figure A7.6: Analytical vs. numerical gradients of the instantaneous economic rate w.r.t. outlet species chemical potentials and the last-node temperature T.